Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises

B. Müller Mario Poneder Alexander Osika Kai Xue Peter Nguyen
David Matscheko Daniel Ari Friedman

November 1, 2026

Paper release: `r2057` **Released:** November 1, 2026

Author affiliations: Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks; Daniel Ari Friedman, Active Inference Institute and FractiAI.

Abstract

Gauge structure is proposed to be the transport of overlap data between observer patches. On twelve ports with icosahedral incidence, a faithful compact commutator-closed response whose proper rechartings act by inner automorphisms of that same response has exactly one compact Lie type, $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Faithfulness fixes the dimension twelve, the single fixed line of the transitive A_5 action bounds the centre by one, the fixed-space count excludes $\mathfrak{su}(2)^4$, and the classification of compact simple Lie algebras, a declared input, leaves the split $11 = 3 + 8$. The dimension arithmetic and the centre bound are machine checked; the fixed-space exclusion and the classification enter Lean as declared hypotheses. A separate categorical route reconstructs a compact group from overlap transport on a declared bosonic sector category.

A conditional matrix realization supplies the current blocks. Its declared spin lift and trace redistribution are the standard $\mathfrak{su}(5)$ embedding, and its declared exterior module is the Georgi–Glashow $\mathbf{10} \oplus \bar{\mathbf{5}}$ generation with exact Standard Model charges, anomaly cancellation and common central kernel $\mathbb{Z}_6$. In the conditional window $3 \leq N_g \leq 5$, whose lower edge is a declared CP-capability clause, a declared band-selection rule and the exact Laplacian costs $5 - \sqrt{5} < 6 < 5 + \sqrt{5}$ select the rank-three band. On the executable carrier the twelve-current law is a specification: the tested native exchange response closes to seventeen dimensions, and no result derives the twelve-dimensional current from unchanged repair. An exact basis map and complete source operations connect the specified matter module to a calibrated field action and colored fermionic exchange. This does not derive the physical current identification, global gauge-group choice or measured couplings from the bare repair rule.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual

quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Coherence, protected

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2057/docs/BASIS.md

codes, radial directions, fields, clocks and couplings may be specified as part of its design; their implementation and observable consequences must be established on that process. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1+3$ scalar description. Complete finite source factors join spatial CAR dynamics, particle creation, repair and a common calibrated clock. Physical validation of that clock and energy normalization, and an interacting chiral $3+1$ limit, remain open.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. A specified coherent ordering memory protects native exchange-history sectors, and a colored extension implements their fermionic exchange. A localized physical defect, its maintenance-to-mass law and a predicted measured spectrum remain unestablished; the earlier double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[2]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a specified and implemented clock calibration relates it to duration. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

From repair topology to a field experiment. A specified repair process admits protected quantum history sectors, fermionic exchange and an executable coupling to field and clock observables. Its ingredients are specified observer, scheduler, repair and memory laws; they need not be uniquely selected to define a mathematical model. What must be established is that the resulting operations share a carrier, retain complete quantum outputs and reproduce the intended observables. The constructions below establish several such joins, including a joint source experiment. They do not identify a complete physical theory.

Protected transport sectors. On the native 66-state binary two-load configuration complex, an exact noncommutative reduction of the face and disjoint-move relations gives $\pi_1 = \langle g \mid g^2 \rangle = \mathbb{Z}/2$. The surviving loop has odd ordered-cover parity. This is topology of the declared transport and deformation domain, not topology of a localized field defect. The complete active branch $*$-algebra on either the unextended 78-state source or its binary shell is the full matrix algebra. A Hermitian source charge commuting with every branch is therefore scalar. This excludes that charge class, not every invariant subspace, expectation-conserved observable or controlled lifetime.

A coherent ordering bit changes the algebra. If a native branch maps x to $f_h(x)$, let $p_h(x)$ be the parity of the transported order and define

$$K_h = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I_{66} \otimes X.$$

Every branch commutes with Q, so its two sharp sectors survive arbitrary finite record-adaptive schedules. An actual odd exchange loop returns the source and applies X to the bit: the memory participates in transport. Sector preservation does not assert preservation of every coherence between the sectors. For the all-seam C or E family, effect separation, a connected movement tree and the odd loop generate exactly $M_{66}(\mathbb{C}) \oplus M_{66}(\mathbb{C})$, with commutant $\mathrm{span}\{I, Q\}$, although the classical ordered cover is connected. The states $I_{66}/66 \otimes |\pm\rangle\langle\pm|$ have identical source and complete classical-record marginals; their distinction is quantum memory. Computational-basis reset

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2057/docs/COMPLETION_PROGRAM.md

or dephasing destroys it. Protection thus depends on the specified memory law preserving these sectors.

Literal source words implement this lift. Complete execution covers all 132 input columns on seam $(0, 5)$, and exact word audits cover all 1024 addresses of the 90 seam/current feedback words. Native outcomes, padding and source-service failures remain in the accounting. The result is a protected sector of the enlarged process; a spatially localized source realization of its charge remains an additional construction.

This same carrier also separates operator action from scalar repair cost. Writing $P_- = (I-X)/2$ and $M_h = \sum_{x\in D_h} |f_h(x)\rangle\langle x|$,

$$A_h = \pi \sum_{x\in D_h} p_h(x)|f_h(x)\rangle\langle f_h(x)| \otimes P_-, \qquad K_h = e^{-iA_h}(M_h \otimes I)/\sqrt{2}.$$

Two closed histories with identical multisets of complete branch labels and identical compiled resource totals induce respectively I and X. An additive scalar cost of those labels cannot determine this response; the displayed operator action does. Its phase is not physical energy per unit time. Nor is the reduced memory autonomous: its one-step flip probability is $26/495$, while the two-step Z multiplier differs from the squared one-step multiplier by $398/245025$. The continuing source must be retained in any proposed Hamiltonian or memory description.

Complete native occupation protection. For the bare twelve-port binary endpoint law, all 4096 occupations give the fixed algebra

$$\mathcal{A}' = \mathbb{C}^{11} \oplus M_2(\mathrm{span}\{|0\rangle, |\mathrm{full}\rangle\}).$$

Exact native effects and signed transport prove this for both C/E laws and exchange characters. A faithful weighted reference gives the full stationary family and fixed-selector limit. Every proper-port-local fixed observable is scalar. The only nontrivial passive quantum multiplicity is the global empty/full span; it has distance one and requires cross-number coherence outside fixed-total V2. Sixteen complete source histories test its preservation and local-error boundary. This classification concerns the binary endpoint instrument, not the distinct protected internal factors or outside-projector field repairs below. It identifies no physical particle or mass; the proof and complete-output checks are in `model/native_fixed_algebra/OCCUPATION.md`.

Internal states and fermionic exchange. Give each occupied port an independent internal space V. The map

$$|\{x < y\}; a, b\rangle \longmapsto c^\dagger_{x,a} c^\dagger_{y,b}|0\rangle$$

fixes the CAR convention. Passing the occupied spectator reverses the internal slots and crosses one creation operator, hence

$$K_h^V = \frac{1}{\sqrt{2}} \sum_{x\in D_h} |f_h(x)\rangle\langle x| \otimes (-\mathrm{SWAP})^{p_h(x)}.$$

The complete effects remain the original domain projectors divided by two. Independent CAR calculations check every branch for $\dim V = 2, 3$; the crossing argument applies to every finite V. The odd loop acts as $-$SWAP, giving -1 on two equal internal states. Global Fock parity on this two-particle sector is $+1$ and cannot replace that exchange.

The ordering memory implements the same operation: for $J = -\mathrm{SWAP}$,

$$E_V\psi = (|0\rangle\psi + |1\rangle J\psi)/\sqrt{2}, \qquad (X \otimes I)E_V = E_V J.$$

Every native branch, and every common $U \otimes U$, intertwines through E_V. The negative exchange character acts on $\mathrm{Sym}^2 V$; the positive one acts on $\Lambda^2 V$. Both are compatible with identical fermions carrying independent internal states.

The finite algebra constrains but does not fix internal encodings. An untouched fiber of dimension m enlarges the repair commutant to $M_m \oplus M_m$. A separate collective-label encoding that puts all fifteen states of $\Lambda^2\mathbb{C}^5 \oplus \overline{\mathbb{C}^5}$ in the negative sector requires $m \geq 15$ and has an explicit inclusion at equality. This is not a universal memory bound for the two-particle construction. No element of the supplied $S(U(3) \times U(2))$ gauge group acts as $-I$ on the whole faithful fifteen-state module: its nonidentity involutions have signature $(7, 8)$. The smaller weak-parity encoding is consequently an internal-grading control, not an all-fermion exchange assignment.

An exact basis permutation joins that exterior module and all twelve currents to the retained chiral action. Specified two-component and three-family factors give $\dim V = 90$; neither family count nor physical Lorentz spin is derived by this basis change. On twelve ports the hard-core two-particle dimension is 534,600, whereas the complete 1080-mode two-particle space has dimension 582,660. The remaining 48,060 states are same-port collisions and must be included when propagation or interactions can reach them.

One calibrated action and its observables. The Weyl–Higgs action uses explicit charged Yukawa, Higgs-background and CKM inputs with recorded matching conventions. Recovering them from its response verifies implementation; it is not an independent mass prediction. The creation-pairing matrix $\Delta = \epsilon_{\mathrm{spin}} \otimes \overline{K}$ is antisymmetric on 90 modes. At rest,

$$B_0 = \begin{pmatrix} 0 & \Delta \\ \Delta^\dagger & 0 \end{pmatrix}, \qquad H = E_{\mathrm{vac}} + \sum_{a=1}^{90} E_a \gamma_a^\dagger \gamma_a.$$

The canonical chart retains 84 positive charged directions, six zero-energy neutrino annihilators and every particle–hole partner. Choosing the annihilator half of the zero block specifies the static reference; it does not supply a positive-energy continuation at nonzero momentum. This minimal action has massless neutrinos; a separate neutrino mass extension is not silently included.

Copying this internal system to twelve ports gives 1080 CAR modes. The static hard-core two-quasiparticle sector supports the colored construction, whereas a bare two-particle sector need not survive pairing. The compiled static convention subtracts $11E_{\mathrm{vac}}$ from the replicated vacuum scalar $12E_{\mathrm{vac}}$; the remaining common E_{vac} does not affect fixed-sector Ramsey gaps. Every extension across particle-number sectors must specify the same reference convention explicitly.

Complete source-controlled Ramsey observations recover an energy difference between equal-charge modes, with an opposite-charge partner. A separate neutral-ancilla phase probe reads the same single-particle charge, retaining all seven charge values and all pointer outcomes. Neither measures an absolute rest energy. The common electromagnetic phase cancels on the neutral pair and is not an electromagnetic-response signal. These executions retain all 128 padded internal addresses, native records and quantum output factors.

Currents are transformed with the Nambu chart, including their anomalous blocks. The neutral Higgs background has a nine-dimensional unbroken stabilizer, color and electromagnetism; three complementary gauge directions are broken. General weak operations cannot therefore be represented solely by $U \otimes U$ within fixed quasiparticle number. A time-dependent Higgs background requires the complete CAR domain.

Spatial dynamics, clock and particle creation. A Hamiltonian cycle of the actual twelve-port seam graph supplies a finite spatial ring. With specified spacing and $c = 1$ units, its bare

action is

$$h = -D \otimes \sigma_z \otimes I_{45}, \qquad D_{r,r+1} = -i/(2a), \quad D_{r+1,r} = i/(2a),$$

with the same local pairing matrix. The static symbol satisfies $E(k)^2 = \sin^2(k)/a^2 + E(0)^2$. All twelve momenta, all 2160 Nambu directions and both kinetic zeros remain. This is a one-dimensional witness, not a chiral $3+1$-dimensional limit. Its 1080 CAR modes retain every occupation sector, including collisions. The pairing graph splits into closed internal components of sizes one, two and six, so the largest spatial component has 72 modes. Complete four-state Fock source gates and fermionic routing determine their full Gaussian unitary maps without allocating arbitrary 2^{1080}-component vectors. A source-prepared localized excess and its bond current have complete occupation readouts, with purifiers retained. The executed finite symmetric product is specified as such; its error relative to the summed Hamiltonian is separate from its source-gate error.

Two source-prepared clock hands, each with faithful eigenvalues $1/6, 5/6$ and a retained purifier, share the reference energy $\Omega_{\mathrm{ref}} = 1/4\,\mathrm{GeV}$. Both hands and every field factor use $\lambda_j = 1 + j/4$ for the actual native current. Complementary readouts infer $\tau_{\mathrm{eff}} = \sum_h \lambda_{j_h} \Delta\tau$ on a declared unaliased interval. The inverse receives complete probabilities and error bounds, not the programmed duration or a species mass. Rational enclosures and finite-shot bounds distinguish ensemble inference from a single clock reading. The energy-unit calibration and $\hbar = h/(2\pi)$ set seconds; repair counts and host runtime do not set this clock.

The particle-changing drive follows the actual common Higgs field $H = (0, ve^{i\phi})$ with fixed gauge background. Its creation phases have opposite signs in the up and down/charged-lepton channels, according to their conjugate-Higgs tensors. In a compatible static chart it comprises 42 neutral two-mode pairs and six massless spectators per site, with

$$H_{E,\phi} = E\cos\phi\,\mathrm{diag}(-1,0,0,1) + E\sin\phi\big(|00\rangle\langle 11| + |11\rangle\langle 00|\big).$$

Literal source gates retain each complete four-state space and all nonadjacent CAR signs. Across twelve sites this gives 504 pairs and 72 spectator modes. An externally driven quadratic field is not a quantized interacting gauge–Higgs theory.

Repair itself extends to every occupation sector. Binary endpoint occupancies follow the original C/E rule; nonbinary endpoints are held. A coherent extension includes that identity block in the existing zero-current Kraus operator. The binary and nonbinary output ranges are orthogonal, so the complete partial-injection effects sum to identity. On states with even occupancy at both endpoints the unread channel is exactly the identity, including coherences and references. The extension is unital before arbitrary record-conditioned controls; its faithful tracial reference has trivial modular flow. General completeness is algebraic. Literal source dilations verify the bounded collision and spectator-crossing cases, not a full 1080-mode dilation.

A separately flagged hold is retained as a control. It measures a vacuum/pair distinction that the coherent hold preserves. If A is the probability of no created pair on the repaired sites, a drive–repair–inverse echo returns $A^2 + (1-A)^2$ for the flagged policy and one for the coherent policy on this even-endpoint domain. For the stated drive at $\tau = 0.002\,\mathrm{GeV}^{-1}$, $j = 0$ and $\phi = \pi/2$, the flagged return is 0.63176674. Coherent collision holding does not remove backaction from the original current or E-parity records.

A joint source experiment. Selected source factors test repair, propagation, particle creation and clock readout together. Four active CAR modes on two sites carry the same charged component. A spatial superposition is source-prepared with a path reference and an opposite-charge spectator at a third site. The quasiparticle-to-bare chart is fixed by both static and driven terms; diagonalizing

the static mass alone does not suffice. Each of two events applies complete native repair, routed bare hopping in that chart, and the local Higgs factors. The same current advances the clock without reset. All 16 active states, spectator states, references and clock purifiers remain, including states with three active particles.

For both C/E laws and both collision policies, all $32^2 \times 4$ outcomes are retained: 16,384 in total. Complete effects are checked on the 64-dimensional field/clock input algebra, together with the untouched reference marginal. The coherent C process has collision population 0.23404814 before the second repair and final three-particle probability 0.53649765, versus 0.36223079 for the flagged control. Their unread final-state trace distance is 0.20709787. E distinguishes the relevant odd binary occupancies, so the two policies agree on this witness within numerical admission. The composed all-outcome total-variation bound is 4.200002×10^{-7}, conditional on the full-column numerical source checks. It is not an interval-certified hardware bound. Boundary hops out of the selected sites are omitted: this experiment joins actual source factors but is not an invariant two-site restriction of the full spatial field.

Energy and modular observations on the same field. The selected four-mode factors admit energy and reference-state calculations on their complete 16-state algebra. Let H be the sum of the static hopping and Higgs terms, with its vacuum scalar retained. The executed ordered product is not silently replaced by e^{-itH}. An explicit piecewise control schedule reproduces that product, separates switching work from energy transfer during complete repair, and uses the same clock duration. The latter transfer is not assigned to heat or controller work without an apparatus model. On the spatial action, onsite terms and half of each incident bond define

$$H = \sum_x h_x, \qquad J_{xy} = i[h_x, h_y] = -J_{yx}, \qquad \frac{dh_x}{dt} = -\sum_y J_{xy} + \partial_t h_x.$$

These identities hold on all occupation sectors. Equality of full unitaries after a scalar energy-reference shift requires the corresponding global source phase; density evolution, currents and gaps do not determine that scalar. Complete, charge-preserving source instruments on all 64 apparatus input columns reconstruct energy, drive-power and ring-current expectations from separate quadratic-group settings. This is expectation readout, not a projective measurement of the summed H.

For the declared $\beta = 1\,\mathrm{GeV}^{-1}$, the same static H gives a faithful $\rho = e^{-\beta H}/Z$. A complete eight-qubit source word prepares its field–purifier state. Signed CAR restrictions and compatible Petz observations compose on every matrix unit, and a native recovery isometry retains all four regional inputs, sixteen field states and four environment states. This is a new Gibbs-input experiment with the same field instrument, not a replacement of the earlier localized input. The modular convention is

$$\rho^{is} A \rho^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \tau_{-\beta s}(A);$$

the dimensionless parameter s does not supply a second physical clock.

The observation is not a regional retraction. The cross-boundary hopping has vanishing partial traces and squared Hilbert–Schmidt norm exactly 4 in the specified units. Its nonadditivity makes the Gibbs density nonproduct for $\beta > 0$, obstructing a reference-preserving UCP retraction onto the full site factor. Compatible non-idempotent Petz observations remain valid. Nor is this Gibbs density stationary under the driven, repaired history. The complete quantum entropy balance is

$$S(\sigma) - S(\rho) = \beta \operatorname{Tr}[H(\sigma - \rho)] - D(\sigma \| \rho)$$

for normalized states; retaining classical records gives an identity involving conditional relative entropies. Numerical trace drift contributes $(\operatorname{Tr}\sigma - 1)\log Z$, and the quadratic Bregman remainder subtracts $\epsilon \operatorname{Tr}\delta$ from $D(\rho + \epsilon\delta\|\rho)$. These reported entropies concern the unnormalized continuing component. First-export abort weights and output families are retained separately; their entropy is not included. A geometric cap kernel, local stress dictionary and generalized-entropy stationarity do not follow from these finite identities.

The Weyl/contact branch and its finite premise closures. The following constructions use the supplied Weyl/contact action with $c = 1/8$ and $U = 1$, separately from the calibrated Weyl–Higgs field above. Its dimer and plaquette restrictions retain their actual reference states. The charged-block service is first defined on decoupled dimers. The three-dimensional cube below instead uses bipartite eta symmetry to join a larger protected interacting sector to the complete field. The common local coefficients do not identify distinct global domains.

A complete charged internal sector. The same lower charged dimer doublets admit a sharper finite native-record attachment than the fixed-column dictionary. Retain both internal columns at each of two occupied blocks among twelve identical dimers, with the same even half-filled ground at every other block. Their carrier has dimension $66 \cdot 4 = 264$. For the declared graded whole-block repair, a movement that crosses the spectator acts internally as minus the swap; a hold acts as the identity. In the singlet–triplet basis every complete native branch is therefore

$$K_{\text{spinful}} = K_+ \otimes I_1 \ \oplus \ K_- \otimes I_3.$$

An integer certificate checks all 720 raw C/E slots and 31,680 internal input columns, including zero slots. The triplet qutrit and its untouched reference correlations survive every admitted complete history without intermediate recovery. For initially correlated spatial/internal inputs, this assertion concerns the complete internal marginal; conditioning on a record can steer it.

The raw effects separate all 66 positions. A connected transition tree and an odd exchange loop then classify the entire passive source commutant as $I_{66} \otimes (M_1 \oplus M_3)$, of complex dimension ten. Thus the three-dimensional factor is maximal under the full branch grammar; singlet–triplet coherences do not give a protected four-state factor. Positive all-seam coverage and the checked invariant maximally mixed state supply the corresponding fixed-algebra premises. This enlarged field carrier is distinct from the bare H66 source with scalar fixed algebra.

Complete extension and an exact spectral exclusion. Let P be the sum of the orthogonal two-doublet/remaining-ground block projectors. The native outcomes have total effect P; adjoining the outside outcome $I - P$ gives a trace-preserving instrument on the full 48-mode Fock space. This is a formal operator construction, with bounded checks on all 4,096 states of three dimers, not a 48-mode source execution. Every definite occupied pair has the unchanged ground on its entire exterior. With the original supplied $K_j = H_j - N_j/2$, $c = 1/8$ and $U = 1$, the same defect has

$$\Delta = \frac{\sqrt{17}}{8} - \frac{1}{16}, \qquad (K_{\text{tot}} - 12E_g I)|_P = 2\Delta I_P.$$

Its excess CAR number is $q = -1$ for the hole and $q = +1$ for the particle; native currents transport q times the defect count. All three protected triplet states have the same gap and total charge $2q$. A common scale or chemical potential cannot split them into three different masses. The triplet is an internal spin-one representation, not three generations.

The same sector is not invariant under the joined uniform four-site Weyl/contact field. For its four charged-pair columns V, exact dyadic arithmetic gives

$$\|(I - VV^*)H_{\text{uniform}}V\|_F^2 = \frac{3}{256} > 0$$

for both particle and hole pairs, whereas the decoupled sum gives zero. Indeed, with $L = (I - VV^*)H_{\text{uniform}}V$, the stronger identity $L^*L = (3P_{\text{singlet}} + P_{\text{triplet}})/512$ excludes every nonzero nonleaking input, including every triplet state. This generator leakage preserves the distinction between the supplied protecting block service and the interacting field law. No binding, topological boundary charge, physical clock, Lorentz spin, observed species or absolute mass follows; the physical particle and common-field gates remain open.

A protected interacting sector on the original three-dimensional cube. The source-bound `cube_eta_protection` packet retains the original open $2\times2\times2$ Weyl/contact action, $c = 1/8$, $U = 1$, on all $2^{16} = 65\,536$ occupation states. In the physical onsite basis define

$$\eta^+ = \sum_x (-1)^{x_1+x_2+x_3} c^\dagger_{x\uparrow} c^\dagger_{x\downarrow}, \qquad \eta^- = (\eta^+)^\dagger, \qquad \eta_z = (N-8)/2.$$

Onsite singlets are invariant under each supplied $i\sigma_i$ link; the bipartite signs cancel the hopping commutator. Thus $[H, \eta^+] = U\eta^+$, checked exactly on the complete field action. For $X = \eta^+|_{N=4} : \mathcal{H}_4 \to \mathcal{H}_6$,

$$X^\dagger X = (\eta^-|_{N=4})^\dagger(\eta^-|_{N=4}) + 4I, \qquad H_6 X = X(H_4 + UI).$$

Consequently X is injective and its polar isometry intertwines the full interacting four-particle parent with a protected candidate of dimension $\binom{16}{4} = 1820$. Its range is precisely the $\eta \geq 2$ subspace at $N = 6$.

The executed repair domain is a finite four-block service: the four x_1-directed dimers, native ports $(0, 1, 5, 7)$, and all five inherited seams $(0, 1), (0, 5), (0, 7), (1, 5), (1, 7)$. Two blocks carry the unchanged lower-hole doublet and two the unchanged dimer ground. This gives a 24-dimensional $N = 6$ support P. Its physical columns are expanded exactly as $A + \sqrt{17}B$, with integer Gaussian matrices A, B; $\eta^- A = \eta^- B = 0$, so the support has $\eta = 1$. The columns are also bound to the consumed normalized analytic dictionary: its exact normalization is $n = 136 - 32\sqrt{17}$, with $(A + \sqrt{17}B)^\dagger(A + \sqrt{17}B) = n^2 I$. Every native C/E coin and raw slot is checked against its actual graded whole-block permutation. With the complete outside outcome $Q = I - P$, all 130 slots are retained: 120 native and ten outside over the ten seam/law choices. Orthogonality gives

$$PX = 0, \qquad L_r X = 0 \quad (r \text{ native}), \qquad QX = X.$$

Every admitted complete repair is therefore exactly the identity on this sector, including arbitrary reference cross terms; interleaved field evolution remains unitary and interacting. This mechanism requires no alignment of the transverse dimer spin frames. It does not assert a twelve-block execution, arbitrary repair invariance, or maximality of the protected subspace.

Exact interacting spectral census and a local fixed-number probe. Let $v = \eta^+|0\rangle$, $v^\dagger v = 8$, and $\Pi_3 = I - vv^\dagger/8$ on the 120-dimensional $N = 2$ space. For $T = (\eta^+)^2|_{N=2}$,

$$T^\dagger T = 60I + 24vv^\dagger, \qquad H_6 T = T(H_2 + 2UI).$$

Hence $T/\sqrt{60}$ is isometric on the rank-119 $\eta = 3$ parent. The hidden-spin census is 35 singlets and 84 triplet states; this is neither Lorentz-spin identification nor a family count. Exact integer characteristic polynomials in the actual onsite Hubbard frame give, with $z = 16(E_{N=6} - 2)$,

$$p_{119}(z) = z^{57}(z^2 - 12)^{21} q(z) c_{64}(z)^3 c_{128}(z)^3,$$
$$q(z) = z^2 - 16z - 12, \qquad c_a(z) = z^3 - 16z^2 - 12z + a.$$

These are eleven distinct energies: multiplicity 57 at zero, 21 at each $\pm\sqrt{12}$, one at each quadratic root and three at each of the six cubic roots. The last twenty states have contact-shifted energies. Squarefreeness, coprimality and rational root isolation are exact; approximate displayed response weights are separate diagnostics.

The $N = 6$ reference and one-site density probe are

$$|R\rangle = \frac{(\eta^+)^3|0\rangle}{\sqrt{2016}}, \qquad H|R\rangle = 3|R\rangle, \qquad |s\rangle = \frac{4n_0 - 3}{\sqrt{15}}|R\rangle.$$

Here $\langle n_0\rangle_R = 3/4$, $\mathrm{Var}_R(n_0) = 15/16$; $|s\rangle$ is an $\eta = 3$ singlet with mean energy 3 and variance 3/112. Its normalized two-particle parent is $(8|\uparrow\downarrow\rangle_0 - v)/\sqrt{56}$. The exact parent Krylov Gram determinant is $14883950729121848391888005329065541569085 44 > 0$. Together with the annihilator $qc_{64}c_{128}$ this proves positive response weight at all eight interacting energies and zero weight at the three free-energy values.

The local number-preserving phase $F_0 = e^{i\pi n_0/2}$ satisfies

$$F_0|R\rangle = \frac{|R\rangle - \sqrt{15}|s\rangle}{4}, \qquad \mathrm{Tr}_0(F_0|R\rangle\langle R|F_0^\dagger) = \mathrm{Tr}_0\,|R\rangle\langle R|.$$

Thus the entire exterior reduced state is unchanged initially, while the global pure-state trace distance is $\sqrt{15}/4$ and the disturbed state has energy variance 45/1792. This is a finite local unitary disturbance of an entangled reference, not a boundary charge or an asymptotic localized particle. Both components remain protected. A fixed-$N = 6$ pointer Ramsey construction using controlled F_0, common field evolution and controlled $F_0^\dagger$ has

$$C(t) = \langle R|F_0^\dagger e^{-it(H-3)} F_0|R\rangle = \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it(z_j/16-1)}, \qquad w_j > 0, \quad \sum_j w_j = 1.$$

No vacuum/charged-sector phase reference is needed in this protocol. The specified H and reference energy fix the frequency origin; these finite spectral gaps are not physical masses.

Native preparation and the remaining detector premise. The actual source prepares the 56 signed three-doublon amplitudes using an eight-bit, 5379-pulse amplitude tree on the even modes, with the odd modes blank. Eight native CNOT words copy into the odd modes in 24 pulses, followed by the three-pulse one-site phase including its scalar phase. The compact tree evaluation is an exact spectator blank isometry; all 65 536 rows are retained for copying and the phase. The independent exact trigonometric and literal-port certificate bounds the complete source-state error by 8.48×10^{-13}, including every source export. All 130 following repair records are retained without success normalization. The native seventeen-bit time-dependent detector and native field continuation are not executed here: the fixed-number response above is analytic, while the reference preparation and local kick are executed. Hardware precision, growing-volume localization, deformed excitation transport, a continuum limit and empirical species identification are not established.

The earlier small-carrier leakage and transverse hidden-spin obstruction remain valid; the larger eta sector supplies a different common protection mechanism.

The preceding preparation packet stops before the time-dependent detector. The following extension executes that detector and field continuation on the same cube, closing this finite readout premise.

An internal observer reads the protected response. The `protected_particle_readout` construction attaches the preceding cube spectrum to complete native pointer records. Its identified object is the eight-dimensional cyclic response generated by the centered local density probe, inside the 119-state $\eta = 3$ sector and the larger 1820-dimensional protected space. It is a neutral, fixed-$N = 6$ response; these three dimensions are not interchangeable particle counts. The construction uses Ramsey interference to read a correlation function, a standard measurement principle, and supplies its actual source implementation for this field.[3]

The apparatus retains sixteen field bits and one pointer, hence all $2^{17} = 131\,072$ field/pointer occupation rows. The unchanged source prepares $|R\rangle$ and a pointer $|+\rangle$, then applies controlled $F_0 = \exp(i\pi n_0/2)$. There are two field segments of duration $1/2$: the first is followed by complete C repair on all five admitted seams, and the second by complete E repair on those seams. Controlled $F_0^\dagger$, a pointer phase $\phi \in \{0, \pi/2\}$, and the inverse preparation rotation precede the complete six-outcome pointer export. Every source first-failure record and all 130 repair records remain present. An outside repair outcome continues; a native repair outcome or source failure terminates in its own retained record. No probability is conditioned on survival.

With exact field evolution, each repair acts identically on both ideal arms. Immediately before the final pointer rotation their field vectors are

$$b_0 = e^{-itH}|R\rangle, \qquad b_1 = e^{i\phi}F_0^\dagger e^{-itH}F_0|R\rangle.$$

Consequently the unconditional ideal pointer-zero probability is

$$p_0(t,\phi) = \tfrac{1}{2}\{1 + \mathrm{Re}[e^{i\phi}C(t)]\},$$
$$C(t) = \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it\omega_j}, \qquad \omega_j = z_j/16 - 1.$$

Thus the two settings read $\mathrm{Re}\, C = 2p_0(t,0) - 1$ and $\mathrm{Im}\, C = 1 - 2p_0(t,\pi/2)$. The phase $3t$ follows from the reference eigenenergy; it is not an adjustable frequency fit. A common scalar added to H, or a common number chemical potential on this fixed number domain, cancels between the arms. Absolute energy zero is therefore not determined by this experiment.

Complete source and field error, without a sector projection. The executor composes success and every first-export failure map of each local word on at most three work bits, then applies these maps to the full field/pointer array. This is finite block contraction of the same native word, not replacement by an ideal gate. Independent tests compare the contracted maps with direct source execution, including order-one failure controls. The exact $N = 6$ restriction is used only by the independent original-H comparator, not by the native execution.

Each half-time field segment uses 32 symmetric product steps. The actual ordered support list has 24 hopping terms of norm $1/16$ and eight contact terms of norm one. Counting only possibly

[3] For many-body Ramsey correlation measurements, see Knap et al., *Phys. Rev. Lett.* 111, 147205 (2013), https://arxiv.org/abs/1307.0006.

nonzero nested commutators gives the all-Fock coefficient $K = 629/256$. The two segments therefore contribute

$$\varepsilon_{\mathrm{split}} = 2\,\frac{629}{256}\,\frac{(1/2)^3}{32^2} = \frac{629}{1048576}.$$

Independent rational trigonometric and literal-port bounds add less than 8.56×10^{-11} for the preparation, controlled pointer operations, and both complete source field words. The complete output-vector error is consequently $\varepsilon < 6.00 \times 10^{-4}$. Every halted record is part of this norm. Complete repair and export preserve the comparison by isometric extension or contraction; the all-input field bounds also hold with untouched references. The initial preparation bound consumes the specified initialized blank.

For normalized complete outputs, half trace distance is at most their vector distance. Each unconditional pointer probability therefore differs from its ideal value by at most ε, and each reconstructed quadrature by at most 2ε. Rational root, residue and trigonometric enclosures at $t = 1$ imply, for the actual specified source,

$$0.993649 < p_0(1, 0) < 0.994850, \qquad 0.501379 < p_0(1, \pi/2) < 0.502580.$$

Both intervals exclude the corresponding no-response values, one and one half. Numerical source residuals have their separate declared admission; they are not substituted for these proved source bounds. The executed construction uses 681 283 native pulses per setting. It gives pointer-zero probabilities approximately 0.9942497011 and 0.5019798894, versus the independent original-H values 0.9942494663 and 0.5019797169. The complete output-vector discrepancy is approximately 2.27×10^{-6}, below the rigorous bound.

The imaginary fringe also distinguishes the contact interaction from the same free cube. Set $U = 0$ while retaining the hopping, reference, local probe and ideal repair schedule. Bipartite occupation parity $\Gamma_{\mathrm{sub}} = (-1)^{\sum_{x\,\mathrm{odd}} n_x}$ satisfies

$$\Gamma_{\mathrm{sub}} H_0 \Gamma_{\mathrm{sub}} = -H_0, \qquad H_0|R\rangle = 0, \qquad \Gamma_{\mathrm{sub}} F_0|R\rangle = F_0|R\rangle.$$

The free characteristic function is consequently real at every time, so $p_{0,\mathrm{free}}(t, \pi/2) = 1/2$ exactly. Both free arms remain in the protected sector. The certified interacting interval therefore excludes this specified contact-free response. This is discrimination between two internal dynamics, not a fit of a physical particle mass.

For independent reset repetitions, Hoeffding's inequality and a union bound give simultaneous sampling error at most 1/2048 in these two probabilities using 2^{24} trials per setting, except with probability less than $4(3/8)^8$.[4] The sampling allowance separates both fringes from no response. Independent resets are the sampling-service premise; a seeded numerical sample is an illustration of that model, not laboratory data. The recorded PCG64 illustration, with seed 20261101, gives respectively 16 680 454 and 8 423 380 pointer-zero outcomes out of 2^{24} trials; the complementary count includes every other complete outcome.

What eight-line resolution would additionally require. The executed two settings attach the known protected spectral measure to an internally accessible record. They do not reconstruct eight unknown frequencies or individually resolve the eight lines. There is, however, a certified prospective experiment on the source-specified support. At times $t_n = 39n$, $n = 0, \ldots, 7$, use both pointer quadratures and set

$$y_n = \frac{16C(39n) - 1}{15}, \qquad V_{nj} = e^{-i39n\omega_j}, \qquad y = Vw.$$

[4] The bounded-independent-sum inequality is imported from Hoeffding, *J. Amer. Statist. Assoc.* 58 (1963), 13–30, https://doi.org/10.1080/01621459.1963.10500830.

Exact interval sine bounds in the Vandermonde product show

$$|\det V| = \prod_{j<k} 2\left|\sin\frac{39(\omega_k-\omega_j)}{2}\right| > 1400.$$

Hadamard's inequality bounds every 7×7 cofactor by $7^{7/2} < 343(8/3)$, so

$$\|V^{-1}\|_\infty < \frac{2744}{525}, \qquad \|\widehat{w}-w\|_\infty < \frac{43904}{2625}\,e,$$

where e bounds the total error of each pointer probability. The latter includes both quadratures and removal of the known elastic atom. Finite precision evaluation of an implemented inverse would need its own error allowance.

A sufficient prospective budget is systematic probability error $1/65536$ and sampling error $1/32768$ at each of the sixteen settings. With $9\cdot 2^{29}$ independent trials per setting, the simultaneous failure probability is less than $32(3/8)^9 < 1/100$ and the weight error is less than $343/448000$. Twice that error is smaller than the least certified weight, so every line is separated from zero. The total is $77\,309\,411\,328$ trials and the largest model time is 273. This long-time accuracy and sampling schedule is not executed here. It identifies weights on the supplied support, not an arbitrary unknown spectrum from finitely many samples.

The finite result is an internally observable protected excitation response of the admitted simulator, with complete-history and finite-time errors. It does not require an external particle name or an SI calibration. It also does not establish that these neutral modes are single odd fermions, that the reference is a vacuum ground state, or that finite spectral lines are relativistic rest masses. Transported excitations, many-body continuum dynamics and comparison with nature remain separate constructions.

The one-time readout above leaves its long-time reconstruction prospective. The following certified accelerated calculation closes that known-support weight-resolution premise while retaining a bound to the full native word.

Full weight resolution by certified accelerated emulation. The preceding sixteen-setting proposal can be executed as a certified emulation of the same source word. The result resolves all eight weights on the source-specified support; it does not infer arbitrary unknown frequencies or identify physical species. The observable remains the eight-dimensional neutral cyclic response of the excited reference, with elastic weight $1/16$.

The actual chronological field word places the two disjoint spin hoppings of each spatial edge consecutively. They commute, and their sum is the complete Weyl edge commuting with physical η^+. The complete contact layer satisfies $D_6T = TD_2 + 2T$, where $T = (\eta^+)^2$. Thus every ideal Strang step has the exact identity

$$S_6(h)T = e^{-2ih}TS_2(h), \qquad S_6(h)|R\rangle = e^{-3ih}|R\rangle.$$

The complete protected repairs act identically on these ideal states. This is a property of the grouped ideal word, not an exact projection identity for its rounded native implementation.

The actual Hubbard-frame singlet chart has dimension36, retaining all35 $\eta = 3$ singlets and the $\eta = 4$ parent. If U is the spatial forward half-hop matrix and $W = \mathrm{Sym}^2 U$ on normalized unordered pairs, the step is $W^T \exp(-ihD)W$. Transpose reverses the hopping order because every individual Hubbard hopping matrix is symmetric. Integer generator checks bind this chart to every original edge and the contact layer. Outward interval square roots supply the distinct-site normalization.

Use the common step $h = 39/2^{19}$. Time $39j$ takes $j2^{19}$ steps, with half before C repair and half before E repair. At time273 this is 3 670 016 steps. Their 38 755 368 960 field pulses are represented by the repeated source word; they are not individually executed. Instead, outward160-bit step entries are rounded to128-bit dyadics and powered by exact Gaussian-integer matrix multiplication. For approximate unitaries with errors a, b, one rounded product has error at most $a+b+ab+36\,2^{-128}$. The longest power needs24 products and has operator error $249561087/2^{128} < 7.34 \times 10^{-31}$.

Independent literal-source templates bound the complete 10 560-pulse step error by 1.323294×10^{-12}. Telescoping includes all native first-export failures, and the full-Fock splitting error is $(629/256)t^3/n^2$. With preparation, pointer operations, complete repairs, certified arithmetic and exact binary64 probability-conversion error, every setting has systematic probability error below $8.568121 \times 10^{-6} < 1/65536$. Individual long-word failure weights are not materialized; their record addresses and complete-output norm bound are retained. No survival normalization is used.

The emulator draws counts from a binary64 proxy to the certified powered ideal-word probabilities, with PCG64 seed20261102. These are surrogate counts, not exact samples of the unmaterialized native failure instrument. There are $9\,2^{29}$ independent reset trials at each of sixteen settings, or77 309 411 328 overall. Zero is contrasted with all other complete outcomes. The stated independent binomial service gives simultaneous sampling error 1/32768 except with probability below $32(3/8)^9 < .01$. The same conditional recovery theorem applies to genuine native counts under that sampling event and the proved native systematic bounds.

Only the source frequency intervals enter the inverse. Rational Lagrange polynomials give a96-bit matrix B for $V_{nj} = e^{-i39n\omega_j}$; outward arithmetic certifies $\|B\|_\infty < 2$ and $\|BV - I\|_\infty < 10^{-18}$. Counts determine $\widehat{y}$ and $\widehat{w} = \mathrm{Re}(B\widehat{y})$. Since the true source weights form a probability vector, the conditional error is bounded by

$$\|\widehat{w} - w\|_\infty \le \|B\|_\infty \frac{16}{5}(\varepsilon_{\text{systematic}} + \varepsilon_{\text{sampling}}) + \|BV - I\|_\infty < 0.000258393828.$$

In ascending frequency order the recovered weights are

$$(.002457894673,\ .005641397395,\ .005890881572,\ .007131698528,$$
$$.004528321839,\ .421578515341,\ .415796456292,\ .136974834359).$$

Every resulting interval is strictly positive and contains the independent exact residue, checked after inference. This closes weight resolution on the given support in the certified emulator. It does not turn $E_H - 3$ into an absolute rest mass, identify an unknown support, or establish physical particle species or native hardware execution.

Comparing the actual interacting regional clocks. The complete four-mode Weyl/contact Gibbs source supplies a useful finite alternative to assuming regional modular covariance. Let $W_S(s) = \exp(isH)$ be its state modular unitary at $\beta = 1$, and let $W_A(s) = \rho_A^{-is}$ be the embedded unitary of an actual CAR-reduced reference, with fixed scalar gauges understood. The relative operations

$$C_{AB}(s) = W_A(s)W_B(s)^*, \qquad C_{AB}(s)C_{BD}(s) = C_{AD}(s)$$

compare these clocks on every input and untouched reference. They obey the time-cocycle identity $C_{AS}(s+t) = C_{AS}(s)W_S(s)C_{AS}(t)W_S(s)^*$. The existing source words implement each global-to-regional correction at $s = 1/2$ with 2,826 pulses and ideal operator splitting bound 11/32768; the regional-to-regional comparison needs twenty pulses. All sixteen Fock columns and first export failures remain present. Such corrections are active: the first changes the common Gibbs reference by half trace distance about 0.00444. They do not assert that both regional flows preserve the same global state.

The stronger identification fails on this reference. For a linear modular intertwiner, an eigenoperator coefficient satisfies $(\nu_j - \omega_i)L_{ji} = 0$. The actual dimer at speed $1/8$, contact $U = 1$, $\beta = 1$ and $\mu = 0$ has three positive regional frequencies, approximately 0.0001258906, 0.9988218345 and 0.9989477251. An exact source characteristic polynomial and nine partial-trace moment identities determine its reduced spectral weights. Rational bounds on the square root, exponential and logarithm then separate all three frequencies from every global energy difference by more than 10^{-4}, in all 243 comparisons. Consequently every same-parameter linear intertwiner has range in the six-dimensional regional modular centralizer and cannot transmit a moving local clock hand. The group and coefficient implications are Lean checked; the concrete spectral and interval attachment is checked separately by exact arithmetic. This excludes a strong identification on one finite reference, while the explicit relative operations provide a functioning comparison. The supplied action, coherent services, reference preparation and dimensionless parameter remain visible inputs. No physical duration, common energy calibration, geometric modular flow or Einstein premise is discharged.

Controlled regional modular energy. The same complete-Fock Weyl/contact action supplies a quantitative regional modular attachment on its four-site open plaquette. All 256 field states and all sixteen states of the two-site region are retained. For the normalized signed partial trace E, put $H_c = H - I$ and $X(\beta) = E(e^{-\beta H_c})$. Exact Gaussian-integer and rational arithmetic constructs the first eight coefficients of $-\log X$. In particular, with $M_j = E(H_c^j)$,

$$C_1 = M_1, \qquad C_2 = \tfrac{1}{2}(M_1^2 - M_2), \qquad C_3 = \tfrac{1}{6}M_3 - \tfrac{1}{4}(M_1M_2 + M_2M_1) + \tfrac{1}{3}M_1^3.$$

For $0 \le x = \beta B < \log 2$, where $B \ge \|H_c\|$, the positive coefficient majorant $f(x) = -\log(2 - e^x) = \sum a_j x^j$ gives

$$\left\| -\log X - \sum_{j=1}^{p} \beta^j C_j \right\| \le f(x) - \sum_{j=1}^{p} a_j x^j.$$

This follows from the norm-convergent noncommutative logarithm series, without replacing the reduced reference by a local Gibbs ansatz. At the declared $\beta = 1/32$ and exact bound $B = 3$, an outward rational evaluation gives eighth-order error below 1.917×10^{-9}. Hence every Hermitian trace-zero regional variation obeys the entropy-tangent error bound $1.917 \times 10^{-9}\|\delta\|_1$. The normalization scalar in $K_A = -\log X + \log \operatorname{Tr}_A X$ cancels only on that tangent space.

The first nonscalar boundary correction is cubic and does not commute with the bare regional energy. Removing the contact interaction removes its interaction-dependent terms; removing the crossing bonds removes all nonscalar corrections. A full-versus-empty regional matrix-element witness and the same rational tail prove distance greater than 2.7×10^{-8} from $\beta C_1 + aI$ for every scalar a on this instance. This supplies both a corrected finite modular generator and a certified boundary to its bare-energy identification. The source preparation below supplies this plaquette reference to a separate controlled error. The action, temperature and lattice remain declared; the norm bound can grow with volume. Geometric cap weights, null stress, fixed-volume entropy–area stationarity, a physical scaling limit and positive Newton normalization remain additional obligations.

A separately prepared interacting Gibbs reference. On the complete 256-state plaquette at $\beta = 1/32$, the degree-eight Gibbs ensemble retains all 256 preparation records and every export abort. Independent outward arithmetic certifies its literal-source preparation error below 1.700×10^{-12} on the initialized input. The actual regional success logarithm, including the exact

normalization scalar, is within 2×10^{-9} of the preceding P_8 comparison. The all-input continuation has a separate source and splitting bound. The proof in `model/certified_reference/README.md` supplies finite Gibbs preparation; it does not assert invariance under the complete repair. The following stationary reference has a different modular generator.

The actual stationary source weight. One identical literal native $Y(-\pi/3)$ request and complete dephasing export per occupation mode prepare $\sigma_M = \mathrm{diag}(a,b)^{\otimes M} = a^M e^{\gamma N}$, where $a = |U_{00}|^2$, $b = |U_{10}|^2$ for the actual represented source unitary and $\gamma = \log(b/a)$. The exact one-mode error is below 1.894×10^{-16}, proving $a, b > 0$ and $b > a$; abort blocks remain separate. Number conservation and complete unital repair preserve this weight exactly. Tracing complete outside outputs before regional compression gives a k-mode logarithm error at most $k\delta/(1/4 - \delta)$, with $\delta = (2+\epsilon)\epsilon$. This charge clock is trivial within fixed number. On the earlier six-site $N = 4, S = 1$ active sector, every common stationary modular generator stays at distance at least $|\beta|\sqrt{13/48}$ from βK, even allowing scalar and protected-spin terms. Thus the stationary reference and interacting Gibbs energy remain distinct. The exact source and reference proofs are in `model/continuing_reference/README.md`; the following construction closes its earlier restricted scheduler premise.

A stationary finite controller for native feedback. The reference can continue under an explicit record-adaptive scheduler. Use two classical working bits (c, d); $c = 0$ selects the complete native C repair on seam $(0, 1)$, and $c = 1$ the complete E repair on $(0, 5)$. Append every raw record r, including zero slots and the outside outcome $Q = I - P$, and update

$$(c, d) \longmapsto (d \,\mathrm{XOR}\, f(r), c), \qquad f(r) = \text{coin XOR (raw mod 2)}, \quad f(Q) = 0.$$

At output (u, v) every incoming term uses the same instrument v. Completeness and unitality of the unchanged repair therefore give

$$\sum_r L_r^{(v)} \frac{\sigma_M}{4} L_r^{(v)*} = \frac{\sigma_M}{4}, \qquad \Omega = \sigma_M \otimes I_4/4, \quad \sigma_M = \mathrm{diag}(a,b)^{\otimes M}.$$

Here a, b are the actual literal-source weights, so the successful block has trace $(a+b)^M$. All initial export-failure sectors are retained and frozen, emitting a bypass record. Separate unitary layers selected by the current working memory and generated by the original number-conserving field Hamiltonian also preserve this marginal. The controller is a declared reversible classical law on $\mathbb{C}^4$, not a claimed quantum pulse circuit.

Two unchanged native events on an explicitly initialized binary spectator, occupied at ports $2, 3$ and repaired on $(0, 1)$, leave it fixed and supply two exact fair coins. Their retained records seed $I_4/4$ without an angle-error assumption. This uses the supplied simulator basis initialization; no hidden interacting-field preparation of that spectator is asserted. The working memory remains two bits, while the archive grows by at most five bits per tick, besides the two seed records and all 6^M source readouts. Stationarity traces these records; it never normalizes the successful field block or declares the growing archive stationary. The modular generator remains $-\gamma N - M \log a + \log 4$, $\gamma = \log(b/a)$, and every complete step intertwines the charge flow $e^{it\gamma N}$. No physical energy-temperature identification follows. Earlier local literal-source bounds remain uniform in volume and ideal adaptive history length; implemented per-step errors must be added.

A complete finite mixing certificate and its boundary. On the invariant three-position highest-spin repair corner, with no joined Hamiltonian step inserted, the field/controller algebra is

$\bigoplus_4 M_3$. Each E outcome has rank at most one. Exact integer enumeration finds at least one eight-tick history encountering E for every one of the 12^2 input/output population pairs, with common probability denominator 2^8. The corresponding rank-one Kraus histories prove

$$\Phi^8 = \frac{3}{64}\mathcal{D} + \frac{61}{64}\mathcal{R}, \qquad \mathcal{D}(X) = \mathrm{Tr}(X) I_{12}/12,$$

where $\mathcal{R}$ is completely positive, trace preserving and unital. Thus $I_{12}/12$ is the unique stationary density on this CQ algebra, and the archive-traced error contracts by $(61/64)^{\lfloor n/8 \rfloor}$, including arbitrary untouched quantum references and all field coherences. This is neither whole-Fock mixing nor mixing with interleaved Hamiltonian evolution. Indeed the full retained history recovers the initial corner basis state and memory. As a negative control, the recordwise reversible one-bit policy $c \mapsto c \,\mathrm{XOR}\, f(r)$ sends the uniform six-state input to $(3, 1, 2, 1, 3, 2)/12$, a trace distance $1/6$ from uniform. The two-register incoming-row balance is a substantive premise, not a consequence of memory reversibility alone.

Exact energy Ward cells under coarse-graining. The same complete-Fock Weyl/contact action supplies a conserved energy column on its standard three-dimensional lattice. With the declared half-bond convention $H = \sum_x h_x$, its range-two currents are $J_{xy} = i[h_x, h_y]$. Distributing each current equally over all shortest nearest-neighbour paths preserves its incidence $\delta_x - \delta_y$: intermediate vertices cancel on every path. The resulting oriented face current $F_{x,i}$ therefore satisfies $\dot{h}_x + \sum_i (F_{x,i} - F_{x-e_i,i}) = 0$ exactly. The rational Clifford certificate checks the actual $3 \times 3 \times 3$ action at $c = 1/8, U = 1$: all 351 site pairs, 81 nonzero currents, 54 faces and 27 local equations, on the complete 54-mode algebra. Its coverage of all 2^{54} Fock states is a symbolic operator identity, not execution of that many amplitudes. Independent complete small-Fock matrices attach all three hopping axes and the interaction to the existing source action.

For a finite time cell, the exact flux is $I_{x,i}(n) = \int_{t_n}^{t_{n+1}} F_{x,i}(t)\, dt$. Hence

$$h_x(t_{n+1}) - h_x(t_n) + \sum_{i=1}^{3} \big(I_{x,i}(n) - I_{x-e_i,i}(n)\big) = 0.$$

For any normalized input and untouched reference, setting $T_{00}(n, x) = \langle h_x(t_n) \rangle$ and $T_{i0}(n, x) = \langle I_{x-e_i,i}(n) \rangle$ supplies the $j = 0$ forward-difference Ward column of the finite tensor interface. These are cell energy and integrated flux, not pointwise continuum stress. Summing spatial blocks cancels internal faces, so $D_{\mathrm{coarse}} P_{\mathrm{faces}} = P_{\mathrm{sites}} D_{\mathrm{fine}}$. Successive block maps compose exactly; successive time cells telescope. All 27 rectangular width choices, including ragged blocks, pass the exact source certificate. Generic finite-sum divergence, continuity, block composition and time telescoping are also Lean-checked; the concrete CAR source binding remains the exact rational certificate.

The existing local source bounds control the approximation uniformly in ambient volume. With $A = U + 3c$, one has $\|h_x\| \le A$ and $\|F_{x,i}\| \le B = 14A^2$: a face receives absolute shortest-path weight at most seven. Its support fits a radius-three cube. If L_h, L_F are the existing normalized locality tails and ϵ is the patch unitary error, an endpoint energy error is at most $A \min(2, L_h + 2\epsilon)$ and an integrated-face error at most $tB \min(2, L_F + 2\epsilon)$. Replacing the integral by m midpoint samples adds the explicit bound $t^2 B\, 343(U + 6c)/(2m)$; endpoint sampling is not exact. A cell residual is bounded by two endpoint errors plus six face errors, with supplied readout errors added separately. The receipt evaluates these resource bounds without claiming to execute the large sample count. Input preparation, hardware errors and controller/failure-space energy remain separate. This closes a finite same-action energy-column attachment, not the three momentum columns, a symmetric Lorentz tensor, geometric modular weights, area stationarity, a continuum scaling theorem or the full set of Einstein premises.

Local momentum balance retains its force. On the open Weyl/contact box, the three local central-difference momenta obey exact balances with kinetic face stress and quartic contact force. All spatial blocks and time cells retain that force. On the 3^3 box at $U = 1$ its total-force Gram matrix is $(9/4)I_3$, so an internal closed divergence cannot absorb it. Conserved number-preserving quadratic charges of the connected action reduce to number and hidden spin. These exact results in `model/lattice_momentum_balance/README.md` concern local quadratic candidates. The following periodic construction supplies conserved global momenta with a separate locality boundary.

Global periodic momentum and the local question. On a declared periodic box, graded translations T_j commute with the full interacting action. Their principal Hermitian logarithms P_j are exactly conserved. They differ from additive one-body momentum by integer many-body winding. On odd-prime rings every number-preserving translation logarithm has nonzero one-particle spin-trace kernels at all nonzero separations. For the principal branch, every range-R density sum obeys $\|P_j - A_R\| \geq 1/(R+1)$ when $L > 2(R+1)$. These full-space statements, proved in `model/periodic_momentum/README.md`, leave controlled restricted state domains available. The following bounds quantify one such domain and retain its actual interaction leakage.

A collision-suppressed local Ward bound. The same periodic Weyl/contact action supplies a controlled local approximation beyond its nonlocal conserved momentum. Let B be a spin-complete set of spatial Fourier modes, $f = |B|/V$, and let Ω restrict the initial state to that band and $N \leq N_{\max}$. Write $D = \sum_x n_{x\uparrow} n_{x\downarrow}$ and $d = \lfloor N_{\max}/2 \rfloor$. Band annihilators have norm $\sqrt{f}$, so $\Omega D \Omega \leq f \min(N_\uparrow, N_\downarrow) \leq fd$. Since $D^2 \leq dD$ on this number sector,

$$\|UD\Omega\| \leq Ud\sqrt{f}, \qquad \|(I-\Omega)e^{-itH}\Omega\| \leq \min\{1, |t|Ud\sqrt{f}\}.$$

The second bound uses free invariance of Ω and Duhamel's formula; the full interacting action is never projected. For two particles, the exact contact principal angles are $\lambda_K = |B \cap (K - B)|/V$, giving escape norm squared $\max_K \lambda_K(1-\lambda_K)$ before multiplication by U^2. Thus projecting the outgoing contact back into B would miss its leading square-root leakage.

Let $s_j(x)$ be the local sine-momentum density. Write $\Pi_{xy,j}$ for the antisymmetric kinetic stress across a nearest-neighbor face. The exact identity is $i[H, s_j(x)] + \sum_y \Pi_{xy,j} = F_j(x)$, with $F_j(x) = i[UD, s_j(x)]$. Normal ordering its quartic contact terms gives $\|F_j(x)\| \leq U$ and $\|F_j(x)\Omega\| \leq 4Uf$. For preparation at zero and $0 \leq t_0 \leq t_1$, the actual integrated Ward residual therefore obeys

$$\left\| \left[s_j(x,t_1) - s_j(x,t_0) + \sum_y \int_{t_0}^{t_1} \Pi_{xy,j}(t)\, dt \right] \Omega \right\|$$
$$\leq \min\left\{ U(t_1 - t_0),\ 4Uf(t_1 - t_0) + \frac{U^2 d\sqrt{f}}{2}(t_1^2 - t_0^2) \right\}.$$

The preparation clock is not restarted in later cells. Spatial blocks cancel their internal faces and sum the site bounds; time cells telescope. The complete 54-mode 3^3 source certificate checks all 81 local equations, 81 kinetic faces per column and 27 rectangular coarse partitions per column. Finite-sum transport of the errors is also Lean-checked. This is symbolic full-Fock coverage, not execution of all its amplitudes.

The two-particle total sine momentum has multiplier $2\sin(K_j/2)\cos(k_j - K_j/2)$. The uniform contact vector gives the sharp complete-fiber force norm $\|i[UD, S_j]\| = \sqrt{2}U|\sin(K_j/2)|$. If initially $|k_j| \leq \kappa$ and $2\kappa < \pi$, total momentum stays within $|K_j| \leq 2\kappa$ even after relative-momentum

scattering. Combining this conserved domain with the sine remainder gives

$$\|(P_j - S_j)e^{-itH}\Omega\| \le \frac{\kappa^3}{3} + 4\kappa \min\{1, |t|U\sqrt{f}\}.$$

For a 64^3 counting example with $B = \{-1, 0, 1\}^3$, $N_{\max} = 2$, $U = 1$ and $t = 1/8$, the certified leakage is below 0.00127, the local Ward defect below 0.000132, and the momentum error after division by κ below 0.0083. The separate eight-site exact-Hamiltonian diagnostic retains a nonzero unforced local defect, so the contact force has not been discarded. These are fixed-number dilute bounds. The critical band-edge scaling $f \sim a^3$, $U \sim a^{-2}$ instead makes $U\sqrt{f} \sim a^{-1/2}$, for which this estimate supplies no vanishing leakage. A finite-density or critically scaled local stress limit, Lorentz symmetry, physical units and the remaining Einstein premises remain separate. Proofs, finite domains and replay are in `model/dilute_momentum_ward/README.md`.

Interacting lattice scattering and its zero threshold. The same Weyl/contact action admits an on-shell two-fermion scattering theorem. On the physical exchange subspace of $L^2(\mathbb{T}^3, \mathbb{C}^4)$ at total momentum K, its free multiplication operator $H_{0,K}$ is bounded self-adjoint and the contact perturbation is $U|d\rangle\langle d|$, with $\|d\| = 1$. The difference is trace class with norm U. The Kato–Rosenblum theorem therefore gives complete wave operators on the absolutely continuous subspaces and a unitary scattering operator.[5] With $g_+(E) = a(E) - i\pi\rho_0(E)$, its contact-channel eigenvalue is

$$S_c(E) = \frac{1 - Ua(E) - i\pi U\rho_0(E)}{1 - Ua(E) + i\pi U\rho_0(E)}.$$

The orthogonal reducing complement of the contact-generated subspace has identity scattering. Point spectrum, including the isolated pair and the free dark zero modes, remains outside the incoming scattering domain. Completeness here concerns two-body absolutely continuous scattering; it is not many-body or chiral-continuum completeness.

At $K = 0$, the singlet resolvent is $z/(z^2 - 4c^2|\sin k|^2)$. Coarea with all eight sine-map preimages gives, for $0 < |E| < 2c$,

$$\rho_0(E) = \frac{E^2}{4\pi^2 c^3} \sum_{n\ge 0} A_n \left(\frac{E}{2c}\right)^{2n}, \qquad A_0 = 1, \quad A_1 = \tfrac{1}{2}, \quad A_2 = \tfrac{11}{40}, \quad 0 < A_n \le 1.$$

The coefficients are exact sphere moments of the binomial Jacobian. After degree M, the dimensionless remainder is at most $r^{2M+2}/(1 - r^2)$, $r = |E|/(2c)$; rational arithmetic and a Machin bound for π give certified density intervals. Local analyticity and evenness give $g_+(E) \to 0$ at zero. Hence $\rho_U(E) = \rho_0(E)/|1 - Ug_+(E)|^2 \sim E^2/(4\pi^2 c^3)$. This supplies nontrivial scattering and a gapless contact spectral support on one interacting lattice action while retaining its isolated pair. Zero model energy is not an identified physical vacuum threshold, so no massless physical particle is assigned. The source-bound proof and imported scattering theorems are recorded in `model/contact_scattering/ONSHELL.md`.

Interacting edge scattering and a moving bound pair. The same two-body source family has a controlled moving bound band. Set $c_a = \sqrt{3}/a^2$ and retain the repulsive coupling

$$U_a^{-1} = \frac{a^2 W_0}{2} - \frac{a^3\gamma}{\pi}, \qquad W_0 = \left\langle \frac{1}{\sum_j \cos^2 k_j} \right\rangle, \qquad \gamma > 0.$$

[5] The imported trace-class and finite-rank scattering results are reviewed in https://arxiv.org/abs/1711.00528 and https://arxiv.org/abs/0902.3568; the attachment here verifies their premises for the stated Weyl/contact exchange fiber.

The normalized full-torus integral satisfies $2/3 \le W_0 < 5$; $a\gamma \le 1$ makes the coupling positive. The single global comparison is $J_a = 6/a^2 - H_a$, explicitly shifting energy and reversing time. Neither γ nor this clock/energy choice is inferred from physical data.

At zero total momentum the threshold estimate is

$$\left| \frac{g_0(6/a^2 + z) - a^2 W_0/2}{a^3} + \frac{\sqrt{z}}{\pi} \right| \le 19a|z|$$

for real $|a^2 z| \le 1/8$, using the upper boundary square root; the exterior constant improves to 11. Coarea for all eight valleys, an inner ball of radius $1/2$ and its complementary measure give the respective Green remainders 18η and $36|\eta|$ before rescaling. Thus, for $0 \le s \le S$, $a^2 S \le 1/8$ and $836aS \le 7\gamma$,

$$S_J(s) \longrightarrow \frac{\gamma - i\sqrt{s}}{\gamma + i\sqrt{s}}, \qquad \sup_{[0,S]} |S_J - S_{J,\mathrm{lim}}| \le \frac{1672aS}{7\gamma}.$$

This is the earlier nonidentity scalar contact-channel limit; its orthogonal free channels remain present. Lattice contact renormalization is an established mechanism, not a new particle identification; compare Koerber–Berkowitz–Luu (2019) and Winkler et al. (2006).

Nonzero total momentum has a controlled pole. Write $K = aP$, $q = k - K/2$, $d_j = \sin q_j \cos(K_j/2)$, $e_j = \cos q_j \sin(K_j/2)$ and $R^2 = \sum_j \cos^2(K_j/2)$. For $|K_j| \le \pi/2$, concavity gives the exact complete free edge $B_a(K) = 2c_a R$, attained at all eight $q_j = \pm\pi/2$. The four-spin singlet resolvent is

$$G_K(q, E) = \frac{E(E^2 - 4c_a^2 e^2)}{E^4 - 4c_a^2(d^2 + e^2)E^2 + 16c_a^4 (d \cdot e)^2}.$$

At $E = B_a(K) + z$, put $t = E/(2c_a)$, $\eta = t^2 - R^2$ and $V = \sum_j \cos K_j \cos^2 q_j$. Equivalently,

$$g_K(E) = \frac{t}{2c_a} \left\langle \frac{1 - e^2/t^2}{V + \eta + (d \cdot e)^2/t^2} \right\rangle.$$

Let $\kappa = \max |K_j| \le 1/2$ and $h = a^2 z \le 1/8$, with $z \ge 0$. Writing $Q = \sum_j \cos^2 q_j$, the denominator lies between $(1 - \kappa^2/2)(Q + h + h^2/12)$ and $Q + h + h^2/12$; $e^2 \le \kappa^2 Q/4$ and $(d \cdot e)^2/t^2 \le e^2$. The complete-torus integral and prefactor therefore give

$$|g_K(B_a(K) + z) - g_0(6/a^2 + z)| \le 3a^2 \kappa^2.$$

Combining this with the exterior threshold estimate proves

$$\left| \frac{g_K(B_a(K) + z) - a^2 W_0/2}{a^3} + \frac{\sqrt{z}}{\pi} \right| \le a(11z + 3P_{\max}^2), \qquad |P_j| \le P_{\max}.$$

Under $256a(\gamma^2 + P_{\max}^2) \le \gamma$, the pole denominator has opposite signs at $z = \gamma^2/4$ and $z = 4\gamma^2$. Strict monotonicity of the exterior Stieltjes transform supplies exactly one upper pole $E_a(K) = B_a(K) + \delta_a(K)$, uniformly on that momentum cube:

$$\frac{\gamma^2}{4} \le \delta_a(K) \le 4\gamma^2, \qquad |\delta_a(K) - \gamma^2| \le \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

This does not invoke the older, stronger sufficient condition $U > B_a$.

An explicit bound-band wavepacket comparison. The same global J_a has bound energy $\epsilon_a(P) = 6/a^2 - B_a(aP) - \delta_a(aP)$. Rationalizing B_a and bounding the sine remainder gives

$$\sup_{|P_j|\le P_{\max}} \left| \epsilon_a(P) - \left(\frac{|P|^2}{4} - \gamma^2 \right) \right| \le \mathcal{E}_a := \frac{a^2 P_{\max}^4}{4} + \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

In particular, subtracting $B_a(K)$ separately at each momentum would incorrectly remove the center-of-mass kinetic term.

Let $R_K = (E_a(K) - H_{0,K})^{-1}$ and $\psi_{a,K} = R_K d/\langle d, R_K^2 d\rangle^{1/2}$. Its positive contact overlap fixes its phase, and the pole equation makes it a normalized, measurable, exchange-antisymmetric eigenvector. With consistent measures $dK/(2\pi)^3$ and $dP/(2\pi)^3$, the formula

$$(W_a f)(K,k) = a^{-3/2} f(K/a)\psi_{a,K}(k)$$

defines an isometry on packets supported in the stated cube. Its range reduces the two-body source action; the remaining source states are retained. If h_{pair} multiplies by $|P|^2/4 - \gamma^2$, then for arbitrary untouched references,

$$\|e^{-itJ_a} W_a - W_a e^{-ith_{\text{pair}}}\| \le \min\{2, |t|\mathcal{E}_a\}.$$

This proves coherent moving-pair propagation with an explicit rational spacing schedule. The limiting inertial mass is twice that of a single free edge in these model coordinates; neither an SI mass nor a relativistic rest energy is identified. The exact bounds, independent four-spin and torus checks, and wrong-subtraction/time-orientation controls are retained in `model/moving_bound_pair/README.md`.

The result concerns one composite sector. Nonzero-K on-shell scattering, a physical vacuum/detector map, many-pair interactions, relativistic massive/massless sectors and Einstein dynamics remain separate. The earlier fixed-zero $c = v/a$ result also remains valid: $\int_{-M}^{M} |S - I|^2 dE \le 1024M^2/(63c)$ for arbitrary real finite $U(a)$ and $M \le c/4$, so every fixed energy-normalized packet becomes transparent while tuned shrinking resonances can persist. Its scaling and energy comparison differ explicitly from the bound-band construction.

Identified model energies and the physical particle boundary. The model energies and inertial mass are identifiable. The cube has eight exactly characterized interacting energies, each reached by the local probe with positive weight. The moving pair has $\epsilon(P) = |P|^2/4 - \gamma^2$ for the reflected generator $J_a = 6/a^2 - H_a$, hence binding energy γ^2 and inertial mass 2 in those model units. If physical energy and length units $E_{\text{u}}, L_{\text{u}}$ are independently attached and $p_{\text{phys}} = \hbar P/L_{\text{u}}$, then $M = 2\hbar^2/(E_{\text{u}} L_{\text{u}}^2)$ and $E_{\text{bind}} = E_{\text{u}}\gamma^2$, with $E_{\text{u}} T_{\text{u}} = \hbar$ for the common clock. These conversion formulas do not select the units or fix an absolute rest-energy offset. Binding energy is not total rest mass. For the cube, the following rounded displays have exact rational energy and weight intervals in the receipt, with the original contact unit $U = 1$:

H energy	$E - 3$	Probe weight
1.816498920	−1.183501080	0.002465366
1.860537501	−1.139462499	0.005644624
1.955137632	−1.044862368	0.005881219
2.108692846	−0.891307154	0.007137664
2.167639734	−0.832360266	0.004528123
3.015861346	+0.015861346	0.421577939
3.030769653	+0.030769653	0.415789141
3.044862368	+0.044862368	0.136975924

Weights refer to the normalized centered density probe; the phase-kicked target has 15/16 times these weights plus an elastic weight 1/16 at energy 3. The lowest displayed line is not the $N = 6$ ground: $-3\sqrt{3}/8 \le E_{0,N=6} \le -\sqrt{3}/8$, by the free one-particle bound and a polarized zero-contact variational state. These are finite energy levels and signed transitions, not eight rest masses. The physical species assignment count remains zero. On the cube the density probe, its phase-kicked reference and all spectral components remain in $N = 6$: the transition operator is number-neutral and even under $\Gamma = (-1)^N$. An isometric dictionary intertwining this grading cannot map that response to a single odd fermion, since an odd creation operator changes parity. This does not exclude other odd model excitations or a fermion–antifermion response; even parity alone does not establish Bose exchange or Lorentz spin. Hidden spin and model number have not been identified with physical spin and electric charge. The physical target conventions are the PDG electron and quark quantum numbers and relativistic kinematics.[6]

The supplied cube reference also fails the vacuum ground-state spectral condition for its own H. Its positive density-probe spectral measure has five negative and three positive frequencies $\omega = z/16 - 1$; all eight weights are strictly positive by the exact Krylov certificate. In particular $z_- = 8 - 2\sqrt{19}$ has weight $(1 - 4/\sqrt{19})/14 > 0$ and gives an $N = 6$ eigenenergy $5/2 - \sqrt{19}/8 < 2$, below the reference energy 3. A number-preserving unitary swapping these two states lowers the energy, so the reference is not passive for the full field unitary algebra. This is a mathematical witness, not an executed extraction protocol: the actual local phase preserves the mean energy. These are frequencies of a positive greater correlation, not the two-sided frequencies of a retarded commutator. A scalar or common number chemical potential leaves the gaps unchanged, and time reversal leaves mixed signs.

The moving-pair limit $\epsilon(P) = |P|^2/4 - \gamma^2$ is a genuine controlled nonrelativistic band. For any fixed affine energy map $E = \alpha\epsilon + \beta$, $\alpha \neq 0$, and fixed invertible linear momentum map $p = LP$, the residual $E^2 - c^2|p|^2 - m^2c^4$ has nonzero quartic coefficient $\alpha^2/16$ along a coordinate line. It cannot vanish on an open momentum set. Thus these maps do not turn this limit into an exact massive or massless Lorentz shell. A low-momentum approximation remains possible: after matching constant and quadratic terms, the difference between $E_0 + Ar^2$ and $\sqrt{E_0^2 + 2E_0Ar^2}$ is $A^2r^4/(E_0 + Ar^2 + \sqrt{E_0^2 + 2E_0Ar^2})$, for rest energy $E_0 > 0$ and $A > 0$.

`model/particle_identification` binds these scoped tests and the remaining vacuum, gauge-charge, spin, clock, detector and mass-shell requirements. The earlier dimer electron mass was a calibration input, not an independent mass prediction; its retrospective equality comparison does not identify the new cube or pair modes. No mass fitting, new eligible physical comparison, or whole-program exclusion is added.

The preceding identity audit leaves a relativistic construction open. The following extension constructs a massless finite-particle limit on the growing lattice. Its carrier and reference remain distinct from the fixed protected cube, and its limiting interaction vanishes.

A controlled massless relativistic sector of the unchanged action. The same Weyl/contact family admits a simultaneous eight-valley continuum comparison on every fixed finite-particle momentum domain. Set

$$H_a = d\Gamma(h_a) + U(a)D, \qquad h_a(k) = -\frac{v}{a}\sum_{j=1}^{3}\sigma_j \sin k_j, \qquad D = \sum_x n_{x\uparrow}n_{x\downarrow},$$

[6] https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-model.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-kinematics.pdf.

with fixed $v > 0$ and finite $U(a) \geq 0$. This declares the scaling $c = v/a$; it adds no mass term, energy subtraction or time reversal. For $\nu = \pi b$, $b \in \{0,1\}^3$, let $C = [-\Lambda, \Lambda]^3$ and $a\Lambda < \pi/2$. With both Fourier measures normalized by $(2\pi)^{-3}$,

$$(W_a f)(\nu + ap) = a^{-3/2} f_b(p), \qquad W_a : \bigoplus_b L^2(C, \mathbb{C}^2) \longrightarrow L^2(\mathbb{T}^3, \mathbb{C}^2)$$

is an isometry, with zero output outside the eight disjoint patches. There is no additional factor $1/\sqrt{8}$. Its exterior powers give an isometry $W_a^{\leq N}$ on all particle numbers through a fixed N, with the vacuum and every valley retained.

Writing $s_j = (-1)^{b_j}$ and $\chi_b = -s_1 s_2 s_3$, the limiting blocks are

$$h_b(p) = -v \sum_j s_j \sigma_j p_j, \qquad E_\pm(p) = \pm v|p|, \qquad E^2 - v^2|p|^2 = 0.$$

All eight continuum masses are zero. Four blocks have each chirality. The proper frame $R_b = -\chi_b \operatorname{diag}(s)$ has a constant $SU(2)$ lift and spin $S_{b,j} = R_{b,jj}\sigma_j/2$, so $h_b = 2\chi_b v S_b \cdot p$. The positive-energy helicity is $\chi_b/2$; the negative-energy helicity is $-\chi_b/2$. At $p = 0$ the two spin states are degenerate and helicity is undefined. In these frames an opposite pair has $\alpha_j = \operatorname{diag}(-\sigma_j, \sigma_j)$. With off-diagonal identity β, the matrices $\gamma = (\beta, \beta\alpha_1, \beta\alpha_2, \beta\alpha_3)$ satisfy the Clifford relations for signature $(1,-1,-1,-1)$, checked exactly. Thus the limiting differential equation is massless Weyl/Dirac in coordinate $x^0 = vt$, with the Clifford relations verified.[7] The fixed cutoff and the lattice regulator are not Lorentz invariant; neither the valley pairing nor this spin frame selects physical species.

The full interacting evolution has an explicit comparison bound. The sine remainder gives, uniformly on C,

$$\|h_a(\nu + ap) - h_b(p)\| \leq \frac{va^2}{6} \sum_j |p_j|^3 \leq \frac{va^2\Lambda^3}{2}.$$

For $a\Lambda \leq 1$ and $p \neq 0$, both spectral-projector errors are at most $a^2\Lambda^2/6$. The squared-energy defect lies between $-v^2a^2 \sum_j p_j^4/3$ and zero. No projector at the degenerate node is assigned.

Let Ω_a project onto the initial Fock band and $d = \lfloor N/2 \rfloor$. On the infinite lattice the site-diagonal band weight is exactly $f_a = 8(a\Lambda/\pi)^3$. The positive countable partial sums defining D are bounded by d, hence converge strongly and satisfy $D^2 \leq dD$. On band states each annihilation has norm at most $\sqrt{f_a}$ and preserves the band for the remaining particles. Summing its squared norms gives $\Omega_a D \Omega_a \leq f_a d$, separately on each up/down number sector, and therefore $\|D\Omega_a\| \leq d\sqrt{f_a}$. This proves the infinite-volume attachment of the finite dilute-band estimate without substituting a continuum fraction for a finite count.

Duhamel's formula yields

$$\|e^{-itH_a} W_a^{\leq N} - W_a^{\leq N} e^{-itd\Gamma(h_W)}\| \leq \min\{2, |t|\epsilon_a\},$$
$$\epsilon_a = \frac{Nva^2\Lambda^3}{2} + |U(a)|d\sqrt{8(a\Lambda/\pi)^3}.$$

The left evolution is the full source Hamiltonian: no band projection is inserted and every outgoing contact state remains present. The estimate also holds with arbitrary untouched references; it

[7] For the standard Weyl equation and helicity convention, see D. Tong, *Quantum Field Theory*, Sections 4.4 and 4.7, https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S4.html.

bounds half trace distance and individual probability errors, capped by one. For fixed N, Λ, v and bounded times it tends to zero whenever $U(a) = o(a^{-3/2})$. At $N \le 1$ the contact is exactly zero for every $U(a)$. Thus the proved limit is free; the critical pair scaling $U \sim a^{-2}$ and finite-density limits are outside this conclusion. For $N = 2$, $\Lambda = v = U = t = 1$ and $a = 1/64$, the complete-vector bound is less than 0.001237 and the projector bound is $1/24576$.

The finite source check retains all 128 one-particle Fourier columns on a periodic 4^3 cube. An eight-valley two-particle state there has exact $\|D\psi\|^2 = 1/8$ and $\|(1-\Omega)D\psi\|^2 = 7/64$, so the contact escape is real. Propagation on all 8,128 two-particle states at $t = 1/8$ gives a numerical vector difference about 0.0441641, below the rigorous $\sqrt{1/8}/8$. These matrix diagnostics and exact source identities do not constitute an additional native preparation or detector execution.

Resulting energies and the physical boundary. At $v = 1$, $a = 1/64$, every valley has the following energies; both signs are retained, and only their common positive magnitude is displayed. The receipt gives rational enclosures for the rounded lattice values.

p	lattice $\lvert E_a\rvert$	continuum $\lvert E\rvert$	continuum mass
$(0,0,0)$	0	0	0
$(1,0,0)$	0.999959310393	1	0
$(1,1,0)$	1.414156018578	$\sqrt{2}$	0
$(1,1,1)$	1.731980331101	$\sqrt{3}$	0

These are original-generator model energies, without an SI calibration. The finite-spacing mass-shell defect is a regulator error, not a tachyonic mass. This growing-lattice branch is distinct from the protected eight-site cube: neither its repair invariance nor the identity of its eight response lines is transferred to the valleys. The empty Fock state is not a ground state: at finite spacing, a one-particle mode at $(\pi/2, 0, 0)$ has energy $-c$, with zero contact for every U. This compares different number sectors; it is not a number-preserving extraction protocol from the empty sector. A filled vacuum, its interacting stability and a physical antiparticle dictionary remain unconstructed. A single 2×2 constant Hermitian mass anticommuting with all three kinetic Pauli matrices is exactly zero; this limited obstruction does not exclude valley coupling or many-body mass generation. The package `model/relativistic_sector_audit` therefore adds a controlled massless relativistic dynamical sector, not massive physical species, a photon assignment, an interacting relativistic vacuum theory or Einstein dynamics.

The preceding continuum construction does not itself execute a detector. The following finite periodic construction attaches model flavor and energy labels to a source-compiled occupation readout, and compares the resulting massless sector with observed particle data.

A detector identifies the model's propagating Weyl flavors. The massless sector has a concrete model-particle readout. On the periodic 8^3 lattice at $c = 1/8$, retain all 1,024 physical CAR modes of the original Weyl/contact action. For each $b \in \{0,1\}^3$, choose momentum labels $\ell_b = (4b_x, 4b_y, 4b_z + 1)$, with $k_b = \pi \ell_b / 4$. For component eigenvalue $r = \pm 1$ of σ_z, the normalized mode is

$$\psi_{b,r}(x,s) = \frac{e^{ik_b \cdot x}}{\sqrt{512}} \delta_{s,(1-r)/2}, \qquad H\psi_{b,r} = E_{b,r}\psi_{b,r}, \qquad E_{b,r} = -r(-1)^{b_z}\frac{\sqrt{2}}{16}.$$

The contact term is exactly zero on the entire one-particle sector for every supplied $U \ge 0$. These sixteen modes are orthonormal, checked in Gaussian integers with amplitudes $(A + \sqrt{2}B)/32$. Their rank-one projectors satisfy $H\Pi_{b,r} = E_{b,r}\Pi_{b,r}$; the remaining one-particle projector commutes with

H. Thus a label gives an actual source-generator energy, not a fitted interpretation of a peak. All modes have $N = 1$, odd CAR parity and fermionic exchange statistics from the supplied CAR algebra.

The operational species are eight Weyl flavors indexed by measured valley momentum. Their common energy magnitude is $\sqrt{2}/16 \simeq 0.0883883476483$ in original model units; it alone does not distinguish the flavors. With $\chi_b = -(-1)^{b_x+b_y+b_z}$, the positive-energy branch has cone helicity $\chi_b/2$ and the negative branch has the opposite helicity.

flavor b	momentum labels ℓ_b	positive-branch component	positive cone helicity
000	$(0, 0, 1)$	1	$-1/2$
001	$(0, 0, 5)$	0	$+1/2$
010	$(0, 4, 1)$	1	$+1/2$
011	$(0, 4, 5)$	0	$-1/2$
100	$(4, 0, 1)$	1	$+1/2$
101	$(4, 0, 5)$	0	$-1/2$
110	$(4, 4, 1)$	1	$-1/2$
111	$(4, 4, 5)$	0	$+1/2$

Here helicity uses the explicit proper valley frame $R_b = -\chi_b \operatorname{diag}((-1)^{b_j})$; it is $(-1)^{b_x+b_y} r/2$ on this family. The raw detector measures momentum and CAR component, so bare $\sigma_z/2$ is not silently identified with this cone spin or with physical particle spin. The total directional current $J_z(k) = -c \cos k_z \, \sigma_z$ has eigenvalue $E_{b,r}$ here, with nonzero current $\pm\sqrt{2}/8192$ on every forward z bond. Current is stationary on these selected modes; $[H, J_z] \neq 0$ at generic momenta. The witness is propagating flux, even for a uniform-density plane wave.

The Fourier readout is compiled with complete native errors. The detector applies the fermionic extension of $F[k, x] = e^{-2\pi i k \cdot x/8}/\sqrt{512}$ separately to each component, then copies all 1,024 occupations to fresh pointers. Its output includes every binary occupation string, the vacuum, all other one-particle addresses, multiple occupations and every export abort. On the chosen sixteen effects it measures the energies above; the other 1,008 one-particle records remain present without that energy assignment. On higher-number inputs these same records measure mode occupations, not the total interacting energy.

An explicit three-axis FFT uses 4,608 two-mode butterflies and 768 bit-reversal transpositions. Each butterfly is a mode phase followed by the native hopping word with coefficient $-i$ and duration $\pi/4$. All nonadjacent operations use graded FSWAP routes, including their double-occupation sign. Exact arithmetic in $\mathbb{Z}[\zeta_8]/(\zeta_8^4 + 1)$ proves the 64 entries and 64 Gram sums of the one-axis transform; tensoring gives the full mode map. One transform requests 1,260,800 adjacent FSWAPs and 32,909,824 native pulses. Native excitation from the vacuum, inverse-transform preparation, forward-transform readout and all occupation pointers total 65,822,721 formal pulses. Original-H propagation is a separate specified model-law service, not an uncounted synthesized field word.

The complete literal-source error for preparation and detection is

$$\epsilon_{\text{source}} < 8.31 \times 10^{-9}.$$

The estimate retains all first-export failures and is valid with untouched references. It does not assume that literal native maps are exactly Gaussian or number preserving. Unitary original-H propagation on the full Fock space does not amplify preparation error. Therefore, at every model time, a prepared catalog member produces its correct native label with unconditional probability

at least $1 - \epsilon_{\text{source}}$; all wrong, outside, vacuum, multiple-occupation and abort outputs together have probability at most ϵ_{source}.

The replay executes the ideal Gaussian comparison on all 1,024 one-particle rows and all sixteen inputs. At time eight its complete output-vector discrepancy from the exact spectral prediction is about 1.94×10^{-15}, separate from the rigorous native bound. A seeded 65,536-record illustration retains all sixteen label counts and every zero outside count for the exact uniform input mixture. These are model-simulated counts, not a literal 2^{1024}-row native execution or laboratory data. No survival normalization is used.

Mass readout and comparison with measured particles. For the periodic family $L = 8 \cdot 2^m$, $a = 2^{-m}$ and $c = (1/8)/a$, the model period stays eight and the momentum offset stays $p_z = \pi/4$. The exact branch energies satisfy

$$E_{\pm}(a) = \pm \frac{\sin(a\pi/4)}{8a} \longrightarrow \pm \frac{\pi}{32}, \qquad m_{\text{continuum}} = 0.$$

This is a direct finite-torus consequence of the original spectrum. The finite-spacing mass-shell defect is a regulator correction, not a tachyonic mass. No SI scale is assigned.

A separate retrospective comparison uses fixed PDG 2025 targets, which never enter the source or detector calculation:

physical target	measured mass in MeV/c^2	present continuum mass
electron	$0.51099895000 \pm 0.00000000015$	0
muon	$105.6583755 \pm 0.0000023$	0
tau	1776.93 ± 0.09	0

No common positive unit conversion maps zero to these positive masses; equal finite-momentum energies are not a rest-mass hierarchy.[8] The photon has spin one, so direct spin-preserving identification with these cone-helicity-1/2 modes also fails. Three such massless flavors give zero mass-squared differences, outside the nonzero ranges of the specified three-neutrino fit in PDG Table 14.7.[9] A single massless neutrino or a composite photon in another sector is not excluded. The model's Weyl flavors are operationally identified; direct identifications with measured physical species remain zero. The empty reference is not a ground state, and the negative branch is retained rather than renamed as a physical antiparticle. Interacting vacuum stability, mass generation, physical charges and the empirical dictionary remain additional work.

The missing ground-reference and mass-generation steps admit further constructions with explicit domains. The original cube has a global interacting ground, and its one-dimensional restriction has an interaction-generated canonical charge gap. A specified refinement of that restriction also permits a controlled relativistic mass calculation. The following results retain the reference and coupling choices rather than combining these distinct domains into an assumed physical sector.

A global interacting ground on the original cube. The unchanged open $2 \times 2 \times 2$ Weyl/contact source, with $c = 1/8$ and $U = 1$, admits a certified comparison across its entire 65 536-dimensional Fock space:

$$E_G = -\frac{\sqrt{3}}{4}, \qquad N_G = 4, \qquad \dim \ker(H - E_G) = 5.$$

[8] PDG 2025 listings: https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf; https://pdg.lbl.gov/2025/listings/rpp2025-list-muon.pdf; https://pdg.lbl.gov/2025/listings/rpp2025-list-tau.pdf.

[9] The photon listing is https://pdg.lbl.gov/2025/listings/rpp2025-list-photon.pdf. The selected normal-ordering fit gives $\Delta m^2_{21} \in [6.92, 8.05] \times 10^{-5}\,\text{eV}^2/c^4$ and $\Delta m^2_{32} \in [2.388, 2.531] \times 10^{-3}\,\text{eV}^2/c^4$; these are reported 3σ fit ranges, not exact bounds: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf.

The five states form one hidden-spin $S = 2$ multiplet. No chemical potential, energy recentering, mass term, or replacement Hamiltonian is introduced. This is a finite interacting ground reference, not a claim of a unique or Lorentz-invariant infinite-volume vacuum.

The existing onsite unitary frame sends the original hopping to the spin-independent signed cube adjacency $A/16$, where

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}, \quad BB^T = 3I_4, \quad K = \begin{pmatrix} \sqrt{3}I_4 \\ -B^T \end{pmatrix}, \quad K^*K = 6I_4.$$

Filling the four negative orbitals $K/\sqrt{6}$ with hidden spin up gives an explicit normalized state G. Its occupation coefficients belong to $\mathbb{Q}(\sqrt{3})$, it has no doublons, and the original physical-frame eigenvalue equation is checked exactly. With $\lambda = \sqrt{3}/16$, the $N = 4$ bound $H \geq -4\lambda$ is saturated precisely by negative-band states annihilated by every local pair annihilator. On the full 70-dimensional four-particle negative-band space, the stacked contact map has spin-weight block dimensions $1, 16, 36, 16, 1$ and ranks $0, 15, 35, 15, 0$. Nonzero integer minors modulo101 certify the rank lower bounds; five explicit spin-lowered vectors certify the matching upper bounds. This proves the complete five-dimensional ground kernel in that number sector.

For $N \leq 3$, positivity gives $H \geq -N\lambda > -4\lambda$. For $N \geq 9$, the occupancy bound $D \geq N - 8$ gives $H \geq -N\lambda + N - 8 > 0$. For $N = 5, 6, 7, 8$, hidden $SU(2)$ symmetry ensures that every eigenvalue occurs in the balanced spin weight. Those block dimensions are $1568, 3136, 3920, 4900$. A numerical factorization proposes an integer matrix Q; only the following subsequently checked integer identity enters the proof:

$$H = 2^{-40}QQ^T - \frac{27}{64}I + R, \qquad \|R\|_2 \leq \|R\|_\infty < \frac{1}{512}.$$

All integer product and row-sum overflow bounds are checked before evaluation. Since QQ^T is positive semidefinite for every proposal, the argument requires no trusted floating eigenvalue, factorization error estimate, or nonzero-pivot premise. The exact comparison

$$\frac{\sqrt{3}}{4} > \frac{441}{1024} > \frac{217}{512}$$

then proves that every $N \neq 4$ lies more than $7/1024$ above E_G. This last bound concerns charged sectors; no numerical lower bound for neutral excitations inside $N = 4$ is asserted. Any density matrix supported on this finite ground space is passive for the stated H.

Positive charged spectral data above this ground. At the origin the onsite frame is the identity, so the probes below are the original physical CAR modes. Their exact local spectral data are

probe or component	N	weight	$\omega = E - E_G$
$c_\uparrow G$	3	1/2	λ
$c_\uparrow^\dagger G$	5	1/2	λ
$c_\downarrow G$	3	0	absent
$S = 5/2$ component of $c_\downarrow^\dagger G$	5	1/10	λ
$\eta = 5/2$ component of $c_\downarrow^\dagger G$	5	1/10	$1 + \lambda$

Thus the resolved model gaps are approximately 0.108253175473 and 1.108253175473 in source energy units. They are finite-volume gaps, not relativistic rest masses. The remaining down-addition weight is $4/5$, with strictly positive spectral support and nonzero variance.

More explicitly, set $v = c_{\downarrow}^{\dagger} G$, with $\|v\| = 1$. The exact vectors

$$v_s = \frac{1}{5} S^- S^+ v, \qquad v_\eta = \frac{1}{5} \eta^+ \eta^- v, \qquad v_r = v - v_s - v_\eta$$

are mutually orthogonal and have squared norms $1/10, 1/10, 4/5$. The first two obey the original Hamiltonian eigenvalue equations at -3λ and $1 - 3\lambda$. Consequently the complete greater addition measure satisfies

$$\mu_\downarrow = \frac{1}{10}\delta_\lambda + \frac{1}{10}\delta_{1+\lambda} + \frac{4}{5}\mu_r, \qquad \operatorname{supp} \mu_r \subset (7/1024, \infty).$$

The displayed $1/10$ values are exact symmetry-resolved contributions; without further resolution of μ_r, coincident residual atoms are not excluded. All moments below are checked over $\mathbb{Q}(\sqrt{3})$:

$$\mathbb{E}\omega = \frac{1}{2}, \qquad \mathbb{E}\omega^2 = \frac{131}{256}, \qquad \operatorname{Var}(\omega) = \frac{67}{256},$$

$$\mathbb{E}_r\omega = \frac{1}{2} - \frac{\sqrt{3}}{64}, \qquad \mathbb{E}_r\omega^2 = \frac{131}{256} - \frac{\sqrt{3}}{64}, \qquad \operatorname{Var}_r(\omega) = \frac{1069}{4096} > 0.$$

The contact expectation in v is $1/2$; deleting that interaction changes its mean gap from $1/2$ to zero. This channel therefore probes the actual contact interaction. Addition and removal carry model number changes $+1$ and -1 and odd CAR grading relative to G; electric charge and physical Lorentz spin do not follow from these labels.

Attachment to the declared repair process. The existing four-dimer repair completion has24 active columns, all entirely in $N = 6$. Its complete130-outcome dictionary therefore acts on $N = 3 \oplus N = 4 \oplus N = 5$ as zero for every active Kraus outcome and as identity for the outside projector. This6748-dimensional statement retains cross-number coherences and arbitrary reference systems. Since the original H preserves number, it also survives every finite interleaving with the ordinary field evolution. This is passive outside-service protection, not active correction or autonomous preparation of the ground.

The new finite ground removes the negative-frequency objection that applied to the earlier excited reference. It does not retroactively turn that earlier preparation into ground-state preparation. A native source and detector for G, the unresolved charged spectrum, a controlled growing-volume interacting vacuum, relativistic massive poles, physical dictionaries and units, and the geometric Einstein premises remain separate obligations. The exact certificate and negative controls are retained in `model/interacting_vacuum`.

An interaction-generated charge scale in the original one-dimensional law. There is a positive many-body charge threshold without inserting a Dirac mass. On the unchanged periodic Weyl/contact geometry $(L, 1, 1)$, $L = 4m$, the onsite frame $G_x = (-1)^x (i\sigma_x)^x$ satisfies $G_x (i\sigma_x) G_{x+1}^{\dagger} = -I$ and has period four. Its CAR lift fixes local doublons, so the original action is exactly the repulsive Hubbard chain with $t = c/2 = 1/16$ and $U = 1$. Four exact local frame identities, source-matrix checks through $L = 16$, and an independent 256-state full-Fock witness bind this join. Unlike the earlier logarithmic-reference alternative Hamiltonian, no coupling is reassigned.

For the *half-filled canonical ground sector*, the imported Lieb–Wu charge-gap theorem gives

$$\Delta_c = \lim_{L\to\infty} [E_L(L+1) + E_L(L-1) - 2E_L(L)] = U - 4t + 8t \sum_{k\geq 1} \binom{1/2}{k} (2t/U)^{2k-1} \eta(2k-1).$$

Here the series is used only for $2t/U < 1$. Twelve binomial terms and 96-term rational Euler bounds for each Dirichlet eta value enclose $\Delta_c = 0.7715517157380375\ldots$ with width below $1.67\,10^{-25}$. The binomial tail is bounded by $8t|\binom{1/2}{K+1}|(2t/U)^{2K+1}/[1-(2t/U)^2]$; the nonnegative Euler coefficients give tail 2^{-96}. Floating quadrature is an independent check only.

Particle–hole symmetry fixes the charge costs for the original H:

$$E_+ = \frac{U+\Delta_c}{2} = 0.885775857869\ldots, \qquad E_- = \frac{\Delta_c - U}{2} = -0.114224142131\ldots.$$

Thus half filling is a specified canonical reference, not the original $\mu = 0$ grand-canonical vacuum. The symmetric threshold $\Delta_c/2 = 0.3857758578690188\ldots$ uses the explicitly centered $K = H - (U/2)N$; Δ_c itself is invariant under any common $-\mu N$. The spin sector remains gapless, contact vanishes in the one-particle sector, and no preparation or complete-repair preservation of this many-body reference is established by the gap theorem.

The fixed-interaction continuum boundary. The equivalent positive integral for the charge gap and the elementary K_1 integral bounds imply, for every $t, U > 0$,

$$\Delta_0 \le \Delta_c \le \Delta_0 \frac{1+3/(8A)}{1-e^{-2A}}, \qquad A = \frac{2\pi t}{U}, \qquad \Delta_0 = \frac{8}{\pi}\sqrt{Ut}\,e^{-A}.$$

Consequently fixed U with $t = v/(2a)$ has $\Delta_c \to 0$ as $a \to 0$: this lattice gap does not itself supply a nonzero continuum mass. A sufficient additional running family is

$$a(A) = \frac{4ve^{-A}}{M\sqrt{2\pi A}}, \qquad t(A) = \frac{v}{2a(A)}, \qquad U(A) = \frac{\pi v}{a(A)A},$$

for a *chosen* $M, v > 0$. It gives $\Delta_0/2 = M$ exactly and $1 \le \Delta_c/(2M) \le [1+3/(8A)]/[1-e^{-2A}] \to 1$. This is a quantitative charge-threshold limit, not a derivation of M from the fixed source, a complete relativistic scaling theory, or a $3+1$-dimensional massive species. The receipt `model/mass_generation_audit/results.json` keeps these reference, coupling-family, and dimensional premises explicit.

The preceding charge-threshold result alone does not prove a mass shell. The next construction supplies energy and momentum together, with a controlled relativistic limit along an explicitly chosen coupling family.

A controlled relativistic charge mass on an explicit running family. The preceding charge gap can be promoted to a relativistic dispersion *in one spatial dimension on a specified coupling trajectory*. The source-bound half-filled Hubbard reference and centered generator $K = H - (U/2)N$ remain explicit. The staggered particle–hole map $d_{x,\downarrow} \mapsto (-1)^x d^\dagger_{x,\downarrow}$, with $d_{x,\uparrow}$ unchanged, preserves hopping and reverses the centered interaction. A complete 256-state CAR witness checks this duality and the translation identity $PTP^\dagger = -(-1)^{N_\downarrow}T$. Consequently the attractive spin dispersion becomes the repulsive charge dispersion, with the appropriate momentum-carrier shift. Here p is relative to that low-energy branch; the onsite source frame also carries its specified uniform internal rotation. The even-ring SO(4) selection rules are retained: an elementary thermodynamic quasiparticle is not automatically an isolated finite-ring state or a local-CAR pole.

Writing $A = 2\pi t/U$, $\theta = A\sin k$, $|k| < \pi/2$, and $v = 2ta$, the imported exact dressed dispersion has the source-normalized form

$$E = \frac{8t}{\pi}\sum_{\substack{m\ge 1\\ m\ \mathrm{odd}}} \frac{K_1(mA)}{m}\cosh(m\theta), \qquad vp = \frac{8t}{\pi}\sum_{\substack{m\ge 1\\ m\ \mathrm{odd}}} \frac{K_0(mA)}{m}\sinh(m\theta).$$

On $a(A) = 4ve^{-A}/[M\sqrt{2\pi A}]$, $t = v/(2a)$, $U = \pi v/(aA)$, the leading Bessel coefficient is exactly M. For $A \geq 1$ and $|\theta| \leq \Theta < A$, put

$$q = e^{-2(A-\Theta)}, \qquad r_E = \frac{1 + 3/(8A)}{1-q} - 1, \qquad r_p = \max\{1/(8A), q/(1-q)\}.$$

The Gamma-integral tangent bounds $1 - 1/(8z) \leq K_0(z)/b(z) \leq 1$ and $1 \leq K_1(z)/b(z) \leq 1 + 3/(8z)$, with $b(z) = \sqrt{\pi/(2z)}e^{-z}$, and the positive odd-mode tail prove

$$|E/M - \cosh\theta| \leq r_E \cosh\Theta, \qquad |vp/M - \sinh\theta| \leq r_p \sinh\Theta,$$

$$\left|(E^2 - v^2p^2)/M^2 - 1\right| \leq \cosh^2\Theta(2r_E + r_E^2) + \sinh^2\Theta(2r_p + r_p^2) \;\longrightarrow\; 0.$$

At $\Theta = 1$, the last bound is below 0.034 for $A = 64$ and 0.017 for $A = 128$. Thus the limiting charge-doublet dispersion has $E = M\cosh\theta$, $p = (M/v)\sinh\theta$, and the calculated rest mass is $m = M/v^2$. Exact rational finite-parameter enclosures and independent Bessel/integral controls are retained in `model/hubbard_relativistic_sector/results.json`.

For a concrete trajectory through the original point, take $A_0 = \pi/8$, $v = 1/8$ and $M_0 = e^{-\pi/8}/\pi$. Then $a(A_0) = 1$, $t(A_0) = 1/16$, $U(A_0) = 1$, and

$$M_0 = 0.2149329913552644\ldots, \qquad m_0 = \frac{64e^{-\pi/8}}{\pi} = 13.75571144673692\ldots \quad \text{in model mass units.}$$

This is the mass of the *chosen running trajectory*, not a unique mass selected by the original bare point. Indeed, for any $M > 0$, multiplying $a_M(A)$ by $1 + (M/M_0 - 1)e^{-(A-A_0)}$ gives the same starting point and limiting rest energy M. The initial lattice half-gap is different, and the fixed-U continuum family has zero limiting charge mass. No native ground preparation, repair-invariant charged pole, $3+1$-dimensional massive species, or empirical particle-mass identification follows from this one-dimensional result.

The original three-dimensional action also permits a growing-volume ground construction at fixed spacing. This closes ground-state existence and positive excitation energy without selecting the one-dimensional canonical reference or its coupling continuation. Its local response is then connected to the existing geometry theorem interfaces.

An original-action infinite-volume ground representation at fixed spacing. The full three-dimensional Weyl/contact action admits a translation-invariant, gauge-invariant ground state at its original zero chemical potential, without choosing a particle-number sector. This is a compactness existence theorem, not a computed or natively prepared state. Keep fixed finite $c > 0$, $U > 0$ and the complete quasi-local CAR algebra with two modes per site. The earlier proved full-Fock locality estimate supplies the strongly continuous dynamics τ_t. On each even-sided cube Λ_L of side $2L$, take the normalized projector onto the absolute minimum of H_{Λ_L}, extend this even state by the exterior vacuum, and average its translates over all sites of Λ_L. A diagonal subsequence of the bounded local matrix entries defines a state ω; positivity, normalization and CAR compatibility pass to the limit. The vanishing boundary-to-volume ratio makes ω translation invariant.

For a local A, including an odd charged probe, the local derivation $[H, A]$ contains only interaction terms touching its support. Whenever the support and its nearest neighbors lie inside the box,

$$\omega_L(A^*[H, A]) = \operatorname{Tr}\rho_L A^*(H_{\Lambda_L} - E_L)A \geq 0.$$

Bad translates occupy a vanishing fraction. Their real-part lower defect and imaginary magnitude are each bounded by $2\|A\|^2|\operatorname{supp} A|(U + 6c)$, so $\omega(A^*[H, A]) \geq 0$ in the limit. This uses the full ground minimum and would fail for a merely canonical ground with an unrestricted charged probe.

For a radius-R buffer, let $B_R(t)$ denote the existing volume-uniform local-dynamics boundary bound. Comparing both finite and infinite dynamics to this intermediate cube gives, for $\|A\| \le 1$,

$$\left|\nu_L(\tau_t(A)) - \nu_L(A)\right| \le \min\Big\{2,\, 2B_R(t) + 2\left[1 - (1 - R/L)^3\right]\Big\}.$$

First $L \to \infty$, then $R \to \infty$, establishes stationarity. The same comparison applies to two-point correlations. Every finite full-ground correlation has a positive spectral measure supported on $[0, \infty)$; compact-time convergence and bounded correlations pass all Schwartz Fourier tests supported on negative energies to the limit. The GNS implementation therefore satisfies

$$U_t \pi(A)\Omega = \pi(\tau_t(A))\Omega, \qquad U_t = e^{itL_\omega}, \qquad L_\omega \ge 0, \quad L_\omega \Omega = 0.$$

The generator measures excitation energy in this representation; no chemical potential has been inserted into the source action.

The state can be chosen at nonzero density. Tiling each even box by the certified four-particle cube vector gives contact zero, energy $-2\sqrt{3}c$ per eight-site tile, and zero expectation for every cross-tile hopping. Hence $E_L/|\Lambda_L| \le -\sqrt{3}c/4$. Cartesian difference operators commute and the Pauli mixed terms cancel, giving $h^2 \le 3c^2 I$ and $H \ge -\sqrt{3}cN$. Thus every such full ground has density at least $1/4$. The exact real-bipartite particle–hole identity $E(2V - N) = E(N) + U(V - N)$ excludes $N > V$ from an absolute minimum when $U > 0$. Spatial averaging before the subsequence passes both bounds:

$$\frac{1}{4} \le \omega(n_{0\uparrow} + n_{0\downarrow}) \le 1.$$

Half-filling and a particular density are not assumed or concluded. The original ground energy per site also has a limit e_∞. For every even cube side ℓ, tiling larger boxes gives the computable bracket, conditional on its finite full-ground energy,

$$\frac{E_\ell}{\ell^3} - \frac{3c}{\ell} \le e_\infty \le \frac{E_\ell}{\ell^3}, \qquad -\sqrt{3}c \le e_\infty \le -\frac{\sqrt{3}c}{4}.$$

The lower surface term counts intertile hopping norms; a product of gauge-invariant tile grounds gives the upper bound. Untiled boundary terms vanish per volume, and the bracket width tends to zero. For the constructed averaged ground, $\omega(h_0) = e_\infty$, where h_0 assigns half of each incident bond to the origin. This supplies an energy-density observable, not a state-convergence rate or rest mass.

The contact interaction remains represented:

$$[H_U - H_0, a_0^* a_2] = U a_0^* a_2 (n_1 - n_3) \ne 0.$$

Every unital GNS representation is injective on each full local matrix algebra, so this quartic operator cannot disappear in representation. This faithfulness of the representation does not assert faithfulness of the ground-state functional. The claim is a nonzero represented interaction derivation, not a scattering or connected-vacuum-correlator theorem.

An exact source countercontrol prevents simply substituting the free filled sea. For a pure Slater projector with $P_{xx} = I_2/2$,

$$\mathrm{Var}(D) = \frac{V}{16} + \sum_{x \ne y} |\det P_{xy}|^2 \ge \frac{V}{16}.$$

If it is also a kinetic eigenstate, its interacting energy variance is $U^2 \,\mathrm{Var}(D) > 0$. On the original cube, the complete 3844-term negative-band filling has $N = 8$, $\langle D \rangle = 2$ and $\mathrm{Var}(H) = 2/3$ at

$U = 1$. The exact checks and proof are retained in `model/thermodynamic_ground/results.json`. For local CAR probes, the same subsequence also supplies the analytic retarded response

$$G_{AB}(z) = -i \int_0^\infty e^{izt} \omega(\{\tau_t(A), B^*\})\, dt, \qquad \operatorname{Im} z > 0.$$

Each time tail after T is at most $2\|A\|\|B\|e^{-\eta T}/\eta$ for $\operatorname{Im} z \geq \eta > 0$. Compact-time locality and weak-star convergence, followed by this tail bound, give uniform convergence on compact subsets of the upper half-plane. The unknown finite-state convergence error is retained; no real-axis pole, local inverse or metric is inferred. The result closes fixed-spacing ground-state existence and the positive-energy representation for the actual interacting lattice. It supplies no selected ground functional, convergence rate, native preparation, repair stationarity, uniform gap, Lorentz continuum, physical particle assignment or Einstein dynamics.

Exact matter response before a geometric identification. The unchanged open $2 \times 2 \times 2$ Weyl/contact cube at $c = 1/8$, $U = 1$ supplies a state-dependent retarded response, including its exact eliminated-state memory. In the existing onsite hidden-spin frame, write $h = A/16$, where A is the real signed cube adjacency and $A^2 = 3I$. The retained full-Fock ground theorem gives the normalized highest-weight reference G, with four occupied negative-band up orbitals, $E_G = -\sqrt{3}/4$, and a strict gap $\delta = 7/1024$ to every different particle-number sector. The second exact stationary reference is $G_+ = \sqrt{2}\, c_{0\uparrow}^\dagger G$, of energy $-3\sqrt{3}/16$. No chemical potential, fitted mass, or new geometry law is inserted. Physical CAR responses are obtained by the existing onsite unitary frame; hidden spin is not assigned a physical Lorentz or electric-charge meaning.

For either reference ψ of energy E, the complete eight-site fermionic response, for $\operatorname{Im} z > 0$, is

$$\begin{aligned} \mathcal{G}_{xy}(z) &= -i \int_0^\infty e^{izt} \langle \{c_x(t), c_y^\dagger\} \rangle_\psi \, dt \\ &= \langle c_x [z - (H - E)]^{-1} c_y^\dagger \rangle_\psi + \langle c_y^\dagger [z + (H - E)]^{-1} c_x \rangle_\psi. \end{aligned}$$

There is an exact isometric dilation

$$\begin{aligned} V_y &= c_y^\dagger \psi \oplus \overline{c_y \psi}, & V^\dagger V &= I, \\ L &= (H_{N+1} - E) \oplus -\overline{(H_{N-1} - E)}, & \mathcal{G}(z) &= V^\dagger (z - L)^{-1} V. \end{aligned}$$

The conjugate removal space fixes the matrix indices; the present source arrays happen to be real. With $P = VV^\dagger$, $Q = I - P$, define $M_1 = V^\dagger L V$, $W = QLV$ and $B = QLQ|_{\operatorname{ran} Q}$. Exact block elimination gives

$$\mathcal{G}(z) = \left[zI - M_1 - \Sigma(z) \right]^{-1}, \qquad \Sigma(z) = W^\dagger (z - B)^{-1} W, \qquad W^\dagger W = M_2 - M_1^2.$$

Both L and B are self-adjoint; consequently $-\operatorname{Im} \mathcal{G}(z) > 0$ and $-\operatorname{Im} \Sigma(z) \geq 0$ in the upper half-plane. The eight observed columns do not replace the source Hilbert space: their exact complement remains in the memory kernel.

A computed change of response at fixed microscopic geometry. For the up channel of both references, the full cyclic identity $LV = Vh$ gives $\mathcal{G}_\uparrow(z) = (z - h)^{-1}$ and $W = 0$. For G, the up dilation retains all 56 addition and 56 removal states. The ground down channel has no removal component and retains all 560 states with $N_\uparrow = 4$, $N_\downarrow = 1$. Exact arithmetic over $\mathbb{Q}(\sqrt{3})$ gives

$$M_{1,\downarrow} = h + \tfrac{1}{2} I, \qquad W_\downarrow^\dagger W_\downarrow = \tfrac{1}{4} I.$$

Thus a static Hartree term alone does not reproduce this interacting response. Indeed $W_x = (n_{x\uparrow} - 1/2)c^\dagger_{x\downarrow}G$; orthogonal down locations and up occupation variance $1/4$ give the displayed Gram matrix directly.

For G_+, the down channel retains its entire 448-state invariant block. In lexicographic order of the eight cube sites, its exact matrices are

$$M^+_{1,\downarrow} - h = \operatorname{diag}(1, \tfrac{2}{3}, \tfrac{2}{3}, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}),$$
$$(W^+_\downarrow)^\dagger W^+_\downarrow = \operatorname{diag}(0, \tfrac{2}{9}, \tfrac{2}{9}, \tfrac{1}{4}, \tfrac{2}{9}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}).$$

The first-moment change is exactly the added orbital density, $(1/2, 1/6, 1/6, 0, 1/6, 0, 0, 0)$. Removing the contact term in the explicit $U = 0$ control removes both this Hartree term and the ground down memory, giving $(z - h)^{-1}$. That comparison does not declare G a full free-Fock ground. The excited reference receives no positive-gap assertion.

The complete local charged spectral measure. For the original physical origin probe $v = c^\dagger_{0\downarrow}G$, the retained calculation resolves all 78 distinct positive spectral lines, including the earlier unresolved weight $4/5$. A monic degree-78 polynomial p over $\mathbb{Q}(\sqrt{3})$ is admitted by the exact source equation $p(16H)v = 0$. A nonzero 78×78 Hankel determinant modulo 97, with $\sqrt{3} \mapsto 10$, proves the matching cyclic rank lower bound. Exact moments determine a numerator q such that

$$\langle v, (z - 16H)^{-1} v \rangle = \frac{q(z)}{1296\, p(z)}.$$

The denominator 1296 is the squared norm of the inherited integer state numerator. Modular reconstruction and a numerical eigensolver only propose coefficients and brackets. Integer annihilation and rank checks, 78 disjoint strict sign changes, and outward 1024-bit interval arithmetic certify every root and its strictly positive residue $q(x)/(1296p'(x))$. Each charged gap is $(x + 4\sqrt{3})/16$.

The smallest gap is approximately 0.029950938342053226, with weight 0.3130113772155167; the largest is approximately 1.8360744654423855, with weight $2.711810625844545 \times 10^{-10}$. The receipt retains every gap and weight in rational intervals rounded outward to 24 decimal places. Polynomial substitution proves that the two lines $\sqrt{3}/16$ and $1 + \sqrt{3}/16$ each have total weight exactly $1/10$. The remaining 76 lines have total weight $4/5$, leaving zero unresolved weight for this local probe. This is the complete local cyclic measure, not the entire 560-state spectrum, a native detector execution, an infinite-volume pole theorem, or a relativistic mass assignment.

Controlled low-frequency coefficients. For the interacting ground down channel, put $K = H_{4,1} - E_G$. The full-ground certificate gives $K > \delta I$. Also, $D \le 1$ in this block, the kinetic norm is at most 5λ, and $-E_G = 4\lambda$, with $\lambda = \sqrt{3}/16$; hence $K \le 1 + 9\lambda < 2$. The same bounds hold for its compression B. For $|z| \le r < \delta$ the exact memory has the convergent expansion

$$\Sigma(z) = -\sum_{n=0}^{p} z^n C_n + R_p(z), \qquad C_n = W^\dagger B^{-n-1} W \ge 0,$$
$$\|R_p(z)\| \le \frac{r^{p+1}}{4\delta^{p+1}(\delta - r)}.$$

These are finite gapped frequency coefficients, not a spatial derivative expansion or a relativistic cone. All retarded moments $M_0, \dots, M_4$ are also computed exactly; the corresponding large-frequency remainder is $2^{p+1}/[R^{p+1}(R-2)]$ for $|z| \ge R > 2$.

The static response is computed with an exact residual certificate. For $C = V^\dagger K^{-1} V = -\mathcal{G}_\downarrow(0)$,

$$\|C - (12.9058381883 I - 4.9026412320 A)\| < 10^{-7},$$
$$\|\Sigma(0) - (-0.3633623064 I - 0.0105943755 A)\| < 10^{-6}.$$

A numerical solve only proposes a matrix with denominator 2^{40}. Integer arithmetic encloses its full $\mathbb{Q}(\sqrt{3})$ residual; division by the proved δ bounds the solution error. The exact isometry then bounds C, and $C > I/2$ bounds its inverse. An independent rational inverse residual certifies $\Sigma(0) = C^{-1} - M_1$. No numerical tolerance admits the theorem, and exact equality to an affine adjacency polynomial is not inferred.

The existing geometry interface and its remaining join. The same-action thermodynamic-ground construction separately supplies analytic local responses along its ground subsequence on compact subsets of $\operatorname{Im} z > 0$. The cube calculation does not supply a rate of convergence to that state, and its finite gap is not transferred to infinite volume. The existing `wave_density_metric` constructor takes a complete local action density $\mathcal{C} = \mu \operatorname{diag}(-w, a)$, with $\mu, w > 0$ and $a > 0$, and recovers $g = \sqrt{-\det \mathcal{C}}\, \mathcal{C}^{-1}$. The existing `source_tetrad` interface in the common matter join then constructs the common coframe and the scalar, Weyl, Dirac and Maxwell principal-symbol identities for its supplied density. The present result supplies an actual matter-dependent inverse response from which such coefficients must be derived. It does not substitute a Hartree diagonal for a spatial principal symbol, identify the cube with the distinct screen source, or complete the common action-normalization join. Fixed microscopic geometry is compatible with this emergent-response route; a new fundamental metric register is not required. Common local limits, modular and area response, full stress, and the same-family Einstein balance premises remain to be proved. No new physical species, native response detector, or completed G1–G6 premise is claimed.

The finite ground and its charged probe can be prepared and read by the declared observer service. The full matrix spectrum extends the local probe calculation. Separately, the infinite ground admits a spatially local inverse response with nonzero interaction memory at complex frequencies separated from the real axis. These constructions supply additional inputs to the existing field and geometry interfaces.

Native preparation of the same ground and its charged response. The original cube ground and its local charged spectral measure have an explicit native source preparation and readout. The exact highest-weight ground G, with $N = 4$ and $E_G = -\sqrt{3}/4$, has 62 nonzero hidden-up occupation amplitudes $(a + b\sqrt{3})/36$, where a, b are integers and $ab = 0$. Its squared amplitudes are therefore exact integers divided by 1296. These source-derived weights and signs determine an eight-bit binary amplitude tree; no fitted spectrum or unknown-state oracle is used. The existing compiler emits 5,379 literal native pulses. An independent rational interval check consumes every event, inverts the Walsh layers, and verifies the sine and cosine of each conditional rotation against the exact signed square-root ratios. Disjoint control blocks give a maximum error per level, and telescoping the levels and complete native load/export maps bounds the full initialized preparation, including aborts.

Eight blank down modes are included by an exact isometric embedding into all 65,536 Fock rows. A further 184 native pulses apply the inverse onsite frame. Each local frame word is checked on its full four-state Fock space, including vacuum and doublon phases, using $e^{is\pi P/4} = (I + isP)/\sqrt{2}$ and Gaussian-integer products. Its literal-angle and native-instrument errors are separately enclosed. The resulting complete preparation-state error is below 10^{-10}; the retained exact bound is approximately 8.67×10^{-13}.

At the origin the hidden and original frames coincide, and the down mode is empty in G. Thus

$$\gamma_1 = Z_0 X_1 = c_1 + c_1^\dagger, \qquad \gamma_1 G = c_1^\dagger G, \qquad \|\gamma_1 G\| = 1.$$

The Jordan–Wigner sign is essential: replacing $Z_0 X_1$ by X_1 produces an orthogonal probe because the exact up occupation is $1/2$. This unitary creation preserves the entire initial exterior CAR state: disjoint even observables commute, and exterior odd expectations vanish in both definite-parity states. The origin-down occupation changes from zero to one and total number from four to five. Its mean added energy is $1/2$ with variance $67/256$; this is a localized initial defect with a conserved number flag, not a persistent localization or mass-eigenstate claim. A 27-pulse pointer word prepares $(|0\rangle G + |1\rangle \gamma_1 G)/\sqrt{2}$. Its controlled Majorana is the product of commuting Pauli rotations in $\exp[i\pi(I - Z_{\text{pointer}})(I - \gamma_1)/4]$. This explicitly supplied native qubit service maintains a phase relation between the $N = 4$ and $N = 5$ arms; it does not derive a physical fermion-parity reference from passive repair.

Both arms evolve under the same original Hamiltonian. With a second controlled Majorana and pointer analyzer of phase ϕ,

$$C(t) = \langle \gamma_1 G, e^{-it(H - E_G)} \gamma_1 G \rangle,$$
$$P_0(\phi, t) = \tfrac{1}{2} \left[1 + \mathrm{Re}(e^{i\phi} C(t)) \right].$$

The ground-energy phase is supplied by the evolving reference arm; no separate scalar offset is inserted into the source law. The two executed settings have $t = 1$ and $\phi = 0, \pi/2$. Each setting uses two half-time segments, each with 32 symmetric splitting steps and 337,920 native pulses, followed by a 29-pulse analyzer. Including its ground preparation and pointer, a complete setting uses 681,459 pulses on the full 17-bit field/pointer register.

Protection and complete output accounting. The five C repair seams follow the first time segment, and the five E seams follow the second. The complete repair theorem acts as identity on $N4 \oplus N5$, including cross-number coherences: all native branches vanish there, and the outside branch is identity. This is passive protection during the waits. Preparation and control words are not claimed to remain in this subspace at every intermediate pulse. The final inverse Majorana can create an $N = 6$ component after the charged particle has moved; no repair is applied after that analyzer.

Every source first-export failure is retained as its own terminal record. The numerical execution also retains all 130 repair outcome slots, including off-subspace outputs from control error, and all six final pointer outcomes. A native repair outcome terminates with that record; continuation uses the outside branch without renormalization. The lossless archive retains the prepared full ground, both full final states and every first-failure weight table. Individual failure vectors remain addressable from their actual source prefixes.

Connection to the resolved spectrum. All field and pointer words are consumed against the exact source templates. Their complete operator errors, the preparation error and the full-Fock Strang errors of both time segments give a complete output error below $3/5000$ for either setting. Contractivity through the complete repair and readout channels preserves the telescoping bound. The same bound controls each unconditioned pointer probability. Independently, the 78 certified spectral intervals evaluate

$$C(t) = \sum_{j=1}^{78} w_j e^{-i\Delta E_j t}$$

with rational sine and cosine enclosures. Adding the source error gives certified pointer intervals, checked against the literal execution and an independent exponential of the original Hamiltonian. The $\pi/2$ outcome is separated from $1/2$; the $U = 0$ comparator has the purely real response $\cos(\sqrt{3}t/16)$. Zero-time native runs check the analyzers as well. These two settings measure the charged characteristic function. They do not independently resolve all 78 lines, select an autonomous ground, prepare a thermodynamic vacuum, establish laboratory units, or turn finite energy gaps into relativistic particle masses.

The complete finite matrix spectrum and its exact spatial algebra. The 78-line local down-addition result extends to all eight source columns $V_x = c^\dagger_{x\downarrow} G$. On the entire 560-state invariant sector, exact arithmetic proves $q(16H)V = 0$ for the same monic degree-78 polynomial q and verifies all 78 matrix moments in $\operatorname{span}\{I, A\}$, where $A = A^{\mathsf{T}}$, $A^2 = 3I$ and $\operatorname{tr} A = 0$. Polynomial division of the resolvent then gives

$$\mathcal{G}(z) = \frac{16\big[p_I(x)I + p_A(x)A\big]}{1296q(x)}, \qquad x = 16z - 4\sqrt{3}.$$

The numerator coefficients are exact convolutions of q and those source moments. This is a rational resolvent identity, rather than a truncated large-frequency expansion.

A second exact polynomial calculation gives $q \mid (p_I^2 - 3p_A^2)$. At each simple root x_i, define $\alpha_i = p_I(x_i)/(1296q'(x_i))$ and $\beta_i = p_A(x_i)/(1296q'(x_i))$. The earlier interval certificates give $\alpha_i > 0$; the new identity gives $3\beta_i^2 = \alpha_i^2$, and strict interval signs determine $s_i = \operatorname{sign}\beta_i$. Therefore

$$R_i = \alpha_i I + \beta_i A = 2\alpha_i P_{s_i}, \qquad P_\pm = \tfrac{1}{2}(I \pm A/\sqrt{3}).$$

Both $P_\pm$ have rank four. There are 40 plus-band lines and 38 minus-band lines; each band's $2\alpha_i$ weights sum to one. The rank of $E_i V$ equals that of its Gram matrix R_i, so the reducing cyclic space generated by all eight probes has dimension $78 \cdot 4 = 312$. The remaining 248 source dimensions are orthogonal to these probes and their time translates; their spectrum is not claimed.

Writing $\mathcal{G} = FI + JA$, both band functions have strictly negative imaginary part for $\operatorname{Im} z > 0$. Hence

$$\mathcal{G}^{-1} = \frac{FI - JA}{F^2 - 3J^2}, \qquad \Sigma = zI - (A/16 + I/2) - \mathcal{G}^{-1}.$$

The full finite inverse response and memory are exactly onsite plus the original signed nearest-neighbor adjacency. This special cube algebra does not supply an infinite-volume spatial limit or a Lorentz metric.

The certified common spectral support lies strictly between $g = 299/10000$ and $b = 37/20$. On the reducing cyclic space, both $K = H - E_G$ and its compression perpendicular to the observed columns obey these bounds. The exact coupling $W = KV - VM_1$ lies in that same space, so the orthogonal unobserved sector contributes zero to the Schur memory. Using $W^\dagger W = I/4$, its order-p Taylor remainder satisfies

$$\|R_p\| \le \frac{r^{p+1}}{4g^{p+1}(g - r)}, \qquad |z| \le r < g.$$

This improves the earlier conservative full-sector disk without assigning the stronger gap to the unresolved sector or to infinite volume. Evaluation at $x = -4\sqrt{3}$ computes $-\mathcal{G}(0)$ and $\Sigma(0)$ exactly over $\mathbb{Q}(\sqrt{3})$, including the earlier bounded coefficients 12.9058381883 and -4.9026412320. No floating linear solve, native detector execution, physical species assignment or relativistic rest-mass inference is used in this extension.

Excluding a complete finite-gap mass dictionary. The full support gives an exact test of assigning these visible single-charge gaps to charged-lepton rest energies under one positive unit factor. Its largest-to-smallest ratio is below 62, including the positive up-channel line. The retained retrospective PDG 2025 benchmark has muon/electron ratio above 206 and tau/electron ratio above 3477, even in the declared central-value plus/minus reported-uncertainty boxes. If a source ratio is at most R and the target ratio at least $Q > R$, simultaneous relative errors at most $\epsilon < 1$ require

$$\frac{1-\epsilon}{1+\epsilon} \le \frac{R}{Q}, \qquad \epsilon \ge \frac{Q-R}{Q+R}.$$

The exact rational calculation therefore forces an error above 54 percent for at least one member of an electron/muon assignment, and above 96 percent for an electron/tau assignment, for every common positive scale and every choice of visible lines. The target values do not enter the source law or spectrum. This rejects the stated finite-gap mass dictionary, not a continuum, an unobserved or composite sector, or OPH as a whole. The finite gaps are not thereby proved to be relativistic rest masses, and no eligible prospective comparison is added.

A local inverse response on the interacting infinite ground. The original zero-chemical-potential Weyl/contact action on $\mathbb{Z}^3$ admits a controlled spatial inverse response on the earlier constructed thermodynamic ground, at fixed spacing and away from real frequency. This does not extrapolate the special two-eigenvalue cube identity. Let ω be the translation-averaged full-ground limit, L_ω its positive GNS generator, and

$$L = L_\omega \oplus -\overline{L_\omega}, \qquad V_{x\sigma} = c^\dagger_{x\sigma}\Omega \oplus \overline{c_{x\sigma}\Omega}.$$

The CAR give an isometry V from $\ell^2(\mathbb{Z}^3;\mathbb{C}^2)$. Define J in the same way from $b_{x\sigma} = n_{x\bar\sigma}c_{x\sigma}$ and write $\rho_{x\sigma\tau} = \omega(c^\dagger_{x\tau}c_{x\sigma})$. Exact onsite CAR identities, with vanishing different-site anticommutators, give

$$R = \bigoplus_x (\mathrm{tr}(\rho_x)I - \rho_x), \qquad V^\dagger J = J^\dagger V = J^\dagger J = R,$$
$$LV = Vh + UJ, \qquad 0 \le R \le I, \quad \|LV\| \le \sqrt{3}c + U.$$

Finite-support approximants and closedness of L extend the last identity to the entire one-particle input space. With $P = VV^\dagger$, $Q = I - P$, the bounded off-diagonal coupling is $W = QLV = U(J - VR)$. Subtracting $WV^\dagger + VW^\dagger$ from L proves self-adjointness of its compressed complement $T = QLQ|_{Q\,\mathrm{Dom}\,L}$. The exact infinite-space Schur identity is therefore

$$\mathcal{G}(z) = V^\dagger(z-L)^{-1}V = [zI - h - UR - \Sigma(z)]^{-1},$$
$$\Sigma(z) = W^\dagger(z-T)^{-1}W, \qquad W^\dagger W = U^2(R - R^2).$$

In particular $\|\Sigma(z)\| \le U^2/(4\,\mathrm{Im}\,z)$.

This ground has strictly nonzero memory for $U > 0$. The source hidden-spin frame obeys $R_{x+t} = \pm R_t R_x$, so translations normalize its global SU(2) symmetry. Every full-box ground projector and the exterior vacuum are invariant; translated averages and weak-star limits retain the symmetry. Hence $\rho_x = rI$, with the earlier density theorem giving $1/8 \le r \le 1/2$, and

$$\frac{7U^2}{64}I \le W^\dagger W = U^2 r(1-r)I \le \frac{U^2}{4}I.$$

The spectral theorem gives $iy\Sigma(iy) \to W^\dagger W$ strongly as $y \to \infty$. Thus the same-mode response cannot be that of a number-conserving static quadratic Hamiltonian with zero memory. Neither a scattering theorem nor faithfulness of the local ground-state density matrices follows.

A quantitative connected four-point witness. The same ground has a non-Gaussian multi-time field, a stronger statement than nonzero two-point memory. Fix $a = c_{x\sigma}$ and $n = n_{x\bar{\sigma}}$, and set

$$A(t) = \{\tau_t(a), a^\dagger\}, \qquad D(t) = A(t) - \omega(A(t))I, \qquad F(t) = \omega(D(t)^\dagger D(t)).$$

The local commutator gives $D(0) = 0$ and $D'(0) = -iU(n - rI)$. Since $n^2 = n$, $F(t)/t^2 \to U^2 r(1-r) \geq 7U^2/64$. An explicit remainder uses $M = 9c^2 + 12cU + U^2$ and $\delta = [H, \cdot]$:

$$\|\delta^2(a)\| \leq 3c(3c+U) + U(6c+3c+U) = M, \qquad \|D(t) + itU(n - rI)\| \leq 2Mt^2.$$

Thus, for $U > 0$,

$$F(t) \geq \frac{U^2 t^2}{64} > 0, \qquad 0 < |t| \leq \frac{U}{16M}.$$

For odd zero-mean operators define the connected fermionic function

$$\begin{aligned}\kappa_4(B_1, B_2, B_3, B_4) = {} & \omega(B_1B_2B_3B_4) - \omega(B_1B_2)\omega(B_3B_4) \\ & + \omega(B_1B_3)\omega(B_2B_4) - \omega(B_1B_4)\omega(B_2B_3).\end{aligned}$$

With $(B_1, B_2, B_3, B_4) = (a, \tau_t(a)^\dagger, \tau_t(a), a^\dagger)$, $F(t)$ is the sum of κ_4 on words $1234, 1243, 2134, 2143$. All crossed contractions, including anomalous ones, cancel pairwise. At least one connected function therefore has magnitude $U^2t^2/256$ or greater on the displayed interval. This excludes a quasifree representation of the complete multi-time field by linear fermionic fields, including pairing and auxiliary Gaussian modes. Indeed any quadratic evolution on an enlarged CAR algebra preserves linearity of an observed mode, making its anticommutator scalar and $D(t) = 0$ in every reference state. Thus even a non-Gaussian reference cannot realize this witness through such a linear quadratic embedding. It does not exclude nonlinear encodings or establish continuum scattering.

The full ordered Euclidean hierarchy. The same positive ground generator $L_\omega \geq 0$ defines, for ordered real times $s_1 \leq \cdots \leq s_n$ and bounded local CAR operators,

$$S(A_1, s_1; \ldots; A_n, s_n) = \langle \Omega, A_1 e^{-(s_2 - s_1)L_\omega} A_2 \cdots e^{-(s_n - s_{n-1})L_\omega} A_n \Omega \rangle, \qquad |S| \leq \prod_j \|A_j\|.$$

Every finite approximant uses one common translated full-ground state and its positive generator $H_\Lambda - E_\Lambda$ for the whole word, with zero exterior Hamiltonian, before averaging over translations. For each positive gap the spectral theorem gives

$$e^{-sL} = \int_{\mathbb{R}} \frac{s}{\pi(s^2 + t^2)} e^{itL}\, dt, \qquad L \geq 0.$$

The product integral is dominated by $\prod_j \|A_j\|$. Its real-time integrand is the expectation of $A_1 \tau_{t_1}(A_2) \tau_{t_1+t_2}(A_3) \cdots$. Local-dynamics bounds, telescoping the product, and the same weak-star ground subsequence give uniform convergence on compact time boxes. The omitted product-measure tail is at most $\sum_j 2s_j/(\pi T_j)$, capped by one; a finite-versus-limit comparison costs twice this tail times the product of operator norms. Dominated convergence therefore supplies every ordered Euclidean n-point limit on this one ground, without another subsequence or a state-convergence rate. Zero gaps are handled by the identity operator.

For a positive-time word $F = (A_1, t_1; \ldots; A_n, t_n)$ define

$$\Psi_F = e^{-t_1 L_\omega} A_1 e^{-(t_2 - t_1)L_\omega} A_2 \cdots e^{-(t_n - t_{n-1})L_\omega} A_n \Omega.$$

The reflected list is $(A_n^*, -t_n; \ldots; A_1^*, -t_1)$, in that order. Concatenation with a second positive-time word gives $S(\theta F, G) = \langle \Psi_F, \Psi_G \rangle$; hence every reflected correlation matrix is positive semidefinite. No CAR factors are reordered. Exact rational two-mode controls check the Gram identity, all 31 principal minors of a five-word example, and a negative odd-word cross entry. This proves ordered time-reflection positivity at fixed spacing. It does not establish $O(4)$ covariance, bosonic permutation symmetry, a spatial continuum, or the full relativistic Euclidean axioms.

Uniform inverse and spatial decay bounds. Fix $\operatorname{Im} z \geq \eta > 0$, $|z| \leq Z$, with $Z \geq \eta$. For a unit vector $v = Vf$, Cauchy–Schwarz applied to $\langle (\bar{z} - L)v, (z - L)^{-1}v \rangle = 1$ yields

$$-\operatorname{Im} \mathcal{G}(z) \geq B_0^{-1} I, \qquad \|\mathcal{G}(z)^{-1}\| \leq B_0, \qquad B_0 = \frac{(Z + 2c + U)^2}{\eta}.$$

Both $\mathcal{G}$ and its adjoint are bounded below; its range is therefore closed and dense, which proves surjectivity in infinite volume. No determinant or finite-volume gap is used.

Let $d = \|x - y\|_\infty$. The original graded locality bound gives

$$\|\mathcal{G}_{xy}(z)\| \leq A(1 + d)^2 e^{-2\mu d}, \qquad \mu = \min\{1/8, \eta/(384c)\}, \qquad A = 192c/\eta^2 + 8/\eta.$$

For $d \geq 2$, approximate the evolved odd operator in the radius $d - 1$ cube, whose anticommutator with the remote mode is zero. Split the Laplace integral at $(d - 1)/(192c)$ and use the previous full-CAR boundary estimate on the first part and the uniform norm bound on its tail. This proves the displayed constants without assuming spatial clustering of ω.

Put $q = (1 + \mu)^{-1}$ and $T_n(q) = \sum_{d \geq 1} d^n q^d$. These are rational functions with positive Eulerian coefficients. Define

$$\widehat{S} = 24T_5(q) + 48T_4(q) + 26T_3(q) + 4T_2(q) + 2T_1(q),$$
$$\alpha = \min\{\mu, (2B_0 A \widehat{S})^{-1}\} > 0.$$

For any bounded real one-Lipschitz function f, the block Schur test gives

$$\|e^{\alpha f} \mathcal{G} e^{-\alpha f} - \mathcal{G}\| \leq \alpha A \widehat{S} \leq (2B_0)^{-1}.$$

A Neumann series bounds the conjugated inverse by $2B_0$. Choosing $f(u) = \min\{\|u - y\|_\infty, R\}$ with $R \geq d$ proves

$$\|(\mathcal{G}^{-1})_{xy}(z)\| \leq 2B_0 e^{-\alpha d}, \qquad \|\Sigma_{xy}(z)\| \leq (Z + U + 2B_0 + 2c) e^{-\alpha d}.$$

Only bounded weights were used. The constants are conservative bounds, not estimates of a physical correlation length.

The controlled spatial symbol and its geometric boundary. Translation invariance and exponential summability give an analytic two-by-two inverse-response symbol $K(k, z)$ for $\|\operatorname{Im} k\|_1 < \alpha$. For real momentum increment p, its total-degree-n Taylor remainder is bounded by

$$\frac{2B_0 \|p\|_1^{n+1}}{(n+1)!} \sum_{x \in \mathbb{Z}^3} \|x\|_\infty^{n+1} e^{-\alpha \|x\|_\infty}.$$

The shell count $24d^2 + 2$ and the same rational T_n functions give explicit upper bounds on this convergent sum. A frequency circle of radius $\eta/2$ gives, for $|w| < \eta/2$, the additional remainder $B_*(2|w|/\eta)^{n+1}/(1 - 2|w|/\eta)$, where B_* is the above inverse bound evaluated at $(\eta/2, Z + \eta/2)$.

These estimates supply actual local spatial coefficients on the interacting infinite ground, with a controlled complex-frequency domain. They do not compute the ground functional or those coefficients, and deteriorate as $\eta \to 0$. A real hyperbolic principal symbol, common matter coefficients and full action normalization are required before applying the existing `wave_density_metric` and `source_tetrad` interfaces. The joint modular, area, stress and Einstein balance requirements remain separate.

For a refinement $c = v/a$ at fixed $v, U, Z, \eta > 0$, the retained estimates have $\alpha = O(a^9)$ and therefore physical momentum width $\alpha/a = O(a^8)$; the connected-witness time window is $O(a^2)$. These shrinking bounds do not prove uniform continuum control or a no-go theorem for the action.

The full ground-family spectrum and its internal spin. The complete five-dimensional $N = 4$ ground space closes a remaining finite spectral ambiguity. Work in the exact hidden-spin frame, with G_0 its spin-two highest-weight ground, $\lambda = \sqrt{3}/16$, and $V_{\uparrow,x} = c^\dagger_{x\uparrow} G_0$, $V_{\downarrow,x} = c^\dagger_{x\downarrow} G_0$. The actual 56×560 raising matrix $T = S_+$ obeys

$$TV_\downarrow = V_\uparrow, \qquad TT^\dagger = 5I, \qquad H_{5,0}T = TH_{4,1}.$$

The full spatial Gram is $V_\uparrow^\dagger V_\uparrow = P_+$, and $(H - E_G)V_\uparrow = \lambda V_\uparrow$. At magnetic weight $3/2$,

$$V_{\text{high}} = T^\dagger V_\uparrow/5, \qquad V_{\text{low}} = V_\downarrow - V_{\text{high}}, \qquad TV_{\text{low}} = 0, \qquad V_{\text{high}}^\dagger V_{\text{high}} = P_+/5.$$

The retained residue at λ is exactly $P_+/5$. Positivity therefore leaves no low-spin contribution at that line: it has total hidden spin $5/2$, while the other 77 visible lines have total hidden spin $3/2$.

This conclusion covers every ground vector. Put $G_k = S_-^k G_0$, with $\|G_k\|^2 = k!4!/(4-k)!$, $0 \le k \le 4$. Exact occupation-space CAR give, for every site,

$$\begin{aligned} c_\downarrow^\dagger G_k &= S_-^{k+1} V_\uparrow/5 + S_-^k V_{\text{low}}, \\ c_\uparrow^\dagger G_k &= (5-k) S_-^k V_\uparrow/5 - k S_-^{k-1} V_{\text{low}}. \end{aligned}$$

The second term vanishes at $k = 0$. All eighty creator columns are checked on the original source; the Hamiltonian commutes with every ladder. No coherent ground superposition or linear CAR probe can introduce another addition frequency.

The full $N = 5$ space has spin multiplicities $56, 504, 1008$ at $S = 5/2, 3/2, 1/2$, respectively. Four reachable copies occur at λ, and four at each of the other 77 lines. Thus the complete reachable dimension is

$$4 \cdot 6 + 77 \cdot 4 \cdot 4 = 1256 \quad \text{inside} \quad \binom{16}{5} = 4368.$$

The dark complement has dimensions $312, 784, 2016$ in those three spin sectors. These are dimensions of source subspaces, not full energy degeneracies; their dark eigenvalues remain uncomputed. All annihilator probes similarly generate sixteen $N = 3$, spin-$3/2$ states with positive removal cost λ, leaving 544 dark dimensions.

The invariant mixed-reference energy readout. For $\rho_G = P_G/5$, the exact addition and retarded matrices are

$$\overline{\mathcal{G}}_{\mathrm{add}}(z) = \frac{1}{2}\left[\mathcal{G}_{\downarrow}(z) + \frac{P_+}{z-\lambda}\right] \otimes I_2,$$
$$\overline{\mathcal{G}}_{\mathrm{ret}}(z) = \frac{1}{2}\left[\mathcal{G}_{\downarrow}(z) + \frac{P_+}{z-\lambda} + \frac{P_-}{z+\lambda}\right] \otimes I_2.$$

At each site and fixed spin, the addition line λ has weight $3/10$, and each other line has half its previous weight. The total addition weight is $3/4$; conditional weights are therefore $2/5$ at λ and $2w_i/3$ elsewhere. Removal has weight $1/4$ at the signed retarded pole $-\lambda$. All 78 positive lines survive. The 79 signed poles have rank-eight residues, giving a minimal mixed-state response realization of dimension 632. This realization keeps sixteen observed columns and differs from the physical reachable range that keeps all five ground inputs distinct.

Direct full-source Gram matrices verify every spin and spatial entry of the zeroth and first moments, including

$$\overline{M}_{0,\mathrm{ret}} = I_{16}, \qquad \overline{M}_{1,\mathrm{ret}} = (A \otimes I_2)/16 + I_{16}/4.$$

The actual physical-frame block is $g_{xy} R_x^\dagger R_y$; intersite spin coherence is retained, while local blocks are scalar. The same positive-energy envelope and finite charged-lepton ratio exclusion hold for every ground choice and linear single-CAR probe. These total internal spins include the spin-two background. They do not identify intrinsic particle or Lorentz spin, a measured species, or a relativistic mass. The result does not exclude composite probes or a continuum spectrum, and does not execute native preparation or complete finite-shot spectroscopy of the mixed reference.

Complete finite-resolution energy readout. For a unit one-particle probe f, the same ground defines the probability measure on signed addition and removal energies

$$\mu_f(B) = \langle Vf, 1_B(L) Vf \rangle.$$

The complete Kraus pair $c(f)^\dagger, c(f)$ labels addition and removal; each branch has a nonnegative physical excitation energy. Only the response-frequency variable assigns a minus sign to removal. The exact identities $LV = Vh + UJ$ and $V^*J = J^*J = rI$ give the operator moments

$$M_0 = I, \qquad M_1 = h + UrI, \qquad M_2 = h^2 + 2Urh + U^2 rI.$$

The second-moment identity is a quadratic-form statement and does not require $Vf \in \mathrm{Dom}\, L^2$. For a local mode at $c = 1/8, U = 1$, the mean lies in $[1/8, 1/2]$, the second moment in $[19/128, 67/128]$, and the variance in $[17/128, 35/128]$. Thus $\mu_f(|E| \geq R) \leq 67/(128R^2)$. More generally, $\|hf\| \leq K$ gives a second moment at most $K^2 + UK + U^2/2$, useful for uniformly controlled physical smearing.

Choose $M \geq 2$ and $\tau > 0$. The positive effects

$$K_k(E) = \left| \frac{1}{M} \sum_{j=0}^{M-1} e^{-ij(\tau E - 2\pi k/M)} \right|^2, \qquad q_k = \int K_k(E)\, d\mu_f(E), \quad 0 \leq k < M,$$

sum exactly to one by finite Fourier orthogonality. Writing $\phi_f(t) = \int e^{-itE}\, d\mu_f(E)$, their complete probabilities are

$$q_k = \frac{1}{M} + \frac{2}{M} \sum_{n=1}^{M-1} \left(1 - \frac{n}{M}\right) \mathrm{Re}(e^{2\pi i k n/M} \phi_f(n\tau)).$$

The principal energy centers are modulo $2\pi/\tau$ and are readout bins, not asserted eigenvalues. For $0 < \tau\Delta \le 3$, circular error has probability at most $\min\{1, 10/(M\tau^2\Delta^2)\}$, improved to $22/(7M\tau\Delta - 44)$ when its denominator is positive. This follows by summing the two inverse-square grid tails. A second-moment bound C adds an alias term at most $C/(3/\tau - \Delta)^2$. Consequently the probability of a real energy error at least Δ is controlled, without a pointwise spectral density or an isolated-pole assumption.

The complete frozen 78-line cube measure gives certified distributions for $M = 8, 16, 64$ at $\tau = 1$. At $\Delta = 1/2$, direct positive-kernel and Fourier interval calculations bound the respective error probabilities by 0.100284517428, 0.055193158232, and 0.020209031434. All outcomes are retained. The earlier executed native time-one quadratures check the first harmonic; higher harmonics here are source calculations, not new native long-time or finite-shot measurements.

Along the existing full-ground subsequence, the finite positive spectral measures converge weakly. The uniform fixed-spacing spatial kernel bound also promotes entrywise convergence to strong convergence of the compressed resolvents on the common one-particle space. No identification of the varying GNS Hilbert spaces is assumed. An explicit finite readout error adds the local-dynamics, native-control, and statistical errors to the unspecified finite-ground state discrepancy; the last term is never set to zero. Sharp interval probabilities require zero limiting mass at their endpoints. These results do not supply physical particle identifications, relativistic rest masses, or an interacting continuum.

An all-valley positive-energy continuum field net. There is a spacing-limit construction beyond fixed-lattice response. Take $c(a) = v/a$, fixed $v > 0$, and $0 \le U(a) \le U_{\max} < \infty$. At each spacing retain the full original Weyl/contact dynamics and a translation-invariant full ground. On centered cells $C_{a,n} = a(n + [-1/2, 1/2)^3)$ define

$$(J_a f)(n) = a^{-3/2} \int_{C_{a,n}} f(x)\, dx, \qquad P_a = J_a^* J_a.$$

Cellwise Cauchy–Schwarz and coordinate averaging give $\|J_a\| \le 1$ and $\|P_a f - f\| \le a \sum_j \|\partial_j f\|$ for $f \in H^1$. Thus $P_a \to I$ strongly. For every $b \in \{0,1\}^3$ set $M_b u(n) = (-1)^{b\cdot n} u(n)$ and $C_{a,b}(f) = c(M_b J_a f)$, with the conventional antilinear annihilator. All eight valleys evolve under the original full Hamiltonian. Writing $D_{a,j} f = [f(\cdot + ae_j) - f(\cdot - ae_j)]/(2a)$ gives

$$h_a M_b J_a f = M_b J_a \left[iv \sum_j (-1)^{b_j} \sigma_j D_{a,j} f \right], \qquad \|h_a M_b J_a f\| \le v \sum_j \|\partial_j f\|.$$

For $b \ne d$, shifting one lattice step in a coordinate where they differ reverses the cross-valley pairing. Subtracting the shifted pairing proves

$$|\langle M_b J_a f, M_d J_a g\rangle| \le \frac{a}{2} \left(\|\partial_j f\| \|g\| + \|f\| \|\partial_j g\| \right).$$

The limiting equal-time fields therefore have the full continuum CAR over $L^2(\mathbb{R}^3; \mathbb{C}^2) \otimes \mathbb{C}^8$.

The necessary time estimate holds in the full CAR operator norm. For $B(u) = \sum_{x\sigma} \overline{u_{x\sigma}} n_{x\bar\sigma} c_{x\sigma}$, the onsite anticommutator is positive and obeys $\{B(u_x), B(u_x)^*\} \le \|u_x\|^2 I$. Different-site odd anticommutators vanish, hence $\|B(u)\| \le \|u\|$. The exact contact commutator consequently gives

$$\|\tau_{a,t}(C_{a,b}(f)) - \tau_{a,s}(C_{a,b}(f))\| \le |t - s| \left(v \sum_j \|\partial_j f\| + U_{\max} \|f\| \right).$$

This avoids an extensive interaction-norm estimate.

A diagonal subsequence of spacings makes every word in a countable dense set of smooth smears, rational times and all eight valleys converge simultaneously. The displayed time and norm estimates extend the limits to all L^2 smears and real times. Polynomial positivity, norm bounds and the CAR pass to the limit and construct bounded fields in one GNS representation. Common time shifts are strongly continuous. Nearest *even* lattice translations approximate every physical translation while keeping all valley phases equal to one, so continuous spatial translations also act on the limiting net. For every multi-time polynomial X, the correlations $\omega_a(X_a^*\tau_{a,t}(X_a))$ have nonnegative energy support and converge uniformly on compact time intervals. Testing against smooth negative-energy spectral functions and using dominated convergence therefore proves a nonnegative generator on the entire limiting GNS space, including all polynomial-field vectors.

Graded locality also survives with an explicit bound. Suppose f is supported in $[-r,r]^3$, g outside $[-R,R]^3$, and put $\delta=(R-r)/2-48vT>0$. For $0<a\le\min\{1,(R-r)/8\}$ and $|t|\le T$, the original full-CAR bound, with lattice radii $\lceil r/a\rceil+1$ and $\lfloor R/a\rfloor-1$, yields

$$\begin{aligned}\|\{\tau_{a,t}(C_{a,b}(f)),C_{a,d}(g)^{\#}\}\| &\le 192vT(2r+5)^3(2R+1)^2\|f\|\|g\|a^{-6}e^{-\delta/a}\\ &\le 192\cdot 7!vT(2r+5)^3(2R+1)^2\|f\|\|g\|\frac{a}{\delta^7}.\end{aligned}$$

Here # allows an adjoint or no adjoint. Inserting this norm estimate between arbitrary polynomial vectors makes the limiting anticommutator the zero operator. A finite partition of unity extends the result to separated compact supports. The continuum net thus has exact graded locality at sup-norm distance greater than $96v|t-s|$. This is a conservative propagation cone, not a Lorentz metric.

There is a further uniform response bound on every physical momentum window $|p|\le P$ around any retained valley. These windows reduce the translation-invariant response and have kinetic norm at most vP. The exact Schur formula gives

$$\|\mathcal{G}_a(z)^{-1}\|_P\le|z|+vP+\frac{U_{\max}}{2}+\frac{U_{\max}^2}{4\operatorname{Im}z},\qquad -\operatorname{Im}\mathcal{G}_a(z)\ge\frac{\operatorname{Im}z}{(|z|+vP+U_{\max})^2}I_P.$$

These bounds are uniform in spacing away from real frequency. The cell-average map differs from the exact band isometry by at most $a^2P^2/24$, so the two conventions have the same limiting correlations. Weak convergence alone does not allow interchanging the inverse and limit.

A norm-controlled free limit and the remaining interacting step. A Duhamel comparison with the free lattice evolution uses only $\|B(u)\|\le\|u\|$ and gives an error at most $|U(a)t|\|u\|$. The central-difference remainder then gives, for $f\in H^3$ and $h_b=iv\sum_j(-1)^{b_j}\sigma_j\partial_j$,

$$\|\tau_{a,t}(C_{a,b}(f))-C_{a,b}(e^{ith_b}f)\|\le|t|\left(U(a)\|f\|+\frac{va^2}{6}\sum_j\|\partial_j^3f\|\right).$$

If $U(a)\to0$, the limiting dynamics is exactly the free eight-valley Weyl dynamics. Positive energy forces positive-band annihilators and negative-band creators to annihilate the ground. The one-particle zero-energy kernel is zero in L^2; spectral-band approximation and the CAR recurrence therefore uniquely determine the filled-sea quasifree state and all its moments. This is a controlled free continuum, not the required interacting one.

For bounded positive limiting U, the construction gives the full positive-energy continuum field net, but it has not proved that a connected interaction survives, selected a unique limiting state,

or derived Lorentz covariance. Its propagation cone and positive energy supply no common physical stress, geometric modular flow, renormalized area stationarity or completed G1–G6 Einstein package.

Ground occupation and energy constraints under refinement. The original zero-chemical-potential ground satisfies a useful uniform constraint before any continuum particle identification. With normalized Brillouin measure, let $\rho(k)$ be its one-body covariance and $P_\pm(k)$ the spectral projectors of the original Weyl hopping. The free filled sea has onsite covariance $I/2$ and double occupation $1/4$. Its finite-box restrictions give the variational bound

$$0 \le e_{\mathrm{kin}} - e_{\mathrm{free}} + Ud = e_U - e_{\mathrm{free}} \le U/4, \qquad d = \omega(n_\uparrow n_\downarrow).$$

Here $e_{\mathrm{free}} = -c\int \sqrt{\sum_j \sin^2 k_j}$; no free reference is substituted for the interacting state. Set

$$w(k) = \mathrm{tr}(P_+\rho) + \mathrm{tr}(P_-(I-\rho)), \qquad D = \int w(k).$$

Then $0 \le w \le 2$ and the kinetic excess is $c\int \sqrt{\sum_j \sin^2 k_j}\, w(k)$. The region with $\sqrt{\sum_j \sin^2 k_j} \le \delta$ has normalized volume at most δ^3 for $0 < \delta \le 1$, since $2\arcsin\delta/\pi \le \delta$. Splitting this integral proves

$$D \le \min\{2, 2\delta^3 + U/(4c\delta)\}, \qquad |n-1| \le D.$$

For $U \le 24c$, choosing $\delta = (U/(24c))^{1/4}$ gives $D \le 8(U/(24c))^{3/4}$. Thus the density tends to one per site, and the per-spin density r tends to $1/2$, when $U/c \to 0$. At $v = 1/8$, $U = 1$, $a = 1/65536$, $c = v/a$ and $\delta = 1/16$, the exact bound is $1023/1024 \le n \le 1$.

The same defect controls every fixed microscopic local word. The free-sea annihilation parts $c(P_+f)$ and $c^\dagger(P_-f)$ of a unit site-basis field have interacting GNS norm at most $\sqrt{D}$. Moving the first annihilation part to the right by CAR gives the free Wick recursion, with two boundary errors at most $\sqrt{D}$. Hence for an even word of length m,

$$|\omega(A_1\cdots A_m) - \omega_{\mathrm{free}}(A_1\cdots A_m)| \le C_m\sqrt{D}, \qquad C_0 = 0, \quad C_m = 2 + (m-1)C_{m-2}.$$

In particular $C_4 = 8$ and $d \to 1/4$. This controls fixed microscopic local observables; shrinking momentum pockets can carry a full normalized continuum packet, so no Gaussian continuum conclusion follows.

For cell-averaged physical H^1 packets and $c = v/a$, bounded $U(a) \to U_0$ fixes the first spectral moment to $\langle f, h_{\mathrm{W}} f\rangle + (U_0/2)\|f\|^2$ in every weak spectral limit. The uniform second-moment bound supplies the required first-moment tail control. The scalar term is retained without a chemical-potential shift. Second moments need not pass: the positive measures $(1-n^{-2})\delta_0 + n^{-2}\delta_n$ converge weakly to δ_0 while their variances tend to one. Thus even a nonzero memory Gram need not survive the limiting response. These constraints supply ground and energy inputs, not relativistic rest masses or an Einstein balance. The proof and exact controls are in `model/continuum_ground_constraints`.

A complete native finite-clock energy measurement. The original zero-chemical-potential cube has a finite native preparation of its invariant ground reference $P_G/5$. Five azimuths $2\pi j/5$ and three polar cosines $-\sqrt{3/5}, 0, \sqrt{3/5}$, with polar weights $5/18, 4/9, 5/18$, twirl the spin-two highest-weight ground exactly. The azimuth sum cancels every nonzero magnetic difference through four, and the three-node quadrature integrates the remaining degree-four binomial weights to $1/5$. A

native four-bit selector has weights $[5, 8, 5]$ repeated five times, followed by zero, divided by 90. Its fifteen conditional full-Fock spin rotations and the original inverse frame are executed; all selector and export outcomes are kept. The complete prepared-state error is below 10^{-9}.

At the origin, a native occupation-copy followed by $\gamma_1 = Z_0 X_1 = c_1 + c_1^\dagger$ gives the complete charge instrument with Kraus maps $c_1^\dagger, c_1$. Both branches are retained. Two clock bits control original-action evolutions for model times one and two, with the independently known phase from $E_G = -\sqrt{3}/4$. The positive Fourier transform produces

$$K_k(E) = \left| \frac{1}{4} \sum_{n=0}^{3} e^{-in(E - 2\pi k/4)} \right|^2, \qquad \sum_{k=0}^{3} K_k(E) = 1.$$

This is a complete positive energy measurement. Both physical charge branches have nonnegative excitation cost; principal bin centers are defined modulo 2π, and the removal tag retains the sign needed for the retarded frequency. For the invariant ground, the local addition measure is $\mu_\downarrow/2 + \delta_\lambda/4$ and the removal measure is $\delta_\lambda/4$, where $\lambda = \sqrt{3}/16$. The complete frozen 78-line measure therefore determines every joint charge/bin probability without fitting.

All eighteen-bit clock rows and thirty classical selector/charge columns are propagated through the actual native local instruments. The exact column decomposition used for computation discards no quantum coherence or outcome. Controlled hopping and contact words use at most three local work bits; no abstract controlled Hamiltonian is supplied as a primitive. With sixteen and thirty-two Strang steps, the full-Fock split errors sum to $1887/65536$. Including every preparation, control, scalar-phase and Fourier error gives complete state and total-variation error below $3/100$. A longest successful history contains 612666 native pulses. All 3240 joint selector/tag/clock slots, the selector stops and every first-export failure are retained. The independent original-Hamiltonian exponential is a comparator only.

This construction executes coherent histories through model time three and closes the stated finite native preparation-and-energy-POVM task. Four broad bins do not separately resolve seventy-eight lines. It neither identifies physical species nor converts a finite excitation energy into a relativistic rest mass. It also does not prepare the separate centered-action reference, establish infinite-ground state convergence, or supply the remaining geometric premises.

A CPT constraint on the unchanged charged fields and clock. For a real, even, unit scalar Schwartz packet ϕ, average the signed addition/removal measure over all sixteen continuum field components. Writing this probability measure as ν_ϕ, spin trace cancels the Weyl first moment and its kinetic cross term. The exact finite-spacing sum rules, $r_a \to 1/2$, and weak lower semicontinuity give

$$\int E \, d\nu_\phi(E) = \mu = U_0/2, \qquad \int E^2 \, d\nu_\phi(E) \le C_\phi = v^2 \|\nabla\phi\|^2 + U_0^2/2.$$

Only the second-moment inequality passes to the limit. Negative response frequency denotes removal; the two charge-tagged excitation energies are nonnegative with respect to the original positive GNS generator L.

Suppose a proposed physical identification supplies an antiunitary Θ preserving the same ground and L, and mapping the sixteen smeared annihilators A_s into their adjoints by a unitary matrix Q:

$$\Theta\Omega = \Omega, \qquad \Theta L \Theta^{-1} = L, \qquad \Theta A_s \Theta^{-1} = \sum_t Q_{st} A_t^*.$$

The real even packet is preserved by the spatial inversion and conjugation in the usual CPT action.

For every spectral set B, antiunitarity and unitarity of Q give

$$\sum_s \|P_L(B)A_s^*\Omega\|^2 = \sum_s \|P_L(B)A_s\Omega\|^2.$$

Thus the averaged addition and removal measures agree, ν_ϕ is reflection symmetric, and its first moment is zero. This precise same-field, same-ground, same-clock CPT identification is therefore impossible for $U_0 > 0$. No CPT theorem is applied without its hypotheses: this does not exclude Lorentz covariance alone or other field dictionaries.

With ν_ϕ^- the reflected measure and total variation defined as $\sup_B |\nu_\phi(B) - \nu_\phi^-(B)|$, Cauchy–Schwarz proves

$$\mathrm{TV}(\nu_\phi, \nu_\phi^-) \geq \frac{\mu^2}{C_\phi}, \qquad W_1(\nu_\phi, \nu_\phi^-) \geq 2\mu.$$

There is a bounded odd witness: for $R = C_\phi/(2\mu)$, clipping E to $[-R, R]$ gives f_R with $|E - f_R(E)| \leq E^2/(4R)$ and $\int f_R \, d\nu_\phi \geq \mu/2$. Gaussian packets of width ℓ have $C_\phi = 3v^2/(2\ell^2) + U_0^2/2$. At $v = 1/8$, $U_0 = 1$ and $\ell = 1$, the total-variation lower bound is $32/67$; broad packets make it tend to $1/2$. These are rigorous bounds, not computed infinite-ground spectra or executed native measurements.

A scalar charged-field rephasing changes the generator to $L - \kappa Q_{\mathrm{charge}}$ and the signed first moment to $\mu - \kappa$. Consequently $\kappa = U_0/2$ is the unique necessary centering for this CPT condition. It is a change of charged dynamics, not an additive energy-origin change or a rest mass. Positivity on the old ground is not automatic: the empty two-mode ground of $H = n_0 + 3n_1$ has a centered response of zero mean at $\kappa = 2$, while $H - 2N$ has eigenvalues $0, -1, 1, 0$. The new reference and its ground property require proof.

The free $U(a) \to 0$ control has physical addition and removal energies $v|p|$, zero rest masses, and four Weyl chiralities of each sign. A massive Dirac control, $H_D = Z \otimes (\sigma \cdot p) + mX \otimes I$, has $H_D^2 = (|p|^2 + m^2)I$ and zero trace: its mass couples chiralities and does not create the scalar signed-energy shift. Exact full-CAR counterexamples and the conditional proof are retained in `model/continuum_cpt_constraint`; no physical species, interacting Lorentz completion or Einstein premise is thereby supplied.

A resolved canonical contact-force channel. The bounded-contact refinement also resolves a nonlinear operator in the original action. At each site set

$$b_\sigma = n_{\bar\sigma} c_\sigma, \qquad d_\sigma = (2n_{\bar\sigma} - 1)c_\sigma = 2b_\sigma - c_\sigma.$$

The onsite unitary $Z = I - 2n_\uparrow n_\downarrow$ obeys $Zc_\sigma Z = -d_\sigma$, so d is itself an exact CAR family, with $\|d(u)\| = \|u\|$. It is a composite of the original fields, not an added microscopic mode. The unchanged Hamiltonian satisfies

$$[H, c(u)] = -c((h + U/2)u) - (U/2)d(u).$$

Thus retaining the scalar term does not justify dropping the composite. For $B(u) = \sum_{x,\sigma} \overline{u_{x\sigma}} b_{x\sigma}$, exact local algebra gives

$$\{B(u), c(v)^*\} = \{B(u), B(v)^*\} = Q(u, v) = \sum_x \mathrm{d}\Gamma(q_x), \qquad q_x = (u_x^* v_x)I - v_x u_x^*.$$

Each q_x has rank at most one and $\|q_x\| = \|q_x\|_{\mathrm{HS}} = \|u_x\|\|v_x\|$. The anomalous mixed anticommutator is the pair sum

$$P(u, v) = \{B(u), c(v)\} = \sum_x \overline{u_{x\downarrow} v_{x\uparrow} - u_{x\uparrow} v_{x\downarrow}} c_{x\uparrow} c_{x\downarrow}.$$

Let D_a be the occupation defect above, $L = \|u\|\|v\|$, and $I_4 = \sum_x \|u_x\|^2\|v_x\|^2$. The uniform four-field Wick comparison, $\omega_a(Q) = r_a\langle u, v\rangle$, and the free covariance identity $\mathrm{Var}_{\mathrm{free}}(\mathrm{d}\Gamma(q)) = \mathrm{Tr}[\Pi q^*(I-\Pi)q] \le I_4$ imply

$$\omega_a(Y_a^*Y_a) \le 4\big[I_4 + (8\sqrt{D_a} + D_a/2)L^2\big]$$

for either $Y_a = 2Q - \langle u, v\rangle I$ or $Y_a = 2P$. For $u = M_bJ_af$, $v = M_eJ_ag$ with bounded smooth compactly supported smears, $I_4 \le a^3\|f\|_\infty^2\|g\|^2$ and the coefficient supremum is at most $a^3\|f\|_\infty\|g\|_\infty$. These ground fluctuations therefore vanish.

The passage is an operator statement on the full field hierarchy. A degree-m CAR polynomial obeys $\|[\mathrm{d}\Gamma(q), A]\| \le m\|q\|\|A\|$ for $q = q^*$: its one-body unitary orbit has frequencies in $[-m\|q\|, m\|q\|]$, and the Bernstein interpolation bound applies. The same argument gives $\|[\sum_x n_{x\uparrow}n_{x\downarrow}, A]\| \le m\|A\|$, with degree at most $m+2$ after the commutator. In the free interaction picture the order-N Dyson remainder is consequently bounded by

$$\|A\|(2U_{\max}|s|)^N \frac{(m/2)_N}{N!}.$$

This is uniform in spacing and volume. Divide $|t| \le T$ into $K = \max(1, \lceil 8U_{\max}T\rceil)$ intervals and choose $N = m + \ell$ at each step. The resulting approximation has degree at most

$$M = 3^K(m_{\mathrm{initial}} + \ell - 1) - (\ell - 1)$$

and relative norm error at most $(1 + 2^{-\ell})^K - 1$. The quadratic coefficient bound then proves $\|[Y_a, X_a]\| \to 0$ for every finite-time field polynomial X_a; choosing $2^{-\ell} \le a^3/(2K)$ gives an explicit $O(a^3\log(1/a))$ bound at fixed time. Therefore

$$\|Y_aX_a\Omega_a\| \le \|[Y_a, X_a]\| + \|X_a\|\,\|Y_a\Omega_a\| \longrightarrow 0.$$

The same argument covers initial composite fields, whose degree is three.

Adjoining their equal-time smearings to the common subsequence constructs operators $D_b(f)$ alongside the original $C_b(f)$, with

$$\{D_b(f), D_e(g)^*\} = \delta_{be}\langle f, g\rangle I, \qquad \{D_b(f), C_e(g)^*\} = \{D_b(f), C_e(g)\} = 0.$$

Together with the original CAR and vanishing same-family anomalous anticommutators, these are two copies of the eight-valley CAR algebra. This is algebraic independence; the state is not asserted to factorize or obey Wick's rule. The original correlations and their cyclic positive-energy representation are preserved. Its continuous time action is not automatically extended to the entire enlarged GNS space: the composite derivative contains kinetic terms of order $1/a$. Thus the result resolves the contact-force channel but does not determine whether its bounded-frequency weight survives, escapes to infinite energy, or produces an interacting continuum. It supplies no physical species, Lorentz stress or geometric modular identification. A positive quadratic two-mode control makes this boundary explicit: for $h = \left(\begin{smallmatrix} E_0 & \kappa \\ \kappa & \Lambda \end{smallmatrix}\right)$, with $E_0\Lambda > \kappa^2$ and $\Lambda > E_0 > 0$, the memory Gram stays κ^2 but $\Sigma(z) = \kappa^2/(z-\Lambda) \to 0$. The field error from free energy E_0 is at most $(2\kappa + \kappa^2|t|)/(\Lambda - E_0)$. The full proof and exact controls are in `model/continuum_composite_channel`.

Regional modular data on the full particle ground. The invariant finite reference $\rho_G = P_G/5$ supplies regional modular data on the same ground used for the particle spectrum. In the hidden spin frame, its highest-weight vector is the Slater determinant of four negative spatial orbitals,

with projector $P_- = (I - A/\sqrt{3})/2$. Averaging a common spin rotation yields ρ_G. For a region R, let λ_j be the eigenvalues of the spatial compression $(P_-)_R$. The Slater Schmidt decomposition and the spin average give, for every occupied natural-orbital subset I,

$$w_I = \prod_{j\in I} \lambda_j \prod_{j\notin I} (1-\lambda_j), \qquad \rho_{R,I} = \frac{w_I}{|I|+1} P_{\mathrm{sym},I}. \tag{1}$$

Here $P_{\mathrm{sym},I}$ projects onto the symmetric spin space of dimension $|I|+1$; all other states have zero weight. Indeed phase integration deletes unequal spin counts, and the remaining diagonal integral is $\binom{k}{l}\int_0^1 t^{k-l}(1-t)^l\,dt = 1/(k+1)$. Thus, retaining the complete kernel,

$$S(\rho_R) = \sum_j h_2(\lambda_j) + \sum_I w_I \log(|I|+1). \tag{2}$$

For $0 < \lambda_j < 1$ the modular generator on the positive support is

$$K_R = -\sum_j \log(1-\lambda_j) I + \sum_j \log\frac{1-\lambda_j}{\lambda_j} N_j + \log(N_R+1). \tag{3}$$

Endpoint occupations are handled as fixed blocks, without taking logarithms of zero. This is a support identity, not a full-CAR faithfulness claim.

Every adjacent two-site region has positive spectrum

$$\left\{ \frac{1}{6}\ [1], \quad \frac{2-\sqrt{3}}{12}\ [2], \quad \frac{2+\sqrt{3}}{12}\ [2], \quad \frac{1}{18}\ [3] \right\}, \tag{4}$$

where brackets denote multiplicities. Its eight other eigenvalues are zero. The support consists of the vacuum, the four one-particle states, and the three two-particle triplet states. With H_R the actual internal regional Hamiltonian, whose contact term vanishes on this support,

$$K_R = \log 6\, I + \beta_R H_R + \log(N_R+1), \qquad \beta_R = 16\log(2+\sqrt{3}). \tag{5}$$

Exact signed CAR traces of all five source ground vectors verify this identity on all twelve cube bonds; the sixteen nonadjacent pairs give $K_R = \log 4\, I + \log(N_R+1)$ instead. The same local spin-frame change conjugates the density, support and Hamiltonian back to the original source coordinates.

For every Hermitian trace-zero variation δ within the positive support, including noncommuting variations,

$$DS_{\rho_R}(\delta) = \beta_R \operatorname{Tr}(\delta H_R) + \operatorname{Tr}\big(\delta \log(N_R+1)\big). \tag{6}$$

The second term is central on the supported number-preserving algebra, not on the full CAR algebra. Since $\lambda^+_{\min} = (2-\sqrt{3})/12 > 1/48$, for $\delta = \sigma - \rho_R$ with σ supported there and $0 \le \epsilon < 1$,

$$0 \le D(\rho_R + \epsilon\delta \| \rho_R) \le \frac{24\epsilon^2 \|\delta\|^2_{\mathrm{HS}}}{1-\epsilon}. \tag{7}$$

In contrast, a density $\sigma_\perp$ in the orthogonal kernel obeys the full-state identity $S((1-\epsilon)\rho_R + \epsilon\sigma_\perp) = (1-\epsilon)S(\rho_R) + \epsilon S(\sigma_\perp) + h_2(\epsilon)$, whose derivative diverges at zero. The kernel is therefore retained in the operational domain; no supported first law is applied to such a disturbance. These are same-ground finite modular and entropy inputs. Identifying their central term with a geometric edge, their energy pairing with a diamond charge, or their coefficient with a physical clock requires the existing Einstein theorem's common geometric and scaling premises.

Deriving a centered clock and constructing its new ground. The CPT obstruction motivates a separate, explicitly changed charge clock. On a bipartite Weyl lattice with V sites, the one-particle antiunitary $J = \eta(i\sigma_y)\mathcal{K}$, where $\eta_x = (-1)^{x_1+x_2+x_3}$ and $\mathcal{K}$ is conjugation, anticommutes with the hopping. Its canonical particle-hole transform satisfies

$$\mathcal{P}(N) = 2V - N, \qquad \mathcal{P}(H) = H + U(V - N), \qquad \mathcal{P}(H - \kappa N) - (H - \kappa N) = (U - 2\kappa)(V - N).$$

Thus $K = H - UN/2$ is the unique scalar charge centering invariant on the full Fock algebra. This changes charged dynamics; the original Hamiltonian and its earlier ground are not silently replaced.

The actual cube gives a rigorous changed-reference test. Its retained all-number theorem states $H \geq -\sqrt{3}/4$. The old $N = 4$ ground has centered energy $-2 - \sqrt{3}/4 > -4$, while a singly occupied $N = 8$ product state has expectation -4. Also, for $N \leq 7$, $K \geq -7/2 - \sqrt{3}/4 > -4$, and particle-hole symmetry transfers this bound to $N \geq 9$. Every new global centered ground therefore has exactly eight particles. A product of four exact dimer grounds on a perfect matching improves the variational bound: inter-dimer hopping expectations vanish because each factor has fixed charge. Hence

$$-4 - \frac{\sqrt{3}}{4} \leq E_K \leq -2 - \frac{\sqrt{17}}{2}, \qquad \inf \operatorname{spec}(K|_{N\neq 8}) - E_K \geq \frac{\sqrt{17}}{2} - \frac{\sqrt{3}}{4} - \frac{3}{2} > \frac{1}{8}.$$

These are exact full-sector bounds, not relativistic rest masses.

We additionally import Lieb's repulsive Hubbard ground-spin theorem, in the form of Theorem 1, printed p. 8, of his 1993 review; see also the 1989 result and its erratum. The actual hidden-frame hopping is real, spin independent and connected, with a balanced bipartition of four plus four sites and strictly positive onsite repulsion. These finite hypotheses are checked from the source; the theorem allows mixed real hopping signs. Together with the separately proved $N = 8$ ground sector, it gives a unique global centered cube ground with hidden spin zero. The review's centered interaction convention differs from K only by the constant 2. No quantitative neutral gap, physical spin dictionary or thermodynamic uniqueness follows here.

The actual cube also has a faithful full-CAR one-site state. Write $K = T + D - 4$ on $N = 8$. The source identity $A^2 = 3I$ gives $\|T\| = \sqrt{3}/2$. If P_0 projects onto the 256 singly occupied configurations, every nonzero hopping leaves this subspace, so $P_0 T P_0 = 0$. The source's signed vertex flips are transitive. Set $p = \omega(1 - P_0) \leq \omega(D)$ and $\delta = \sqrt{17}/2 - 2 > 3/50$. The block Cauchy–Schwarz estimate gives

$$|\omega(T)| \leq 2\|T\|\sqrt{p(1-p)} + \|T\|p \leq 3\|T\|\sqrt{\omega(D)}, \qquad \omega(T) + \omega(D) \leq -\delta.$$

Thus the common site doublon expectation $d = \omega(D)/8$ satisfies $d \geq \delta^2/54 > 1/15000$, while $d \leq (\|T\| - \delta)/8 < 1/8$. Gauge, hidden-spin and particle-hole invariance determine the complete one-site state and its modular operator:

$$\rho_x = \operatorname{diag}(d, \tfrac{1}{2} - d, \tfrac{1}{2} - d, d), \qquad K_x = -\log(\tfrac{1}{2} - d)I + \log\left(\frac{\frac{1}{2} - d}{d}\right)(N_x - 1)^2.$$

In particular, $\lambda_{\min}(\rho_x) > 1/15000$ and $\|K_x\| < \log 15000$. For every same-site density σ, put $\Delta = \sigma - \rho_x$ and $\rho_\epsilon = \rho_x + \epsilon\Delta$. The full-site entropy first law and remainder are

$$\left.\frac{dS(\rho_\epsilon)}{d\epsilon}\right|_0 = \operatorname{Tr}(\Delta K_x), \qquad 0 \leq D(\rho_\epsilon\|\rho_x) \leq \frac{7500\epsilon^2}{1-\epsilon}\|\Delta\|_{\mathrm{HS}}^2, \quad 0 \leq \epsilon < 1.$$

This finite result uses no fitted ground vector and covers noncommuting variations. It is not an infinite-ground faithfulness theorem or an identification of the modular operator with a geometric charge.

For a dimer with hopping $t > 0$, let $s = \sqrt{U^2 + 16t^2}$. Its unique full-Fock centered ground is a singlet/doublon superposition with energy $-(U + s)/2$. Its one-site density is

$$\rho_{\text{site}} = \operatorname{diag}(d, \tfrac{1}{2} - d, \tfrac{1}{2} - d, d), \qquad d = (1 - U/s)/4 > 0.$$

Thus this changed reference is faithful on the full local CAR algebra. For a fixed local spin, both addition and removal have energies and weights

$$g_\pm = s/2 \pm t, \qquad w_\pm = (1 \mp 4t/s)/4.$$

Each charge branch has total weight $1/2$; zero weights are retained. At the actual bond parameters $U = 1, t = 1/16$, the positive energies are $(2\sqrt{17} \pm 1)/16$, and the signed second moment is $65/256$.

Normalized full centered-ground projectors, with a tracial exterior, are particle-hole, gauge and hidden-spin invariant. Translation averaging and the existing boundary/volume and local-dynamics estimates give a new infinite-volume ground with positive centered GNS energy and exactly one particle per site in expectation. The old unshifted state is not reused. The exact canonical composite $d = (2n_{\text{opposite}} - 1)c$ gives

$$[K, c(f)] = -c(hf) - (U/2)d(f), \qquad \|d(f)\| = \|f\|.$$

Consequently the same all-word compactness construction supplies a centered eight-valley continuum net. Its moment identities are $M_1 = h$, $M_2 = h^2 + U^2/4$; only the corresponding second-moment upper bound is passed to the continuum. Particle-hole symmetry pairs the complementary valleys and equates averaged addition/removal measures. This resolves the earlier first-moment obstruction, without establishing full spacetime CPT or Lorentz covariance.

There is also a stronger microscopic occupation bound. On a positive kinetic band with $\epsilon \geq \delta > 0$, the GNS maps $Cf = c(P_+ f)\Omega$, $Df = d(P_+ f)\Omega$ obey $LC + C\epsilon = -(U/2)D$. Integrating $e^{-tL} C e^{-t\epsilon}$ and using $L \geq 0$, $\|D\| \leq 1$ gives $\|C\| \leq U/(2\delta)$. Translation invariance, arbitrary measurable band windows and rational thresholds yield $w(k) \leq U^2/(2c^2 \sum_j \sin^2 k_j)$, including the negative-band hole estimate. Folding into the node cells and integrating over a circumscribed ball gives

$$\int \frac{d^3k}{(2\pi)^3 \sum_j \sin^2 k_j} \leq \frac{\pi\sqrt{3}}{2} < \frac{11}{4}, \qquad D \leq \min\{2, 11(U/c)^2/8\}.$$

The same argument on the original uncentered ground uses $\|B\Omega\| \leq \sqrt{r}$ and $r \leq 1/2$, giving the separate bound $D \leq \min\{2, 11(U/c)^2/4\}$. Neither bound is uniform on a normalized packet in a shrinking momentum region. Native preparation of the new ground, surviving continuum interactions, measured particle identities and the common geometric Einstein premises remain further requirements. The exact source checks and proofs are in `model/centered_source_ground`.

The centered bounded-contact limit is a free field theory. The preceding centered construction admits a stronger conclusion. For $c(a) = v/a$, fixed $v > 0$ and $0 \leq U(a) \leq U_{\max} < \infty$, every retained continuum net of the canonical charged fields is the free eight-valley Weyl vacuum net. This determines every order of the field hierarchy. Its particles and holes have $E(p) = v|p|$ and zero rest masses. These modes are not identified with measured species; the bounded-contact branch does not supply the required interacting theory or its massive sectors.

The proof begins with the exact canonical composite $d = (2n_{\text{opposite}} - 1)c$ and the centered occupation bound $\Delta \le 11u^2/8$, where $u = U/c$. At microscopic time s/c, write $F_c(s; f) = \omega(\{\tau_{s/c}(d(f)), d(f)^*\})$, and let F_0 denote the free filled-sea response at normalized hopping speed one. A degree-m CAR polynomial has centered onsite interaction frequencies in $[-m/2, m/2]$. Bernstein's bound and Duhamel give the degree-three dynamical error. For the state error, the cubic/cubic anticommutator has nine single contractions leaving degree-four words; the two cubic/linear terms leave degree-two words. Their local Wick errors are at most $8\sqrt{\Delta}$ and $2\sqrt{\Delta}$. Absolute row and column sums of the free propagator are at most $e^{3|s|}$, and the contracted factor retains the Schur kernel joining the two smearing indices. Consequently, uniformly over all lattice smearings,

$$|F_c(s; f) - F_0(s; f)| \le \left(3u|s| + 312\sqrt{\Delta}\, e^{9|s|}\right)\|f\|^2.$$

No divergent $\|f\|_1^2$ estimate is used.

The free onsite spin covariance is $I_2/2$, so normal ordering gives $d = 2{:}n_{\text{opposite}}c{:}$ with no linear component. It creates three positive-energy excitations, whose total energy is the sum of three values $e(k) = (\sum_j \sin^2 k_j)^{1/2}$. The spin and antisymmetrization bounds give coefficient sixteen per charge branch. Two independent internal momenta must have $e(k) \le \epsilon$; each such set has normalized volume at most ϵ^3. Thus the complete signed free measure obeys

$$\mu_{0,f}([-\epsilon, \epsilon]) \le 32\epsilon^6\|f\|^2.$$

All eight nodes and both positive physical excitation branches are retained. Negative response frequency labels removal, not a negative ground excitation energy.

Gaussian smoothing bounds the actual centered composite weight in any fixed physical energy window:

$$\mu_{c,f}([-R, R]) \le \eta_c(R)^2\|f\|^2,$$

$$\eta_c(R)^2 \le \min\left\{1, e^{R^2/(2c^2\epsilon^2)}\left(1536\epsilon^6 + \frac{3u}{\epsilon} + 624\sqrt{\Delta}\, e^{81/(2\epsilon^2)}\right)\right\}.$$

First send $a \to 0$ at fixed ϵ, then $\epsilon \to 0$. This gives $\eta_c(R) \to 0$ uniformly in unit smearings. The composite retains its equal-time CAR norm while its ground spectral weight escapes every finite energy interval.

The next steps use both charge branches in the positive ground GNS space. For the annihilation map $Af = c(f)\Omega$ and composite map $Df = d(f)\Omega$, the source equation is $LA + Ah = -(U/2)D$, with conjugate input space understood. On $h \ge \delta > 0$, integration against the two contraction semigroups gives, after splitting output energies at R,

$$\|\mathbf{1}_{[0,R]}(L)A\| \le \frac{U_{\max}\eta_c(R)}{2\delta}, \qquad \|\mathbf{1}_{(R,\infty)}(L)A\| \le \frac{U_{\max}}{2(R+\delta)}.$$

The negative-band creator has the same bound. Taking the two limits and using the CAR determines the entire initial-time state as the pure free filled sea. On a general bounded kinetic window $\|h\| \le P$, Duhamel followed by integration by parts above R gives

$$\|[C_a(f, t) - C_a(e^{ith_a}f, 0)]\Omega_a\| \le \frac{U_{\max}}{2}\left(|t|\eta_c(R) + \frac{2 + |t|P}{R}\right)\|f\|.$$

The adjoint difference satisfies the same estimate. Passing to the continuum therefore gives both vacuum identities for the free Weyl generator $h_b = iv\sum_j (-1)^{b_j}\sigma_j\partial_j$.

Vacuum identities alone would not prove a free hierarchy. Here spatial locality supplies the missing implication. The free filled-sea vacuum is cyclic for the initial CAR algebra outside any

bounded spatial region. Indeed, a vector in either Weyl energy band orthogonal to all outside test functions would have compact support. Its Fourier transform would be entire, but the band equation at fixed nonzero transverse momentum contains $\sqrt{z^2+r^2}$. Continuation around one of its simple branch points reverses that square root while the entire vector and polynomial Weyl matrix remain single-valued, forcing the vector to vanish. This proves density of the projected outside test functions in both bands, separately in every valley. Their particle and hole creation vectors span the one-particle space; Wick recursion then spans all finite exterior products.

For compact smooth f, let $A_b(t,f) = C_b(f,t) - C_b(e^{ith_b}f, 0)$. The retained finite propagation cone and free Weyl finite propagation localize both terms inside one bounded enlargement of the support. Graded locality and $A_b(t,f)\Omega = A_b(t,f)^*\Omega = 0$ make both differences vanish on the dense outside-polynomial vacuum vectors. The initial CAR cyclic space therefore reduces every time field, with free action. Full all-time GNS cyclicity makes it the entire Hilbert space. Thus $A_b(t,f) = 0$ as an operator and the complete canonical charged-field hierarchy is free. No Lorentz axiom or external renormalization theorem was assumed in this argument.

This excludes a surviving interaction and positive rest masses on the stated centered bounded-U branch. It does not decide unbounded coupling trajectories, different field scalings, physical species identifications or the common gauge and Einstein requirements. The full proof, exact bound calculators and countercontrols are in `model/continuum_free_limit`.

Other finite interfaces. The full-CAR locality bounds used above control dynamics at fixed lattice spacing; a chiral spatial continuum remains a further limit. For partition expectations E_c and positive rates,

$$L = \sum_c \gamma_c (E_c - I), \qquad \operatorname{Tr}(X^* L X) = -\sum_c \gamma_c \|X - E_c X\|_{\mathrm{HS}}^2$$

identifies $\ker L$ with the intersection of the partition commutants, without assuming commuting pinchings or independent no-cancellation. Applying this theorem requires the stated generator. Equivariant rational chain maps and homotopies on the supplied finite support tower are homology maps, not quantum channels or a dynamical refinement limit.

What the negative controls exclude. Several finite benchmarks constrain possible identifications. A supplied Weyl/contact action has an isolated moving bound-pair band, complete contact residues and an optical identity, but its positive contact section makes the band topologically trivial. Its certified curvature fails the rest/kinetic mass relation with the same speed parameter. Contact interactions also couple proper Weyl momentum pockets to their complements. The lattice scattering theorem, finite thermal positivity and KMS identities do not remove those obstructions or supply physical asymptotic detectors, a chiral continuum or calibrated cross sections.

The centered repulsive dimer gives a sharper family-wide exclusion. For $t = c/2 > 0$, $U \geq 0$, $\mu = U/2$ and $d = \sqrt{U^2 + 16t^2}$, its two charge-one doublets have gaps $g_\pm = d/2 \pm t$, hence

$$1 < g_+/g_- \leq 3.$$

No common positive linear conversion yields the charged-lepton hierarchy or three distinct same-charge families there. Allowing another chemical potential can evade the ratio bound but does not create a third doublet. The fixed complete charged-frequency catalogue also fails its broader lepton-ratio comparison. The periodic four-site extension fails its declared $E(k)^2 = m^2 + c^2 \sin^2 k$ benchmark at $c = 1/8$; a two-parameter fit to two distinct momentum values does not rescue that

test. These exclude stated finite dictionaries, not every larger action or a renormalized continuum limit. Electron/positron masses set by one common measured calibration, and retrospective parameter comparisons, provide no independent mass prediction.

Finite current response also has a precise scope. An open-bond phase is pure gauge and measures no magnetic moment; the contact kernel is not a Coulomb kernel. On the complete periodic ring, a genuine loop phase has gauge-invariant current and total curvature,

$$E_0' = \langle \partial_\phi H \rangle, \qquad E_0'' = \langle \partial_\phi^2 H \rangle - 2 \sum_{n \neq 0} \frac{|\langle n | \partial_\phi H | 0 \rangle|^2}{E_n - E_0},$$

under the isolated-ground hypothesis. Degenerate free and atomic controls lie outside that formula. The calculation retains all 256 Fock states but does not identify a photon or physical Lorentz spin.

Clock and memory exclusions are similarly conditional. A frozen real-operation grammar has a normalized trace-distance lower bound $1/2$ for a specified nontrivial modular action of its nontracial reference. Finite independent copies of the original native reference also fail the required field modular-gap ratios. These do not exclude the explicitly enlarged clock and control laws above. The earlier pair/dark-reference encoding covers a fixed corner, not the whole spatial field. Finite stationary observer references concern source/buffer marginals at the stated repair rate and on their recurrent support; delayed variants need their own construction, and growing exported histories are not made stationary by these marginal references. Actively corrected coherent memories can have flat logical energy and a fixed core, so memory lifetime alone does not establish a moving massive particle. Properties proved on distinct carriers require an explicit intertwiner before they can be combined.

Physical target and remaining obligations. The target is an interacting quantum field theory with gapless photons and experimentally matched massive sectors, not a positive global mass gap. A gapless photon rules out such a global gap. An exact nontrivial positive-energy conformal theory in a flat-space vacuum representation with unbroken unitary dilations also cannot have a positive flat-space vacuum mass gap: for a spectral value $E > 0$ and proposed gap $g > 0$, scale covariance also puts $g/2$ in the spectrum. This does not exclude a gap in scaling dimensions or discrete finite-radius cylinder levels; the cylinder radius supplies an energy scale. Conformal behavior may describe a critical or short-distance regime, but a deformation or symmetry-breaking mechanism responsible for massive flat-space sectors must then be specified and implemented.

The finite results leave concrete physical joins. A localized defect must carry the protected charge and the same measured field observables. Spatial refinement must control the additional lattice modes and yield the required Lorentz propagation, while keeping the interactions and reference state. The electromagnetic field and gauge–Higgs interactions must be dynamical, with charge constraints, detector coupling and their infrared behavior treated on that same theory. Massive excitations need a common energy and clock normalization, including the relation $E''(0) = c^2/E_0$ for a relativistic branch with $E_0 > 0$, together with the appropriate stress and proper-time relations. The neutrino sector and other measured particle properties require their own matching tests. Neither a successful finite source compilation nor a calibrated spectrum alone discharges these obligations.

This paper's hypothesis and place in the tower

This paper is the gauge half of layer 5. Its hypothesis is that gauge structure is the transport of overlap data between observer patches: on the twelve-port icosahedral carrier, a complete faithful

commutator-closed port response on which every proper carrier recharting acts endogenously, by an inner automorphism of the same response, is the local gauge current, and its compact Lie type is $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$.

What is proved. The Lie-type theorem (Theorem 3.1) is a finite implication from the response clauses of premise A1, the endogenous-transport clause of premise A2 and the classification of compact simple Lie algebras. The categorical route proves a fixed-cutoff bosonic sector category unconditionally and a compact Tannaka group on any refinement tail that carries the stated receipt; a Doplicher–Roberts field net requires further declared hypotheses. The matrix current, spin lift, trace redistribution, exterior matter module, band selection, bracket discriminator and local action are conditional realizations under displayed declarations.

Open joins. The executable model must realize the complete response of A1 and the endogenous transport of A2 from its own repair and memory law, with the port generators, commutators and implementers reconstructed from ordered source histories. The tested native exchange response gives seventeen currents and fails A1; a specified coherent control law gives the twelve closed currents on a declared control menu (Section 4). Physical spin and statistics, a physical Weyl field, the global quotient, the family attachment and all couplings are further joins. The matter hypothesis, that stable massive matter is a localized topological defect of native overlap transport whose boundary holonomy takes values in this gauge structure, is developed in the particle paper [21].

Refutation. The complete accessible current of a specified executable law, if it is not a faithful closed twelve-dimensional response with inner carrier action, excludes that law's realization of the branch. A proof that no admissible local repair and memory law on the carrier supplies the A1 and A2 response clauses would remove this gauge branch. The finite failure conditions are listed in Section 12.

Contents

1 Two reconstruction routes **56**

2 Premises **56**

3 The finite-current Lie-type theorem **59**

4 The current on the executable carrier **60**
4.1 Carrier rechartings, the shared triplet and physical rotations 62

5 The categorical route **62**
5.1 Transport from overlap gluing 62
5.2 The fixed-cutoff sector category 63
5.3 Refinement and compact-group reconstruction 64

6 Conditional matter realization **66**
6.1 The matrix current, spin lift and exterior module 66
6.2 Global form 68

7 Families and the generation window **69**

8 The equivariant bracket variety and a face discriminator **70**
8.1 An assembled structural action 71

9 Refinement stability and relevant transverse perturbations **72**

10 Edge-sector weights and coupling extraction **72**

11 Classical carrier modes and the quantum-particle criterion **74**

12 Failure conditions **75**

A Local action, Cartan reduction and fermion current **75**
A.1 A local gauge-covariant action on the declared matter grammar 75
A.2 A consistent local Cartan–scalar reduction and current response 77
A.3 A continuous charged-current and separated-detector bound 79
A.4 Fermionic hypercharge transport on prepared source addresses 80
A.5 A flux-dressed fermion state satisfying abelian quantum Gauss 83

B A coupled fermion current in a declared hybrid gauge action **84**
B.1 Spin transport and the information needed to select exchange statistics 87
B.2 Conditional exclusion from a stable generator and a scalar deformation law 91
B.3 What finite source operations must supply for exclusion 93

1 Two reconstruction routes

The first route is structural:

$$\text{overlap gluing and transport} \Longrightarrow \text{rigid symmetric sector category} \Longrightarrow \mathrm{Aut}_{\otimes}(\mathcal{F}).$$

It reconstructs a compact group on the declared bosonic branch (Section 5). It does not select the Standard Model without a realized sector witness.

The second route is finite and carrier-specific:

$$\begin{aligned}&\text{twelve-port incidence and complete reversible response}\\ &\Longrightarrow \text{internal proper-carrier holonomy} \Longrightarrow \mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)\\ &\Longrightarrow \text{conditional Standard Model matter image.}\end{aligned}$$

Incidence supplies the twelve-dimensional A_5-module and its single fixed line. Premise A1 supplies the faithful compact commutator-closed response on that module. Premise A2 makes the proper carrier action inner on the same response. Incidence without the response clauses leaves many equivariant laws. Response naturality without endogeneity leaves the abelian $\mathfrak{u}(1)^{12}$ and the mixed $\mathfrak{su}(2)^2\oplus\mathfrak{u}(1)^6$ branches available; endogeneity bounds the centre by the single fixed line, and the fixed-line count then excludes $\mathfrak{su}(2)^4$. The two routes have separate premise sets. No theorem identifies the finite current with the Tannaka group; that equality requires a same-source current construction and a commuting-square identification.

2 Premises

The premises A1–A3 are shared with the other layers. The Lie-type theorem uses only the response clauses of A1 and the endogenous-transport clause of A2. A3 enters only the edge-sector weighting of Section 10. Branch-local hypotheses are stated at the results that consume them.

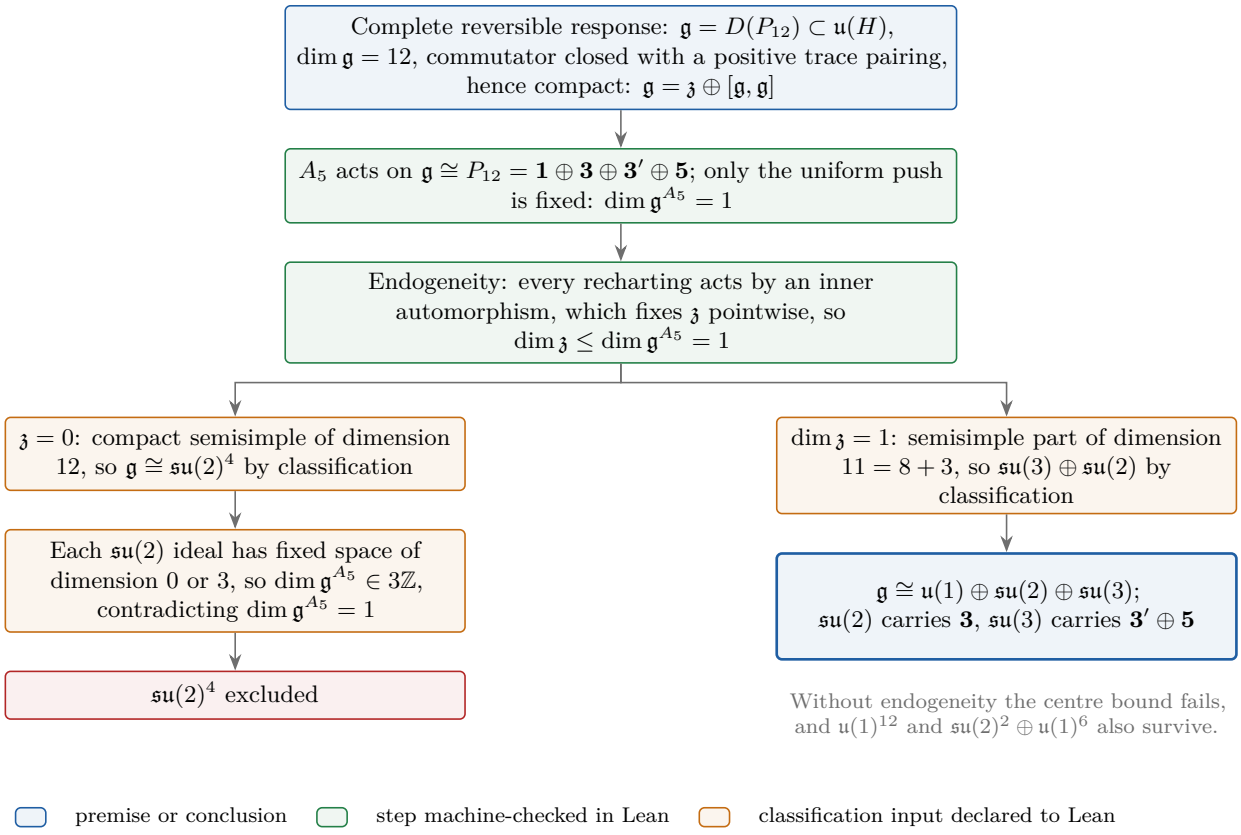


Figure 1: Steps of the gauge Lie-type theorem. Green steps are machine-checked in the Lean library; orange steps use the classification of compact simple Lie algebras and the fixed-space argument against $\mathfrak{su}(2)^4$, which enter Lean as declared hypotheses. The port count fixes the dimension twelve by declaration; the content of the theorem is the exclusion of $\mathfrak{su}(2)^4$ and a one-dimensional centre.

A1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space

$H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. A1 does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

A2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. A2 constrains accepted shared data only, and the domain it quantifies over is supplied by the A1 interfaces rather than by A2 itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, its closed paths $\mathrm{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\mathrm{Aut}^+(K_{r,i})$. For every proper carrier automorphism a, A2 supplies a closed path γ_a and one projective implementer $[U_a] \in \mathrm{PU}(H_{r,i})$ satisfying

$$\mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. A2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

A3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by A1 and A2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an

A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. A3 selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal statement of A1–A3, including its constraint map and non-implications, is given in the premise reference.

3 The finite-current Lie-type theorem

Theorem 3.1 (Finite-current fixed-space theorem)**.** *Let $\mathfrak{g} = D(\mathbb{R}^{12}) \subset \mathfrak{u}(H)$ be the image of the complete reversible twelve-port response of premise A1, and let the proper carrier rotations act on it by inner automorphisms of the same response, as premise A2 requires. Then*

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

The implication assumes no ambient continuous gauge group and is independent of the categorical refinement receipt.

Proof. Faithfulness gives $\dim \mathfrak{g} = 12$. Commutator closure inside $\mathfrak{u}(H)$ makes $\mathfrak{g}$ compact reductive, so $\mathfrak{g} = Z(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$. The transitive action of the alternating group A_5 on the twelve ports gives $\dim \mathfrak{g}^{A_5} = 1$. Inner automorphisms fix the centre pointwise, so $Z(\mathfrak{g}) \subset \mathfrak{g}^{A_5}$ and $\dim Z(\mathfrak{g}) \leq 1$. If the centre vanished, $\mathfrak{g}$ would be semisimple of dimension twelve; the compact simple algebras of dimension at most twelve have dimensions 3, 8 and 10 [77], so $\mathfrak{g} \cong \mathfrak{su}(2)^4$. The inner action preserves each simple factor of $\mathfrak{su}(2)^4$; simplicity of A_5 makes each restricted action trivial or faithful, with fixed dimension three or zero. No sum of such terms equals one, so this branch is excluded. Hence $\dim Z(\mathfrak{g}) = 1$, the semisimple part has dimension eleven, and compact-simple classification gives the unique split $11 = 3 + 8$. □

The port count fixes the dimension twelve by declaration; the content of the theorem is the exclusion of $\mathfrak{su}(2)^4$ and a one-dimensional centre. The dimension arithmetic, the centre bound and the dimension-six exclusion below are machine checked in the accompanying Lean development [29]. The $\mathfrak{su}(2)^4$ fixed-space step and the compact-Lie classification enter Lean as declared hypotheses, and no source current or full Lie-algebra isomorphism is constructed there. The formal interface keeps the inputs separate: an injective port-response map supplies dimension twelve; an inner carrier action places the centre inside the one-dimensional fixed space; the fixed-space exclusion supplies a nonzero centre; and compact-simple classification supplies the allowed simple dimensions $3, 8, 10$. The final Lean selection theorem takes the last three constraints as hypotheses and proves

the split. An observable relative phase alone establishes neither centrality nor trivial carrier action on the whole centre. The trivial carrier action on the centre is a sufficient premise for this Lie type; the proof supplies no independent physical origin for it.

What incidence alone supplies. The certified incidence and inverse-port pairing determine J, with

$$10J = A^3 - 4A^2 - 5A + 10I,$$

where A is the port adjacency. The equivariant linear-response commutant is four-dimensional, and a target-blind impulse and readback protocol produces $R = -J$. Its restrictions to $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$ have fixed relative signs; simultaneous reversal is an algebraic response convention and is not identified with matter charge conjugation. These facts constrain the response without constructing the current derivative.

Exact negative controls. The registered single-recurrence interface generates the commutative algebra $\mathrm{span}\{I, A, A^2, A^3\}$. It contains neither twelve independent current generators nor a nonzero bracket, and its only proper recharting is the identity; this bound applies to that registered packet. The target-free diagonal phase lift has twelve independent first derivatives and an abelian bracket. Adding the connected adjacency tangent generates $\mathfrak{u}(12)$, with derived rank 143. The canonical scalar seam equalizers do not supply the current either. On the eleven-dimensional zero-sum port space,

$$\mathrm{rank}\langle [G_e, G_f] \rangle = 49, \qquad \mathrm{Lie}\langle [G_e, G_f] \rangle = \mathfrak{so}(11), \qquad \mathrm{Alg}\langle G_e \rangle = \mathrm{End}(\mathbb{R}^{11}),$$

of dimensions 49, 55 and 121. The strict-descent repair maps therefore cannot be relabelled as the twelve-dimensional compact reversible current, and the repair law acquires no gauge content by definition.

Carrier specificity. Identify the twelve ports in antipodal pairs. The quotient triangulates the real projective plane with six vertices, fifteen edges and ten triangular faces; its edge graph K_6 has twenty 3-cycles. Its transitive A_5 action retains one fixed line. If a six-dimensional compact current carries that action internally, a one-dimensional centre leaves the impossible semisimple dimension five, whereas a zero-dimensional centre gives $\mathfrak{su}(2)^2$ with fixed dimension zero, three or six. No current satisfies all those premises. The adjacency spectrum $5, -1$, with multiplicities one and five, is the antipode-even part of the icosahedral spectrum; the two $\pm\sqrt{5}$ triplets are antipode-odd. Antipodal identification changes topology and port count; forgetting a chosen orientation alone leaves the twelve-port sphere.

4 The current on the executable carrier

The Lie-type theorem consumes a complete response; it does not produce one. On the executable carrier the twelve-current law is a specification; no result derives it from the unchanged native repair. Fixed unrecorded native seam channels annihilate nonzero operators and cannot supply invertible currents. Active correction has a different domain, examined here on one protected native object [26, 29].

A persistent geometric object. The thirty-six nonedge pairs of the native binary load-two carrier admit two source-derived equivariant perfect-matching colourings by the six antipodal axes.

Each normalized matching code has dimension six, carries the actual port-permutation representation $1 \oplus 5$, and corrects every complete recorded C/E branch on all thirty native seams with the declared coherent encoder and record-dependent recovery. Its sum-zero quintet is protected. Two such corrected objects contain a unique geometric $3 \oplus 3'$ sector under simultaneous port rotations, with an exact intertwiner inside the full thirty-six-dimensional product code. This is a persistent geometric carrier; it supplies no spin-half lift, exterior matter module or autonomous selection of that sector.

The native exchange response fails the complete-response clause. On opposite colouring choices, the twelve corresponding-port exchanges on all 10626 native load-four configurations yield six independent Hermitian mixed responses that preserve the geometric sector. They use the declared joint-code phase control and are not copied from any target current. Their exact real Lie closure is

$$\mathfrak{g}_{\text{native}} = \mathfrak{su}(3) \oplus \mathfrak{su}(3) \oplus \mathbb{R}\, iI_6, \qquad \dim \mathfrak{g}_{\text{native}} = 17.$$

The proper port rotations belong to this connected response group at the effective-algebra level, so the response realizes an endogeneity bridge on one object. It does not instantiate A1. Conjugation on $\mathrm{End}(\mathbb{C}^6)$ sees sixteen directions, since the common scalar phase is silent there. On the full product code the same preserving menu has nine independent Hermitian operators with a three-dimensional restriction kernel, so its Lie closure has dimension at least twenty and its conjugation-visible response on $\mathrm{End}(\mathbb{C}^{36})$ has dimension at least nineteen. The response lacks the relative phase between 3 and $3'$ and allows five traceless phase directions on each triplet. The exchange law and the specified law below intersect in eleven dimensions, and exact quotients and rotation-fixing channels cannot delete the unwanted quintet while supplying the missing centre. This seventeen-versus-twelve mismatch excludes the exchange grammar; it does not exclude other admissible current laws.

A specified twelve-current law. A distinct specified coherent law realizes the current interface on the same protected object without changing its ports or C/E instruments. For the native port vectors v_p, Galois conjugation σ and cross-product matrix $\widehat{v}$, set

$$D_p = \mathrm{diag}\left(\widehat{v_p/2} + iQ_p,\ \widehat{\sigma(v_p)/2}\right), \qquad Q_p = \tfrac{1}{2} v_p v_p^T - \tfrac{1}{6}\|v_p\|^2 I_3 + \tfrac{1}{12} I_3.$$

The native near-fivefold class sum distinguishes the two triplets before the complex-versus-real control choice is applied. Exact checks give rank twelve, all sixty-six bracket closures and all seven hundred twenty port covariance identities. Proper carrier rotations belong to the same connected current group. The sum of the currents is $\mathrm{diag}(iI_3, 0)$, whose relative phase is observable by cross-triplet readout. The algebra is $\mathfrak{u}(3) \oplus \mathfrak{so}(3) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Source words execute its primitive currents, composed with the geometric preparation, native repair, record-assisted recovery and pointer outputs. Completeness holds over the specified current-control menu. The coherent compiler and detector services are declared apparatus, and excluding them from the menu does not establish their physical inaccessibility, full physical A1 closure or gauge redundancy.

Controlled limits. A positive native class-sum bias, fresh-bath coupling and a complete eighteen-coefficient disturbance-gradient controller give a source-executed bounded-drive joint limit that approaches twelve observable currents. Every finite filter is invertible on all eighteen input currents; fixed-bias long feedback instead leaves six observable currents. Reference cooling alone leaves every work current available. The bias, baths, actuator and feedback rule are architecture choices that unchanged C/E repair does not select.

4.1 Carrier rechartings, the shared triplet and physical rotations

One three-dimensional A_5 representation on the port vectors appears in three roles. The vectors v_p span the adjacency eigenspace with eigenvalue $\sqrt{5}$, the Laplacian band $L = 5I - A$ of cost $5 - \sqrt{5}$. This band is the rank-three Gram frame and slow repair band that supplies the spatial frame triplet [20, 17]; it is the band selected as the family candidate in Section 7; and the colour block of the specified current acts on it, each generator containing the frame rotation $\widehat{v_p/2}$. Its Galois conjugate $\sigma(v_p)$ spans the $-\sqrt{5}$ band $3'$ on which the weak $\mathfrak{so}(3)$ block acts.

The paper keeps these roles distinct. Premise A2 places the finite carrier rechartings inside the connected internal current group, up to centralizer elements and phases; this is a finite covariance statement and it is a premise. No result here identifies those rechartings with spatial rotations of a recovered interacting field, and representation equivalence alone proves neither an identification nor an independence of physical colour, family and spatial actions. Equivalent triplets can occur on different tensor factors: the specified family-feedback construction uses a gauge-trivial ancillary triplet rather than the coloured current triplet. The local rank-three Gram band and the refining global low band are also distinct: the former has a fixed nonzero carrier energy, whereas the latter concentrates on cell averages.

The conditional spin and matter join constructs commuting actions on $\mathbb{C}^2 \otimes \mathbb{C}^{15}$, with physical-spin central element $-I$ on all thirty states. The diagonal locked action instead has fourteen minus and sixteen plus central eigenvalues and fails that fermionic test. An explicitly chosen compensating internal transport realizes the independent spin action. These are finite existence and obstruction results under a chosen tensor attachment, not a source-selected spacetime connection or a joint charged propagator.

Identifying carrier rechartings with physical rotations while A2 keeps them internal would combine spacetime and internal symmetry. In a relativistic interacting scattering theory that combination is constrained by the Coleman–Mandula theorem under its Poincaré, spectrum, analyticity and nontrivial-scattering hypotheses.[10] The paper makes no such identification, and those hypotheses are not established for the finite interface. The required test identifies rotations operationally on the same clocked field response, establishes their independence from internal currents up to the declared gauge equivalence, including any allowed compensators, and checks that the dynamics preserves that identification.

5 The categorical route

5.1 Transport from overlap gluing

At a fixed regulator the overlap rechartings of a connected cut Σ form a compact unitary transition system. We use chronological multiplication of comparison maps: U_{ij} maps components in chart j to components in chart i, so an overlap path $p = (i_0 \to \cdots \to i_m)$ has $U_p = U_{i_0 i_1} \cdots U_{i_{m-1} i_m}$. Thus U_p compares the terminal chart to the initial chart; forward transport of column components is $U_p^\dagger$. If $p * q$ means first following p, then q, the conventions are $U_{p*q} = U_p U_q$ and $U_{\overline{p}} = U_p^{-1}$ for the reversed path $\overline{p}$. Gauge-as-gluing, the tree and loop lemmas and the transportability theorem are proved in the spacetime paper [17]. The gauge route consumes the following exact criterion.

On the central-defect branch, identify the overlap centres with one fixed abelian unitary coefficient group Z_Σ on which overlap transport acts trivially. With $U_{ji} = U_{ij}^{-1}$, define the implementer-

[10] S. Coleman and J. Mandula, "All Possible Symmetries of the S Matrix," *Physical Review* **159**, 1251–1256 (1967), https://doi.org/10.1103/PhysRev.159.1251.

level multiplier

$$\Omega_{ijk} = U_{ij}U_{jk}U_{ki} = z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. The multiplier is defined at the implementer level: $\mathrm{Ad}(z_{ijk})$ is the identity and does not record the defect.

Theorem 5.1 (Central defect strictification and residual holonomy)**.** *The family $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in Z_Σ, and its class $[z] \in \check{H}^2(N_\Sigma, Z_\Sigma)$ is gauge invariant. A central edge rephasing by a cochain in $C^1(N_\Sigma, Z_\Sigma)$ removes every triangle defect if and only if $[z] = 0$. After such a rephasing, let U^{str} be the resulting edge 1-cocycle and $\mathcal{C}_\alpha$ the full transported charge block, including carrier and multiplicity data. Loop-coherent endpoint-only transport exists on that block if and only if, in addition,*

$$\mathrm{Hol}_{\alpha,U^{\mathrm{str}}}([\gamma]) = U^{\mathrm{str}}_\gamma|_{\mathcal{C}_\alpha} = \mathbf{1}_{\mathcal{C}_\alpha} \qquad \text{for every } [\gamma] \in \pi_1(|N_\Sigma|, i_*).$$

Proof. Conjugation by Ω_{ijk} fixes every triple-overlap observable, so the multiplier is central there; the fixed-coefficient and trivial-action hypotheses place it in Z_Σ. Comparing $(U_{ij}U_{jk})U_{kl}$ with $U_{ij}(U_{jk}U_{kl})$ on a quadruple overlap gives $z_{jkl}z_{ikl}^{-1}z_{ijl}z_{ijk}^{-1} = 1$. A vertex-frame change $U_{ij} \mapsto V_iU_{ij}V_j^\dagger$ conjugates z_{ijk} and leaves a central multiplier unchanged. A central edge rephasing $U_{ij} \mapsto a_{ij}^{-1}U_{ij}$ replaces z by the corresponding coboundary, so $[z] = 0$ is equivalent to removing all triangle multipliers. For two paths p, p' with common endpoints, the residual comparison $U_pU_{p'}^{-1}$ is the holonomy of the closed loop $p * \overline{p'}$, based at their common initial chart. It is the identity for every pair exactly when the displayed holonomy representation is trivial. Conversely, strict path independence makes every triangle comparison and every represented loop action trivial. □

If overlap transport acts nontrivially on a local system of centres, the comparison carries the transported factor and requires the twisted Čech differential. On the genuinely noncentral branch the coefficient data is a crossed module $H_\Sigma \to G_\Sigma$. The vanishing $o_\Sigma^{(2)} = 0$ strictifies the higher associator, after which at least one allowed strict G_Σ-valued representative must have trivial represented holonomy. The full orbit q_Σ need not determine a unique ordinary H^1 class and is not itself the strictification test. A nonzero $o_\Sigma^{(2)}$ remains a higher-gauge sector, and an orbit whose every strict representative has residual loop action lies outside the ordinary path-independent sector. The class $[z]$ is structurally analogous to an effective 't Hooft anomaly class; a map between them requires a separate anomaly-descent construction, so $[z] = 0$ is used here only as the internal strictifiability condition. The overlap obstruction calculus classifies and routes sectors; it does not select the Standard Model.

5.2 The fixed-cutoff sector category

At cutoff r, choose one common stagewise strict edge cocycle U_r^{str} from the allowed representatives, and retain only seeds on which this representative acts trivially around every closed overlap loop. Sectors requiring incompatible strictifications define alternative fixed-stage categories; they are not unioned before tensor closure.

Definition 5.2 (Fixed-cutoff tensor-generated sector category)**.** *Let S_r be the set of visible collar carriers $W_{\alpha,r}$ with trivial holonomy under the common U_r^{str}. Define*

$$\mathsf{Sect}_r^{\mathrm{bos}} = \langle \mathbf{1}, S_r, S_r^* \rangle_{\oplus,\otimes,\text{subobjects,isomorphism}} \subset \mathrm{Rep}_{\mathrm{fd}}(\widehat{K}_{\Sigma,r}),$$

the replete full additive Karoubi rigid tensor subcategory generated by the finitely many visible carriers. Its simple objects are the irreducible summands of finite tensor words in seeds and conjugates; there may be infinitely many of them although every object and Hom space is finite-dimensional.

Theorem 5.3 (Fixed-cutoff bosonic sector category)**.** *At every fixed regulator cutoff, on the ordinary or central-defect zero-obstruction branch and in the bosonic internal-gauge sector of the* $3+1$*-dimensional effective regime,* $\mathsf{Sect}_r^{\mathrm{bos}}$ *is a semisimple rigid symmetric* C^**-tensor category with one monoidally coherent path-independent transport. The carrier-forgetful functor* $F_r : \mathsf{Sect}_r^{\mathrm{bos}} \to \mathsf{Hilb}_{\mathrm{fd}}$ *is faithful,* **-preserving and symmetric strong monoidal.*

Proof. Complete reducibility of finite-dimensional representations of the compact group gives semisimplicity and subobjects; the operator adjoint and norm give the C^*-structure; conjugate representations give duals; the flip gives the bosonic symmetry. Around a closed loop U_r^{str} acts as the identity on every retained seed, hence on direct sums, tensor products, conjugates and invariant subobjects. Collar concatenation realizes tensor words whenever the finite tensor-realization receipt is supplied. □

Fermionic signs, spinorial matter and chirality are not part of this bosonic category; they belong to the matter lift of Section 6.

5.3 Refinement and compact-group reconstruction

Definition 5.4 (Compact-gauge refinement receipt)**.** *On a cofinal tail, the receipt supplies: a surjectivity certificate for each finite-state restriction* $\rho_{sr} : Q_s \to Q_r$*; multiplicities* $m_{\beta\alpha}^{rs}$ *and unitaries* V_β^{rs} *with* $d_{\beta,s} = \sum_\alpha m_{\beta\alpha}^{rs} d_{\alpha,r}$*, defining the injective block embedding*

$$\big(\jmath_{rs}(a)\big)_\beta = V_\beta^{rs}\left[\bigoplus_\alpha (a_\alpha \otimes \mathbf{1}_{m_{\beta\alpha}^{rs}})\right](V_\beta^{rs})^*$$

of the collar algebras, coherent under composition; continuous surjections $\pi_{sr}^\Sigma : \widehat{K}_{\Sigma,s} \twoheadrightarrow \widehat{K}_{\Sigma,r}$ *with* $\pi_{tr}^\Sigma = \pi_{sr}^\Sigma \circ \pi_{ts}^\Sigma$*; one common strict representative at each stage, intertwined by the pullbacks; and compatible finite concatenated-collar realizations of every tensor word and invariant subobject projector. The finite-state consensus maps alone imply none of these data.*

Theorem 5.5 (Refinement functor and fibre descent)**.** *On a tail carrying the receipt,* $U_{rs} := (\pi_{sr}^\Sigma)^* : \mathsf{Sect}_r^{\mathrm{bos}} \to \mathsf{Sect}_s^{\mathrm{bos}}$ *is fully faithful, symmetric strong monoidal and* **-preserving; it preserves the unit, duals, irreducibles and the full zero-obstruction criterion, with* $U_{st}U_{rs} = U_{rt}$ *and* $F_sU_{rs} = F_r$ *on carrier spaces.*

Proof. Set $U_{rs}(H_X, \varrho_X) = (H_X, \varrho_X \circ \pi_{sr}^\Sigma)$ and $U_{rs}(f) = f$. Surjectivity gives

$$\mathrm{Hom}_{\widehat{K}_{\Sigma,s}}(U_{rs}X, U_{rs}Y) = \mathrm{Hom}_{\widehat{K}_{\Sigma,r}}(X, Y),$$

so pullback is fully faithful and a coarse simple cannot split. Pullback commutes with tensor products, unit, conjugates, duality maps and the flip. The receipt intertwines the stagewise strict representatives, so zero holonomy persists for the whole category. Pullback changes only the action, so the fibre comparison is the identity. □

On such a tail, the directed colimit $\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}_r^{\mathrm{bos}}$ retains the sectors and intertwiners that persist, and the stagewise carriers descend to a faithful bosonic fibre functor $\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$. A realized zero-obstruction class with representatives satisfying $U_{st}\alpha_s \simeq \alpha_t$ determines a unique object of the colimit. Without the receipt only the fixed-stage categories are constructed.

Theorem 5.6 (Receipt-conditional Tannaka reconstruction)**.** *On the ordinary or central zero-obstruction bosonic branch with a cofinal tail carrying the receipt,* Sect_∞ *is a rigid symmetric* C^**-tensor category with faithful bosonic fibre functor* $\mathcal{F}$*,* $G := \mathrm{Aut}_\otimes(\mathcal{F})$ *is a compact group, and* $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$*, with* G *unique up to isomorphism. If the category is trivial,* G *is trivial.*

Proof. The refinement functors preserve tensor products, unit, duals and *-structure, so the colimit inherits a rigid symmetric C^*-tensor structure and the carriers descend to $\mathcal{F}$. A unitary symmetric monoidal natural automorphism η embeds as $(\eta_X)_X$ in $\prod_X U(\mathcal{F}(X))$. Naturality, $\mathcal{F}(f)\eta_X = \eta_Y\mathcal{F}(f)$, and monoidality, $\eta_{X\otimes Y} = J_{X,Y}(\eta_X \otimes \eta_Y)J_{X,Y}^{-1}$ with $\eta_{\mathbf{1}} = \mathrm{id}$, are closed relations, so G is a closed subgroup of a compact product. Doplicher–Roberts and Tannaka reconstruction give $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$ [66, 67], and two groups reconstructing the same category with compatible fibre functors are both the automorphism group of that functor. □

The local-Gibbs and collar-mixing package is not used to prove nontriviality or to supply the receipt. The theorem is neutral on whether Sect_∞ is trivial: it returns whatever compact group the tensor-fibre data reconstructs and does not select $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.

Definition 5.7 (Field-net realization hypotheses)**.** *(R1) An isotonous net* $O \mapsto \mathcal{A}(O)$ *of* C^**-algebras over the small-region collar localizations, with a faithful vacuum representation. (R2) Spacelike commutativity, Haag duality or its declared essential variant, and Property B. (R3) Each persistent class* $[\alpha]$ *is realized by a localized endomorphism* ρ^α_O*. (R4) Transportability by unitary intertwiners in* $\mathcal{A}$*; vanishing collar holonomy is a necessary input to (R4) and is not equivalent to it. (R5) Finite permutation-symmetric statistics and conjugates matching* Sect_∞*. (R6) The assignment* $[\alpha] \mapsto [\rho^\alpha]$ *is an equivalence onto the localized transportable finite-statistics endomorphisms.*

Corollary 5.8 (Two-level reconstruction)**.** *Level (i), proved: on the receipt-certified branch,* $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$ *with no observable net involved. Level (ii), conditional: under (R1)–(R6) the Doplicher–Roberts theorem [66, 67] supplies a bosonic field net* $\mathcal{F}_{\mathrm{net}}$ *carrying a* G*-action whose field operators implement every persistent sector, with* $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$*.*

Proof. Level (i) is Theorem 5.6. Under (R1)–(R6) the localized transportable endomorphisms form a category equivalent to Sect_∞ by (R6), hence to $\mathrm{Rep}(G)$; the Doplicher–Roberts inputs are exactly (R2)–(R6) on the net (R1). □

This paper constructs the categorical side only; no observable net is specified on which the persistent sectors act, so every fixed-point identity $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^G$ is conditional on (R1)–(R6).

Lemma 5.9 (Product factorization)**.** *If* $\mathsf{Sect} \simeq \mathsf{Sect}_1 \boxtimes \mathsf{Sect}_2$ *identifies the reconstructed fibre functor with* $\mathcal{F}_1 \boxtimes \mathcal{F}_2$*, where each* $\mathcal{F}_i$ *is a faithful symmetric tensor fibre functor, then* $\mathrm{Aut}^\otimes(\mathcal{F}) \cong G_1 \times G_2$ *with* $G_i = \mathrm{Aut}^\otimes(\mathcal{F}_i)$ *and* $\mathsf{Sect}_i \simeq \mathrm{Rep}(G_i)$*.*

Proof. A tensor-natural automorphism of the external-product functor restricts to each factor by evaluation on $(X, \mathbf{1})$ and $(\mathbf{1}, Y)$, and the two restrictions determine it on every (X, Y) by tensor naturality. Reconstruction on each factor gives the representation equivalences. □

A bare categorical factorization without compatible fibre functors does not establish the product statement, and factorized heat-kernel weights on one state do not reconstruct a product group. If, independently, the finite Lie-type theorem certifies the two nonabelian factors and the charge theorem certifies one connected U(1), the connected realization is $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ up to the separately computed finite kernel.

6 Conditional matter realization

6.1 The matrix current, spin lift and exterior module

The charged-double-triplet matrices are an exact conditional realization of the premise-implied Lie type. A separate geometric-response construction probes all matrix units of $M_6(\mathbb{C})$ and resolves twelve current generators. The declared spin lift sends a six-state generator $D = \mathrm{diag}(A, \widehat{w})$ to

$$L(D) = \mathrm{diag}\left(A, -\frac{i}{2} w \cdot \sigma - \frac{\mathrm{tr}\, A}{2} I_2\right).$$

Together with the declared central trace redistribution this is the standard embedding $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3) \subset \mathfrak{su}(5)$. The nonvacuum even exterior module $M = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5$ has dimension fifteen and is the Georgi–Glashow $\mathbf{10} \oplus \bar{\mathbf{5}}$ generation [117]. Inside the declared ten-component exterior menu, the exhaustive 1024-subset anomaly scan selects $P_{\mathrm{even}} - P_{\mathrm{vac}}$ and $P_{\mathrm{odd}} - P_{\mathrm{top}}$ as the unique charge-conjugate pair of rank-15 chiral projectors. Source-computed charges and Casimirs identify Q, u^c, d^c, L, e^c. One declared Higgs doublet gives three invariant interaction lines Qu^cH, Qd^cH^* and Le^cH^*, and with three family copies these admit three arbitrary complex 3×3 Yukawa matrices. The spin cover, exterior-sector choice and physical fermion field are separate attachments; even auxiliary exterior parity does not establish physical fermion statistics, and the decoder of these labels is a representation check rather than a particle identification [21].

Lemma 6.1 (Factor identification)**.** *Let H be a positive-dimensional compact connected Lie group. (a) If H has a faithful irreducible two-dimensional pseudoreal unitary representation, the connected derived subgroup of the image is* $\mathrm{SU}(2)$ *acting as the doublet. (b) If H has a faithful irreducible three-dimensional representation whose nonabelian factor acts irreducibly and inequivalently to its conjugate, the connected derived subgroup of the image is conjugate to* $\mathrm{SU}(3)$ *on the fundamental, up to finite central kernel. (c) A connected abelian charge factor acting nontrivially on the coupled carrier gives a one-torus in the abelianized group.*

Proof. By Schur the centre acts by scalars, so the derived subgroup acts irreducibly. On $\mathbb{C}^2$ the only connected compact groups acting irreducibly have derived subgroup $\mathrm{SU}(2)$. On $\mathbb{C}^3$, two nontrivial simple factors would force a tensor decomposition of dimension at least four; among compact simple algebras only $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ have nontrivial irreducibles of dimension at most three, and the three-dimensional representation of $\mathfrak{su}(2)$ is real. Part (c) is immediate. □

Irreducibility in (a) is load-bearing: $\mathrm{U}(1)$ acting by $\mathrm{diag}(z, z^{-1})$ is faithful, pseudoreal and reducible. The intrinsically complex hypothesis in (b) is load-bearing: $\mathrm{U}(1) \times \mathrm{SO}(3)$ acting by zR satisfies every other hypothesis. None of these lemmas enters Theorem 3.1.

Proposition 6.2 (Conditional matter image)**.** *Assume the certified twelve-port carrier, the response and endogenous transport clauses, the declared matrix current, the declared fermionic spin category and the resulting conjugate projector pair. Anomaly-forced determinant balance and tensor descent on every realized matter tensor give: either projector supplies the one-generation chiral module with weak-doublet and colour-triplet roles on a common carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$; the exact Standard Model hypercharge lattice up to simultaneous charge reversal; $N_c = 3$; the conjugation-insensitive kernel $\mathbb{Z}_6$; and compatible scalar charges and Yukawa channels. The maximal faithful matter image is*

$$G_{\mathrm{packet}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

The Lie type uses the carrier, the response clauses and classification; the other conclusions use the declared matrix and matter premises. The response-sign convention and the choice between conjugate projectors are separate, and no theorem identifies them. The cover and its $\mathbb{Z}_2$ and $\mathbb{Z}_3$ quotients carry the same local tensors, so local data do not select the physical global form (Section 6.2). The proposition attaches no laboratory current and does not identify G_{packet} with the Tannaka group.

Theorem 6.3 (Hypercharge from anomaly freedom and Yukawas)**.** *Assume* $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$*, one generation of left-handed Weyl fermions* (Q, u^c, d^c, L, e^c)*, and the Yukawa channels* QHu^c*,* $QH^\dagger d^c$*,* $LH^\dagger e^c$*. Then anomaly freedom and Yukawa invariance fix the hypercharges up to normalization; with* $Q = T_3 + Y$ *and* $Q(\nu_L) = 0$*,*

$$Y_Q = \tfrac{1}{2N_c}, \qquad Y_L = -\tfrac{1}{2}, \quad Y_e = 1, \quad Y_u = -\tfrac{N_c+1}{2N_c}, \quad Y_d = \tfrac{N_c-1}{2N_c}, \quad Y_H = \tfrac{1}{2}.$$

For $N_c = 3$*:* $Y_Q = \frac{1}{6}$*,* $Y_u = -\frac{2}{3}$*,* $Y_d = \frac{1}{3}$*.*

Proof. Yukawa invariance gives $Y_u = -(Y_Q + Y_H)$, $Y_d = -Y_Q + Y_H$, $Y_e = -Y_L + Y_H$. The $\mathrm{SU}(2)^2\mathrm{U}(1)$ condition $N_c Y_Q + Y_L = 0$ and the mixed gravitational condition $2N_c Y_Q + N_c Y_u + N_c Y_d + 2Y_L + Y_e = 0$ give $Y_L = -N_c Y_Q$, $Y_H = N_c Y_Q$, $Y_u = -(N_c+1)Y_Q$, $Y_d = (N_c-1)Y_Q$, $Y_e = 2N_c Y_Q$; the $\mathrm{SU}(N_c)^2\mathrm{U}(1)$ and cubic anomalies then vanish. The normalization gives $Y_Q = 1/(2N_c)$. Without the Yukawa channels the cubic anomaly leaves two branches exchanging Y_u and Y_d. □

This is the textbook argument applied to the declared packet; the exact rationals reproduce the Standard Model because every discrete input is the Standard Model's. The channel scan selects the compatible scalar charge branch, not scalar attachment, multiplicity or a one-Higgs completion. Per generation there are $N_c + 1 = 4$ weak doublets, so the global SU(2) anomaly [72] is absent on the realized branch; this is a consistency check on $N_c = 3$, not a selector.

Theorem 6.4 (Conditional charge quantization)**.** *If the maximal faithful image is* G_{packet} *and* $Q = T_3 + Y$ *on a separately assumed electroweak branch, every colour-singlet representation of the packet has integer charge.*

Proof. The $\mathbb{Z}_6$ quotient identifies $(e^{2\pi i/3}, -1, e^{i\pi/3})$ with the identity. On a colour singlet with weak spin j and $n = 6Y$, this requires $(-1)^{2j} e^{i\pi n/3} = 1$, so Y is an integer for integer j and a half-integer for half-integer j. In both cases $T_3 + Y \in \mathbb{Z}$. □

The statement concerns packet representations, not a quantum field realization. It is consistent with the bound $|q_p + q_e|/e < 1 \times 10^{-21}$ on matter neutrality, with bulk fractional-charge abundance limits near 10^{-22} per nucleon, and with collider exclusions of stable particles with $q \in [e/3, 0.9e]$ up to masses of about 640 GeV at 95% CL; these searches do not discharge its premises.

Corollary 6.5 (No simple-GUT X/Y gauge channel)**.** *On the connected* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *realization and the twelve-dimensional current algebra, the adjoint is* $(8,1,0) \oplus (1,3,0) \oplus (1,1,0)$*. There are no gauge generators in* $(3, 2, \pm 5/6)$*, so no ordinary* X, Y *exchange inside the current algebra.*

Heavier modes outside the algebra, scalar mediators and other ultraviolet mechanisms are not excluded. Conditional on the one-generation table, its accidental baryon and lepton labels and the declared local dimension-six grammar, an exhaustive census of the 715 four-field multisets admits four Hermitian-conjugacy classes, $QQQL$, $QQu_R e_R$, $QLu_R d_R$ and $u_R u_R d_R e_R$, whose exact exterior contractions remain nonzero after the Schouten, Fierz and identical-field Pauli relations. The

diagonal $\mathbb{Z}_6$ generator fixes every field, so the gauge structure and the quotient do not prove proton stability; the census supplies no coefficient or lifetime. In the same way the mixed hypercharge anomaly gives $k_R^{YWW} = N_g(3Y_Q + Y_L) = 0$, so the $\mathbb{Z}_6$ deck phase supplies no direct $\Theta_R W\widetilde{W}$ baryogenesis coupling; a global $B + L$ attachment would give $k_R = 2N_g$.

6.2 Global form

Four central global forms act consistently on the same local tensors: the cover and its $\mathbb{Z}_2$, $\mathbb{Z}_3$ and $\mathbb{Z}_6$ quotients. The kernel theorem identifies the maximal faithful local matter image. On the finite weight lattice of hypercharge, triality and duality labels in the $q = 6Y$ normalization, a label is fixed by every element of the diagonal $\mathbb{Z}_6$ kernel exactly when

$$2t + 3d + q \equiv 0 \pmod 6,$$

and the kernel acts on any label by multiples of this centre character. Every row of the matter table satisfies the congruence, including both carrier blocks and the scalar; a fractionally charged singlet is blocked by the generator; and exactly one sixth of the thirty-six lattice labels descend, each character value being carried by six labels. A machine-checked receipt composes this onto the selected matter mask, so every selected component descends through the full kernel. The six-axis artifact computes an order-six flux menu after its coefficient relations are declared; the menu matches the $\mathbb{Z}_6$ quotient and carries the declared matter single-valuedly.

The order-six seam classification keeps two actions separate. Every pure-hypercharge character is enumerated: the fixed-space dimension menus are $\{15, 7\}$ at order two, $\{15, 3\}$ at order three and $\{15, 7, 3, 1\}$ at order six, while the canonical diagonal $\mathbb{Z}_6$ kernel acts trivially on all fifteen matter states. These are different representations of the same abstract cyclic order, and refinement preserves the classified actions. The calculation selects no physical seam character, flux action or 2-representation and does not exhaust every seam grammar. Selection of one physical quotient among the four requires a source-derived complete character category, a same-source loop-to-kernel identity, four-dimensional instanton normalization and theta periodicity, monopole dynamics and a laboratory flux attachment; the congruence derives no physical global form.

Corollary 6.6 (Structural electroweak package)**.** *On a separately declared one-Higgs branch, the electroweak factor has Lie algebra* $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$*. A nonzero neutral vacuum vector* $\phi_0 = \frac{v}{\sqrt{2}}\binom{0}{1}$ *of* $H = (1, 2)_{1/2}$ *has* $Q\phi_0 = (T_3 + Y)\phi_0 = 0$ *and stabilizer* $\mathrm{U}(1)_Q$*; the package contains* $W^{\pm} = (W^1 \mp iW^2)/\sqrt{2}$*, Z, A and the exact charge operator* $Q = T_3 + Y$*.*

Since $e^{i\alpha T_3}e^{i\beta Y}\phi_0 = e^{i(\beta-\alpha)/2}\phi_0$, vector invariance ties $\beta = \alpha$, while the projective ray $[\phi_0]$ has the two-torus stabilizer $\mathrm{U}(1)_{T_3} \times \mathrm{U}(1)_Y$; projectivization alone does not select the electromagnetic diagonal. A machine-checked composition on the committed doublet carrier proves, for declared $\lambda, v, g, g' > 0$ and Yukawa coefficient y, the minimizer shell $|h|^2 = v^2$, one unbroken line with three Goldstone directions, broken mass values $g^2v^2/4$ and $(g^2 + g'^2)v^2/4$, and a Yukawa bilinear of coefficient yv. The mixing angle, v and the W/Z masses belong to the running and matching surface. On the unbroken factor, a supplied Lorentzian Maxwell action $-\frac{1}{2g_Q^2}\int F_Q \wedge *F_Q + \int A_Q \wedge *J_Q$ with $F_Q = dA_Q$ gives $dF_Q = 0$ and $d * F_Q = g_Q^2 * J_Q$; the reconstruction does not supply that action or its coefficient. In a tensor-network realization a faithful representation of $\mathrm{SU}(3) \times \mathrm{SU}(2)$ needs edge dimension $\chi \geq 5$, and $\chi \geq 6$ only under the additional declared requirement of a single irreducible tensor-product carrier.

7 Families and the generation window

Anomaly cancellation is generation by generation and does not fix the number of generations.

Proposition 7.1 (Conditional generation window)**.** *On a separately declared one-Higgs quark branch over the matter image of Proposition 6.2, with $N_c = 3$ and the declared clauses of intrinsic CKM CP capability and weak-sector ultraviolet completability, the admissible window is $3 \le N_g \le 5$. No count inside the window is selected.*

Proof. An $N_g \times N_g$ CKM matrix has $(N_g - 1)(N_g - 2)/2$ physical CP phases, which vanish for $N_g \le 2$; the declared CP-capability clause therefore gives $N_g \ge 3$. The one-loop coefficient $b_{\mathrm{SU}(2)} = \frac{22}{3} - \frac{1}{3}N_g(N_c+1) - \frac{1}{6}$, with the last term from one complex Higgs doublet, is positive exactly when $N_g(N_c+1) < 43/2$; with $N_c = 3$ this gives $N_g \le 5$. □

The lower edge is the declared CP-capability clause; the window is the complete numerical content of the two clauses and is not forced by anomaly cancellation, the A_5 graph or the target-free source reduct.

Proposition 7.2 (Conditional screen-band selection)**.** *The twelve-port coefficient module decomposes into bands of ranks $1, 3, 3', 5$, with exact operational costs of the nontrivial bands*

$$5 - \sqrt{5} < 6 < 5 + \sqrt{5}.$$

Assume that one complete faithful band in the declared rank window $3 \le r \le 5$ realizes the family carrier and that the lower cost selects between admissible complete bands. Then the rank-three band of cost $5 - \sqrt{5}$ is unique. Tensoring it with the rank-15 generation gives a conditional rank-45 candidate whose chirality and diagonal $\mathbb{Z}_6$ action come from the generation table.

These implications are kernel-decided in the Lean screen certificate. A response receipt using the graph Laplacian as a unitary generator recovers the same cost bands and places the rank-three residue at the lowest positive generator frequency; the channel conserves mode norms, so this is a frequency statement rather than a relaxation statement. For the four-entry band table, faithfulness excludes the dimension-one band and leaves dimensions $3, 3, 5$; removing the window while keeping faithfulness changes no admissible candidate, so the window does not itself explain triplication. The load-bearing identification is that the family object is a single complete faithful band, compared by the declared cost rule and realized in the matter response. A specified source-executed cooling law computes this rank-three band on the native carrier; selecting that rule and identifying its levels with physical families are inputs.

Generation names require no universally conserved generation charge. Within a sector whose charges and statistics are established, ordering nondegenerate masses is a naming convention, and a physical comparison must retain the eigenvectors, residues and charged-current matrix of the same response. The mismatch of the two left quark diagonalizers determines CKM mixing; a generation label conserved by every charged weak transition would exclude that mixing.[11]

Corollary 7.3 (Declared minimal completion)**.** *Assume Proposition 6.2, the receipt-certified bosonic branch, a one-Higgs scalar attachment, the CP and weak-sector clauses, and a declared physical attachment of the rank-three band to the matter packet. Then the completed packet has maximal faithful image data $(G_{\mathrm{packet}}, \mathcal{R}_{\mathrm{SM}}, H, \mathcal{Y}_{\mathrm{SM}})$ with $N_c = 3$, $N_g = 3$ and*

$$Q_i = (3,2)_{1/6}, \quad u_i^c = (\bar{3},1)_{-2/3}, \quad d_i^c = (\bar{3},1)_{1/3}, \quad L_i = (1,2)_{-1/2}, \quad e_i^c = (1,1)_1, \quad H = (1,2)_{1/2},$$

[11] A. Ceccucci, Z. Ligeti and Y. Sakai, "CKM Quark-Mixing Matrix," in the Particle Data Group review (2024), Section 12.1, https://pdg.lbl.gov/2024/reviews/rpp2024-rev-ckm-matrix.pdf.

$i = 1, 2, 3$.

The Lie type uses no clause of the completion; the matter type, charges and $N_c = 3$ are exact inside the declared fixture; $N_g = 3$ and the one-Higgs scalar enter through the completion declaration. Under the fixed-cutoff heat-kernel law of Section 10 every finite-dimensional witness label has positive weight; that positivity concerns bosonic sector labels and supplies no light chiral multiplet. Physical identification of the rank-45 candidate requires a source-derived matter-pole map, continuum spin and locality, physical seam selection, residues, excluded-band and refinement receipts; the absence of other light sectors is unproved.

8 The equivariant bracket variety and a face discriminator

Conditional on the canonical oriented carrier, the target-free space $\mathrm{Hom}_{A_5}(\Lambda^2\mathbb{Q}^{12}, \mathbb{Q}^{12})$ has dimension fourteen and an exact rational Reynolds basis. It is a search space; no bracket is selected. Substitution of the general bracket into Jacobi gives 2,640 coordinate equations in 44 target orbits with coefficient-row rank 38. Over $\mathbb{Q}(\sqrt{5})$ that rowspace splits into eleven fixed-line derivation-weight products and a rank-27 residual system that applies on every weight flat. A three-parameter derivation-only Jacobi family supplies no compactness or source selection. Conditional on three named textbook lemmas, the compact real locus of the fourteen-parameter equivariant variety is classified exactly over $\mathbb{Q}(\sqrt{5})$ into three certified bracket families F, G, P. For each family the coroot lattices, integral carrier weights and carrier-admitted global forms are computed; the selected forms are adjoint with zero abelian charge lattice, so the abelian sector is undetermined at carrier level. Noncompact and residual components outside the compact locus are unclassified.

Applying the declared equal-weight cyclic rule to the twenty oriented faces gives $B_{\mathrm{face}} = 60R_{13}$. This bracket is A_5-equivariant and fails Jacobi in exactly 240 of the 2640 coordinates, 120 at $+1$ and 120 at -1. Under the added Euclidean metric on upper-triangular structure constants, orthogonal projection onto the compact locus has the unique nearest family G:

$$d_G^2 = \frac{615 - 123\sqrt{5}}{22}, \qquad d_F^2 = \frac{615 + 123\sqrt{5}}{22}, \qquad d_P^2 = 45.$$

Both F and G projections lie in their open compact strata. On the same 792-coordinate convention, exact primal–dual certificates give

$$d_1(G) = 30(\sqrt{5} - 1) < \inf d_1(F) = d_1(P) = 60,$$

$$d_\infty(G) = \frac{5 - \sqrt{5}}{10} < d_\infty(F) = \frac{\sqrt{5}}{5} < d_\infty(P) = \frac{1}{2}.$$

The ℓ_1 value for F is an infimum approached under scaling, not an attained minimum. The Python replay proves the primal–dual optimization and the Lean theorem checks the radical values and their order. The family verdict holds across the ℓ_1, ℓ_2 and ℓ_∞ scores on the classified compact locus once the coordinate norm is supplied.

The quadratic verdict is robust over the invariant class. The carrier splits multiplicity-free as $1 + 3 + 3' + 5$, the commutant of the port action is spanned by the four spectral projectors, and every invariant carrier inner product is $\alpha P_1 + \beta P_3 + \gamma P_{3'} + \delta P_5$. The induced distances are

$$d_P^2 = \frac{15 - 3\sqrt{5}}{\beta} + \frac{15 + 3\sqrt{5}}{\gamma} + \frac{15 - 3\sqrt{5}}{2}\frac{\beta}{\delta^2} + \frac{15 + 3\sqrt{5}}{2}\frac{\gamma}{\delta^2},$$

$$d_F^2 = \frac{15 + 3\sqrt{5}}{2}\frac{\gamma}{\delta^2} + \frac{15 + 3\sqrt{5}}{\gamma} + \frac{60 + 12\sqrt{5}}{11\beta}, \qquad d_G^2 = \frac{15 - 3\sqrt{5}}{2}\frac{\beta}{\delta^2} + \frac{15 - 3\sqrt{5}}{\beta} + \frac{60 - 12\sqrt{5}}{11\gamma},$$

with α absent. Both gaps $d_P^2 - d_G^2$ and $d_P^2 - d_F^2$ are coefficient-positive Laurent polynomials, so P is excluded for every invariant metric. At $\gamma = \beta$ the gap $d_F^2 - d_G^2 = 3\sqrt{5}\,\beta/\delta^2 + (90\sqrt{5}/11)/\beta$ is positive, and for $\beta/\delta \in [1/50, 6]$ the family G is uniquely nearest for all γ, δ. The mirror family F is nearer on the other side of the exact three-scale tie surface, with witness $(\beta, \gamma, \delta) = (8, 1, 1)$. Conjugating $\sqrt{5}$ and swapping $\beta \leftrightarrow \gamma$ maps d_G to d_F. Reversing every face orientation negates B_{face} and leaves the distances unchanged. A channel-diagonal invariant metric that is not carrier-induced reverses the balanced-point selection, so the carrier-induced restriction is load-bearing. On the pinned incidence table, equal normalized face weights and the declared barycentric one-third share give port-dual weight $1/12$ at every port; under the declared diagonal measure-to-metric rule this uniform measure induces the scalar carrier inner product at $(1/12, 1/12, 1/12, 1/12)$, where G is uniquely nearest with $d_G^2 = (3690 - 738\sqrt{5})/11$, $d_F^2 = (3690 + 738\sqrt{5})/11$ and $d_P^2 = 540$.

The equal-weight rule, the coordinate norms, the nearest-point law and the measure-to-metric rule are declared inputs, not consequences of A1–A3, so the verdict is a conditional discriminator obtained without reference to a measured value. A composed machine-checked surface bundles these premises with the classification inputs and the exterior selection, and proves that the premise-implied type has a unique type-matched nearest family at the balanced point and that the fifteen-state count, the anomaly forms and the $\mathbb{Z}_6$ stabilizer are carried by the bundled selection mask. Among the three family labels and the declared 2^{10} exterior table, the admissible pairs are G with either of two conjugate masks. This does not count points of the continuous G-family or classify physical Standard Model structures.

Over the certified brackets, ad-invariance fixes the kinetic shape block-diagonally with zero cross terms. On carrier-projector forms, families F and G reduce the three-dimensional invariant space to two dimensions, $F: \; w_{3+} = \sqrt{5}\, w_5$ and $G: \; w_{3-} = \sqrt{5}\, w_5$, with P the two-to-two control; simultaneous invariance for F and G leaves the ray $(w_{3+}, w_{3-}, w_5) = t(\sqrt{5}, \sqrt{5}, 1)$, an extra premise. The two-factor history theorem constructs a Gibbs kernel from $aq_1 + bq_2$ with exact factorization; equality of two such kernels identifies λa and λb, so only common scaling survives and a/b is not multiplier gauge. No theorem supplies relative factor coefficients or coupling scales [28, 29].

8.1 An assembled structural action

On one typed premise bundle, the gauge kinetic term (one declared magnitude per invariant carrier sector $3^+ + 3^- + 5$, weighted in the two-parameter G-invariant normal form), the quartic potential $\lambda(|h|^2 - v^2)^2$, the gauge mass form at the chosen minimum and the realified Yukawa line with coefficient y sum to one real function of one field tuple. Exact restriction theorems recover each sector on its slice; the gauge coefficients satisfy the G ad-invariance row and collapse to the mirror ray under the extra F row; the cross-sector generators realizing those rows have coefficient ratio $(1, -\sqrt{5})$, and the control weight $(1, 1, 1)$ breaks invariance with exact value $1 - \sqrt{5}$. A fourth summand carries one complex amplitude per row of the ten-component table; its bilinear $\overline{\psi_i}\psi_j$ is phase-invariant on the selected rows exactly when the charges $q = 6Y$ balance, and the charge column forces the pairing to be diagonal. On the shared doublet and singlet rows the extended action evaluates to the mass bilinear with coefficient yv. Every coefficient is a declared parameter; the construction supplies no source-produced spacetime action, measured coupling, family attachment, global-form selection, chirality or spin attachment. The local covariant action, its Cartan reduction and the fermion current are given in the appendices.

9 Refinement stability and relevant transverse perturbations

The state-side refinement stability used by the gauge branch is a separate optimizer-pushforward and refinement-tail interface: it keeps the finite local constraint family across cutoffs, so that a refinement channel acts on the realized Gibbs family $\omega_\ell(\lambda) \propto \exp(-\sum_{x,a} \lambda_a O_a(x) - \sum_b \mu_b Q_b)$ by an induced map $\omega_L(R_{\ell\to L}(\lambda))$. Coarse-graining a finite-range Gibbs family generically leaves any fixed finite list, so this is a substantive premise; premise A3 supplies the fixed-stage optimizer and not this cross-scale statement or the refinement receipt. The square defects that control reconciliation under coarse-graining are treated in the consensus paper [19].

Proposition 9.1 (Conditional growth of a relevant transverse component)**.** *Assume the local finite-constraint branch of A3, the refinement-tail interface, and a differentiable extension of the refinement maps to an enlarged coupling space with transverse coordinate $g_{\mathcal{O}}$ for a gauge-invariant Lorentz-scalar deformation $\mathcal{O}$. If $\delta g_{n+1} = \Lambda_n \delta g_n$ with $|\Lambda_n| \geq q > 1$, every nonzero infinitesimal transverse component grows by at least q^m after m steps. If the zero-coupling surface is invariant and the trajectory starts on it, relevance alone does not generate $\mathcal{O}$.*

Proof. Iteration gives $|\delta g_{n+m}| = \prod_j |\Lambda_{n+j}||\delta g_n| \geq q^m |\delta g_n|$. Generation from zero concerns the transverse part of the full extended map, which is not a coordinate of $R_{\ell\to L}$ and is not supplied by A3. Exact tuning, an invariant manifold, vanishing operator mixing, cancellations, an additional symmetry or fixed-point dynamics can keep the component zero, so the statement is not an exhaustive protection theorem. □

A gauge-invariant Dirac mass is a relevant scalar when the representations allow it. With nonzero mixing into that coordinate, a nonzero mass perturbation is linearly amplified unless a protecting mechanism intervenes. Chiral matter forbids the Dirac mass by symmetry; vector-like branches with vanishing mixing, tuning or fixed-point protection are not excluded, so the diagnostic does not select chirality by itself.

10 Edge-sector weights and coupling extraction

The edge-centre completion yields sector probabilities p_R on collar boundaries. A conditional heat-kernel weighting relates them to a gauge coupling.

Theorem 10.1 (Fixed-cutoff edge-sector law)**.** *Under premises A1–A3, the fixed-cutoff overlap-gauge realization, a quasi-local local-Gibbs generator on the faithful interior branch, and hypotheses (EH-1) and (EH-2), the regulated same-overlap sector distribution is*

$$p_R = \frac{d_R e^{-t\lambda_R}}{\sum_{R'} d_{R'} e^{-t\lambda_{R'}}},$$

with λ_R the Laplacian eigenvalue on the R-isotypic component and t the collar Gibbs parameter.

(EH-1) The edge generator is the factorwise group Laplacian, $H_{\text{edge}} = \Delta_G$, or a positive combination $\sum_i c_i \Delta_{G_i}$ for a product group; the separate Gibbs parameter is t. For finite groups the declared Cayley generating multiset is conjugacy-invariant and inverse-symmetric, so $\lambda_R = |S| - d_R^{-1} \sum_{s\in S} \chi_R(s)$. Without centrality the Fourier transform is a matrix on each irreducible block and the character average is only its normalized trace, not a sector eigenvalue. This is a declared input: gauge invariance and the Gibbs form force only $H_{\text{edge}} = \sum_R h_R P_R$, and any function of the sector label is gauge invariant and realizable by a local Hamiltonian at fixed cutoff.

Within second-order bi-invariant operators the Laplacian is unique up to scale, and one declared Markov chain with a Casimir acceptance rule derives (EH-1); its universality is unproved. (EH-2) The probability space is the one-sided electric-centre edge algebra $\bigoplus_R B(V_R)$, on which the sector projector has rank d_R. On the two-sided space $L^2(G)$ the projector has rank d_R^2, and a two-sided Gibbs trace gives weights $d_R^2 e^{-t\lambda_R}$; the d_R law is a statement about the one-sided algebra.

Proof. The edge degrees of freedom at Σ carry a G-action, modelled for the refinement lift by $L^2(G)$ with its Peter–Weyl decomposition [76]. By (EH-2) the physical one-sided observables are $\bigoplus_R B(V_R)$; by (EH-1) the generator is $\sum_R \lambda_R P_R$. Maximum entropy selects $\rho = Z^{-1}\sum_R e^{-t\lambda_R} P_R$, and $\operatorname{Tr} P_R = d_R$ gives the stated law. □

For $\mathbb{Z}_n$ the Laplacian eigenvalues are $\lambda_q = 4\sin^2(\pi q/n)$. The heat-kernel ansatz assigns the sector probabilities $p_q \propto e^{-t\lambda_q}$, so $t = -\log(p_q/p_0)/\lambda_q$. For SU($N$) the conditional ansatz assigns the probabilities $p_j \propto d_j e^{-tC_2(j)}$, so $t = (\log(p_0/d_0) - \log(p_j/d_j))/C_2(j)$ and $g^2_{\text{ent}} = t/(2\pi)$. In two-dimensional Yang–Mills the Hamiltonian is the group Laplacian, $H = -\frac{g^2}{2}\Delta_G$ with $E_R = \frac{g^2}{2}C_2(R)$, and a cylinder of area A gives $t = g^2 A/2$ exactly; gluing the open-edge weights gives the closed partition sum $\sum_R d_R^2 e^{-tC_2(R)} = K_t(1)$. The Casimir log-gaps $\Delta_R = \ln(p_0/d_0) - \ln(p_R/d_R) = tC_2(R)$ have parameter-free ratios: for SU(2), $\Delta_1/\Delta_{1/2} = 8/3$, $\Delta_{3/2}/\Delta_{1/2} = 5$; for SU(3), $\Delta_8/\Delta_3 = 9/4$, $\Delta_6/\Delta_3 = 5/2$, $\Delta_{10}/\Delta_3 = 9/2$, $\Delta_{15}/\Delta_3 = 4$, $\Delta_{27}/\Delta_3 = 6$. These are algebraic consequences of the weighting, not source-selected coupling predictions. The repair-gap identification on the compact-gauge branch is the subject of the Yang–Mills supplement [23].

Finite diagnostics. In two-dimensional $\mathbb{Z}_n$ lattice gauge models on a 2×2 torus with $H = -K\sum_p \mathrm{Re}B_p - h\sum_\ell \mathrm{Re}X_\ell - \Gamma\sum_v \mathrm{Re}A_v$, $K = 1$, $\Gamma = 5$, the electric-centre edge charge at a boundary vertex is the restricted star. For $\mathbb{Z}_2$ one eigenvalue allows an exact fit, and for $\mathbb{Z}_3$ the two extractions agree by charge conjugation and eigenvalue degeneracy; these are implementation checks. $\mathbb{Z}_5$ has two distinct nonzero eigenvalues with $\lambda_2/\lambda_1 = \phi^2 \approx 2.618$. Fitting t on $q = 1$ and holding out $q = 2$, direct diagonalization gives the following.

h	$\ln(p_2/p_0)/\ln(p_1/p_0)$	deviation from ϕ^2
0.50	2.876	9.9%
0.20	2.750	5.1%
0.10	2.671	2.0%
0.05	2.643	1.0%

The deviation decreases monotonically across these samples; no limit proof or interval enclosure is supplied. A symmetry-reduced electric calculation with n^3 states reproduces the link-basis values.

For the one-plaquette S_3 model with transposition Cayley eigenvalues $\lambda_{\text{triv}} = 0$, $\lambda_{\text{sign}} = 6$, $\lambda_{\text{std}} = 3$, Gauss' law leaves the three character states, and $p_R = |c_R|^2$ exactly. With $K = 1$ and four electric links the reduced Hamiltonian is

$$H(h) = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 24h & -1 \\ -1 & -1 & 12h - 1 \end{pmatrix}, \qquad h \geq 0,$$

and the Gauss penalty is a common shift. Fitting t on the standard sector and holding out the sign sector gives the following values.

h	p_triv	p_sign	p_std	t (sign)	t (std)	Δt/t	log-ratio
0.5	0.962927	2.476×10^{-4}	0.036825	1.37766	1.31898	4.3518%	2.088972
1.0	0.991861	1.400×10^{-5}	0.008125	1.86139	1.83261	1.5580%	2.031406
2.0	0.998116	8.158×10^{-7}	0.001883	2.33619	2.32203	0.6080%	2.012197
5.0	0.999713	1.993×10^{-8}	2.871×10^{-4}	2.95511	2.94950	0.1897%	2.003798
12	0.999951	5.895×10^{-10}	4.890×10^{-5}	3.54195	3.53963	0.0656%	2.001312
100	0.999999	1.208×10^{-13}	6.956×10^{-7}	4.95749	4.95721	0.0056%	2.000112

Here $\Delta t/\bar{t} = (t_{\rm sign} - t_{\rm std})/\bar{t}$ and the log-ratio is $\log(p_{\rm sign}/p_0)/\log(p_{\rm std}/(2p_0))$, which equals 2 under exact heat-kernel weights. Writing the ground energy as $-x < 0$, the secular equation is $x + 12h - 1 - 1/x - 1/(24h + x) = 0$, with a unique positive root decreasing from $x(0) = 2$. With $z = x + 12h$ and $p_{\rm sign}^{\rm pred} = p_{\rm std}^2/(4p_{\rm triv})$,

$$\frac{p_{\rm sign}}{p_{\rm sign}^{\rm pred}} = \left(\frac{z-1}{z}\right)^2 < 1,$$

so the standard-sector fit overpredicts the sign sector at every finite h, and the ratio increases strictly from 1/4 toward one as $h \to \infty$. The fitted time $t_{\rm std} = \log(2/x^2)/3$ is positive exactly for $h > h_* = (1+\sqrt{3}-\sqrt{2})/24 \simeq 0.0549098852$. At $h = h_*$, the relative log residual and log-ratio divide by zero; below it the fitted time is negative and outside the positive-diffusion ansatz. On $h > h_*$ the relative residual and log-ratio minus two decrease strictly to zero; at $h = 100$ the held-out relative log residual has magnitude about 0.0056%, with the fit overpredicting the sign sector. The identity proves a strict held-out mismatch at every finite coupling inside one three-dimensional model and transfers nothing to either continuous group. Neither diagnostic validates the law generally. A held-out refit script is part of the public code [29].

11 Classical carrier modes and the quantum-particle criterion

A compact group identifies a symmetry type and a classical field equation a classical branch; neither constructs a propagating quantum particle.

Definition 11.1 (Carrier-mode and quantum-particle receipts)**.** *Fix the invariant conversion speed $c_\star$; its value* 299 792 458 $\mathrm{m\,s^{-1}}$ *is exact by the SI definition of the metre. A classical massless carrier-mode receipt for q_X consists of (C1) a stated background and phase, (C2) a quadratic action with positive kinetic coefficient, (C3) a gauge fixing or constraint reduction with physical projector Π_X, and (C4) a positive reduced Hamiltonian, with physical Hessian $K_X = Z_X(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_X$, $Z_X > 0$. A quantum-particle receipt additionally requires (Q1) a positive-energy quantization with physical Hilbert space, (Q2) a two-point function whose Källén–Lehmann measure contains $Z_X^{\rm pole}\delta(\mu^2)$ with $Z_X^{\rm pole} > 0$, and (Q3) the asymptotic-state or LSZ hypotheses of the phase, including a deconfined sector for a coloured carrier.*

Theorem 11.2 (Action-level modes)**.** *(i) An unbroken $\mathrm{U}(1)_Q$ connection with the Lorentzian Maxwell action $S_Q = -\frac{1}{4g_Q^2}\int F_{Q,\mu\nu}F_Q^{\mu\nu}$, $0 < g_Q^2 < \infty$, and no mass operator, has in radiation gauge $K_{Q,T} = g_Q^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T$ with $\operatorname{rank}\Pi_T = 2$. (ii) A compact G with the two-derivative pure Yang–Mills action about the trivial background in a perturbative or deconfined phase has $2\dim G$ transverse modes with $K_{\mathrm{YM},T}^{ab} = \delta^{ab}g^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T$ and no hard mass at quadratic order.*

Proof. Gauss' law removes the longitudinal pair, leaving two transverse polarizations with positive Hamiltonian and dispersion $\omega^2 = c_\star^2|\mathbf{k}|^2$. The Yang–Mills Hessian is one Maxwell Hessian per generator; interactions begin at cubic order. These calculations establish (C1)–(C4) and none of (Q1)–(Q3). □

Calling these modes photons or gluons requires the quantum-particle receipt. The perturbative mode does not produce a gauge-invariant asymptotic gluon, and in a confining phase the coloured free pole need not occur in the physical spectrum. The analogous transverse-traceless calculation for a supplied Einstein–Hilbert action gives two classical spin-two modes and no graviton state. Vanishing quadratic mass parameters are action-level statements; Higgs or Stueckelberg completions, medium self-energies, confinement and additional massive fields require their own field-content and phase hypotheses.

12 Failure conditions

The finite chain is exact on its stated premises, and each step has a registered negative control. The Lie-type theorem fails if a faithful compact commutator-closed twelve-port response with inner carrier action is exhibited outside $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$; the conditional matter results fail if the incidence identity, the target-blind response, the equivariant lift, the anomaly scan, the tensor-character enumeration or the refinement persistence receipt fails its control. The categorical route applies only inside its combined strictification, trivial-holonomy, localization and refinement and fibre receipts.

On the executable model the branch is tested by computing the complete accessible current of a specified law, with every admitted control and apparatus service in the menu. A complete accessible response that is not faithful, closed and twelve-dimensional with inner carrier action excludes that law's realization of A1 and A2, as the seventeen-dimensional exchange response excludes its grammar. A physical attachment that forced the carrier rechartings to be spatial rotations of an interacting relativistic field while they remain internal would be constrained by the Coleman–Mandula theorem. The particle paper [21] tests whether localized defects of the same transport carry holonomy valued in this structure.

A Local action, Cartan reduction and fermion current

The appendices carry the local covariant action on the declared matter grammar and the finite current constructions built on it.

A.1 A local gauge-covariant action on the declared matter grammar

Fix an oriented flat Lorentzian chart with metric $\eta = \mathrm{diag}(1, -1, -1, -1)$, its two-component Spin convention, and smooth compactly supported variations. These geometric and Spin data are inputs. Take the declared algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ on its fundamental, dual and trivial carriers. The exterior grammar supplies

$$Q : (3,2)_{1/6}, \quad u^c : (\bar{3},1)_{-2/3}, \quad d^c : (\bar{3},1)_{1/3}, \quad L : (1,2)_{-1/2}, \quad e^c : (1,1)_1.$$

The Higgs choice $H : (1,2)_{1/2}$, family multiplicity, minimal first-derivative kinetic law, scalar polynomial truncation and Yukawa intertwiners are declared. No coupling, mass or Yukawa entry is selected.

Use anti-Hermitian connections, with coupling normalization absorbed into their coordinates, and set

$$D_\mu = \partial_\mu + \rho(A_\mu), \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \tag{8}$$

Fermion values and first derivatives generate a complex exterior algebra. Conjugation is the antilinear anti-automorphism exchanging fermions with independent conjugate generators and reversing products. Scalar and connection coefficients are even. Write $\bar\sigma^\mu = (1, -\sigma^1, -\sigma^2, -\sigma^3)$ and $\epsilon_{01} = 1$. With color index i, weak indices a, b, spin indices α, β and family indices r, s, define

$$T^u_{rs} = \epsilon_{\alpha\beta}\epsilon_{ab} Q^{ia\alpha}_r H^b (u^c)^\beta_{s,i}, \qquad T^d_{rs} = \epsilon_{\alpha\beta} Q^{ia\alpha}_r \overline{H}_a (d^c)^\beta_{s,i}, \qquad T^e_{rs} = \epsilon_{\alpha\beta} L^{a\alpha}_r \overline{H}_a (e^c)^\beta_s. \tag{9}$$

The superscript c denotes charge conjugation. These are all-left-Weyl expressions, not four-component barred expressions with unchanged charges.

Theorem A.1 (Conditional local action composition)**.** *Under the stated representation, Spin and kinetic assumptions, the density*

$$\begin{aligned}\mathcal{L} = {} & \frac{1}{2}\sum_{s=3,2,1} \kappa_s \operatorname{Tr}(F^s_{\mu\nu} F_s^{\mu\nu}) + \sum_{r,\psi} \frac{i}{2}\left[\psi^\dagger_r \bar\sigma^\mu D_\mu \psi_r - (D_\mu \psi_r)^\dagger \bar\sigma^\mu \psi_r\right] \\ & + (D_\mu H)^\dagger D^\mu H - m^2 H^\dagger H - \lambda (H^\dagger H)^2 - \sum_{r,s,\chi=u,d,e} \left(Y^\chi_{rs} T^\chi_{rs} + \overline{Y^\chi_{rs}} (T^\chi_{rs})^\dagger\right)\end{aligned} \tag{10}$$

is Grassmann even, Hermitian and invariant under arbitrary infinitesimal local gauge transformations. The real m^2, λ, positive κ_s and complex Y^χ_{rs} remain free coefficients. The construction composes varying-field gauge, scalar and fermion kinetic terms and all three Yukawa channels.

Proof. For an anti-Hermitian local parameter θ, set $\delta\psi = \rho(\theta)\psi$, $\delta H = \rho_H(\theta) H$ and $\delta A_\mu = -\partial_\mu \theta + [\theta, A_\mu]$. The product rule gives

$$\delta(D_\mu \psi) = \rho(\theta) D_\mu \psi, \qquad \delta F_{\mu\nu} = [\theta, F_{\mu\nu}].$$

The first identity cancels the two opposite $(\partial_\mu\theta)\psi$ terms. The second uses symmetry of the second parameter jet and cancels the first-derivative terms against the connection commutators. No equation of motion is used. Cyclic trace proves gauge-kinetic invariance; anti-Hermiticity and commutation of internal generators with $\bar\sigma^\mu$ prove kinetic invariance. The trace forms are ad-invariant, with the free normalizations permitted by the invariant-form classification.

The color fundamental–dual pairing and weak determinant tensor are invariant. The hypercharge sums are $1/6 + 1/2 - 2/3 = 0$, $1/6 - 1/2 + 1/3 = 0$ and $-1/2 - 1/2 + 1 = 0$. The supplied Spin epsilon contracts anticommuting Weyl components. Each T^χ_{rs} and its adjoint are gauge neutral, separately for every family pair. Every term has even Grassmann degree and the displayed conjugate combinations are Hermitian. The symmetrized kinetic expression differs from $i\psi^\dagger\bar\sigma^\mu D_\mu\psi$ by a total derivative. □

Proposition A.2 (Matter current of the same local action)**.** *For an anti-Hermitian basis element T_a, define $\delta\mathcal{L}_{\text{matter}} = j^\mu_a \delta A^a_\mu$ with matter jets fixed. Then*

$$j^\mu_a = i\sum_{r,\psi} \psi^\dagger_r \bar\sigma^\mu \rho_\psi(T_a)\psi_r + (\rho_H(T_a)H)^\dagger D^\mu H + (D^\mu H)^\dagger \rho_H(T_a) H. \tag{11}$$

Its covariant divergence vanishes on the matter Euler–Lagrange equations, using the dual adjoint connection on the current index.

Proof. Differentiate the two covariant kinetic terms. Anti-Hermiticity doubles the fermion contribution in the symmetrized expression; the potential and Yukawa tensors have no connection dependence. For a compactly supported gauge variation, the matter equations remove matter variations from the first variation of the action. Substituting $\delta A_\mu = -\partial_\mu\theta + [\theta, A_\mu]$ and integrating by parts gives the dual covariant-divergence identity. □

Exact local checks and scope. The executable fixture retains three families, 900 independent Grassmann value, derivative and conjugate generators, and all 27 complex Yukawa slots. Connection and scalar jets are exact Gaussian-rational test values, not physical coupling constants or measurements. Independent arithmetic checks reconstruct the density, retain first and symmetric second jets of the gauge parameter, and reject omitted inhomogeneous connection terms, omitted curvature commutators, wrong dual color action and mixed Yukawa charges. Grassmann anticommutation is checked separately from neutrality. Lean proves the noncommutative first- and second-jet identities, the cyclic-trace Ward identity and the spin-sign control. The full action and on-shell current argument are analytic, not full Lean formalizations.

The flat chart, Spin interpretation, kinetic choice, Higgs/Yukawa content, relative normalizations, family multiplicity and local connection remain declared. Identifying the connection with one reconstructed from source response histories is separate. There is no source-produced Standard Model action or current, broken-vacuum selection, gauge fixing or ghost construction, quantum anomaly cancellation from a fermionic measure, renormalization, continuum quantum-field existence theorem or numerical particle prediction. Global gauge quotient and physical line-spectrum attachments remain separate. The local action admits a consistent classical charged-scalar slice, with both Cartan gauge equations and their variational current retained. Its finite source-address execution uses a declared logarithmic plaquette action and model clock, as follows.

A.2 A consistent local Cartan–scalar reduction and current response

Work on the flat Spin chart and local classical action of Theorem A.1. All kinetic coefficients are positive constants; the mass, quartic and Yukawa coefficients remain declared. Use the standard weak Cartan generator $i\sigma_3/2$ and central generator i. Writing the two real Cartan connection coordinates as $a = (a_2, a_1)$ gives

$$q = (1/2, 1/2)^T, \qquad K = \mathrm{diag}(\kappa_2, 2\kappa_1), \qquad d = q^T K^{-1} q, \qquad w = K^{-1} q/d.$$

Here the gauge density is $-\frac{1}{4} f^T K f$: the fundamental traces are $\mathrm{Tr}((i\sigma_3/2)^2) = -1/2$ and $\mathrm{Tr}(i^2) = -1$. The executable weak basis uses $i\sigma_3$, in which $q = (1, 1/2)^T$ and $K = \mathrm{diag}(4\kappa_2, 2\kappa_1)$; d agrees.

Theorem A.3 (Local classical solution embedding)**.** *Set all fermions and their independent conjugates, color connections and off-diagonal weak connections to zero. Let $H = (\varphi, 0)$ and $a = wA$. Every smooth solution, including the temporal gauge equation, of*

$$\mathcal{L}_{\mathrm{red}} = -\frac{F_{\mu\nu}F^{\mu\nu}}{4d} + |(\partial_\mu + iA_\mu)\varphi|^2 - m^2|\varphi|^2 - \lambda|\varphi|^4 \tag{12}$$

lifts to a solution of the full local classical equations, with ansatz-compatible initial and boundary data. All complex Yukawa entries remain unrestricted.

Proof. The identities $q^T w = 1$, $Kw = q/d$ and $w^T K w = 1/d$ give the density and both Cartan equations. In the current convention $\delta\mathcal{L}_{\mathrm{matter}} = j^\mu \delta A_\mu$, these equations are $q_a[d^{-1}\partial_\nu F^{\nu\mu} + j^\mu] = 0$. If $q^T n = 0$, the decomposition $a = wA + nB$ has no scalar coupling to B and no kinetic cross

term, since $w^T K n = 0$; $B = 0$ solves its free equation. The Higgs and its covariant derivative lie in the upper weak line, so both off-diagonal weak currents vanish. The gauge covariant divergence of a Cartan curvature is Cartan. Color curvature and sources vanish. The diagonal connection preserves the lower Higgs line, whose potential derivative vanishes at zero. Every fermion equation contains a fermion factor; every Yukawa contribution to a Higgs equation contains two. Thus all discarded variations vanish without setting any Yukawa coefficient to zero. The temporal Cartan equation is included, so Gauss's law is retained. □

This is a local real-connection statement. Generic kinetic coefficients give a direction incommensurate with the full charge lattice. No compact subgroup, global quotient, large-gauge identification, electromagnetic vacuum, perturbative stability or quantum truncation follows. Retaining a real unwrapped connection is essential: its exponential alone need not determine the embedded connections in every representation.

A compatible spatial discretization. The same reduction holds for a declared semidiscrete gauge–scalar action with covariant link differences, invariant electric metric, and the local logarithmic plaquette energy. For the weak holonomy use its principal matrix logarithm on the smooth identity chart. On the Cartan slice, $\log U_f = i\sigma_3 w_{\mathrm{w}}(\mathrm{d}A)_f/2$ when $|w_{\mathrm{w}}(\mathrm{d}A)_f| < 2\pi$, where w_{w} is the weak component of w. The hypercharge curl is real and unwrapped. Consequently the plaquette energy restricts to the quadratic Maxwell curl with coefficient $1/d$. The first variation of $\mathrm{Tr}(\log U)^2$ is $2\,\mathrm{Tr}(\log U\, U^{-1}\delta U)$; discarded weak variations are off-diagonal and have zero trace pairing. Scalar link currents also have zero off-diagonal component. Electric transport along the Cartan line is geodesic for the invariant metric. These facts give the same omitted equations and weighted Cartan square. This statement requires the logarithm chart throughout the history. A Wilson cosine plaquette does not supply this quadratic reduction for arbitrary w.

On the prepared golden $q = 5$ tensor family use dimensionless positions x/L, $L = 2/\sqrt{\phi+2}$. The ordered one-dimensional nodes are $0, 2\phi-3, 4\phi-6, \phi-1, 3\phi-4, 1$. The 4^3 interior nodes have lumped masses v_i; an internal edge has conductance c_e equal to its transverse dual area divided by its gap. The fixed zero scalar boundary contributes the usual nonnegative onsite stiffness b_i. Declare $\kappa_1 = \kappa_2 = m^2 = 1$, $\lambda = 1/4$, so $d = 3/8$. For each internal plaquette set its Maxwell weight γ_f to $1/d$ times normal dual length divided by area, and set $\epsilon_e = c_e/d$. These are action quadratures, not causal counting weights.

With $\pi_i = v_i D_0 \psi_i$ and canonical real link momentum Π_e, the Hamiltonian is

$$\begin{aligned}\mathcal{H} = &\sum_i \frac{|\pi_i|^2}{v_i} + \sum_e \frac{\Pi_e^2}{2\epsilon_e} + \sum_{e=i\to j} c_e |e^{ia_e}\psi_j - \psi_i|^2 \\ &+ \sum_i [(v_i + b_i)|\psi_i|^2 + v_i|\psi_i|^4/4] + \tfrac{1}{2}\sum_f \gamma_f (\mathrm{d}a)_f^2. \end{aligned} \tag{13}$$

The complex scalar symplectic form is $2\,\mathrm{Re}(\mathrm{d}\overline{\pi}_i \wedge \mathrm{d}\psi_i)$. Differentiating this same energy gives

$$J_{i\to j} = -2c_e\,\mathrm{Im}(\overline{\psi}_i e^{ia_e}\psi_j), \qquad \rho_i = 2\,\mathrm{Im}(\overline{\psi}_i \pi_i), \qquad G_i = \rho_i + (\mathrm{div}\,\Pi)_i.$$

Here divergence is outward; $\dot\rho + \mathrm{div}\, J = 0$ and $\dot\Pi = J - \mathrm{d}^T\gamma \mathrm{d}a$ imply $\dot G = 0$.

Proposition A.4 (Local split evolution preserves Gauss constraints)**.** *Scalar kinetic drift, electric kinetic drift and every individual edge, onsite and plaquette potential kick preserve each G_i in ideal arithmetic. Their symmetric composition is symplectic and preserves the initial Gauss constraints. Disjoint local subflows commute.*

Proof. Scalar drift adds a real multiple of π_i to ψ_i, leaving ρ_i unchanged; electric drift changes only a. An edge kick changes the two charges by $-hJ_e, +hJ_e$ and Π_e by hJ_e. Onsite kicks add a real multiple of ψ_i to π_i. A plaquette kick adds a curl transpose to Π, whose divergence vanishes. Each map is its Hamiltonian subflow; disjoint subflows have disjoint dynamical variables. The potential kicks also commute because their coordinates are fixed during the kicks. □

Finite current and feedback control. The executable packet uses 64 scalar sites, 144 internal links and 108 plaquettes, with two steps of declared model duration 1/200. It compares the real preparation $\psi_i = 1/5$, $\pi_i = a_e = \Pi_e = 0$ with a local phase intervention $(3 + 4i)/5$ and a genuine gauge copy, which also changes the incident unwrapped connections. The intervention leaves charge and Gauss constraints unchanged but changes the edge current. At the selected edge $c_e = 1/4$ exactly, so $J_e = 2/125$ and its first electric half-kick is 1/25000. The gauge copy preserves invariant readouts. Complete operation records bind every dynamic read to its version's actual writer; decoded checkpoints contain scalar intensity, charge, edge current and electric feedback. Independent high-precision replay checks these binary64 records, including a nonempty off-influence control. This numerical agreement is separate from the exact local identities and gives no continuous-history error enclosure. The prepared addresses, local execution law, action, boundary and model tick remain supplied. No causal-count clock attachment, inherited fine-grid detector error, selected physical current or full Standard Model quantum evolution is asserted.

A.3 A continuous charged-current and separated-detector bound

The same finite Hamiltonian admits a useful continuous-time comparison without increasing the charged execution. Work with its exact golden weights and exact initial scalar magnitude $s_0 = 1/5$, zero scalar and electric momenta, and zero initial connections. Compare the baseline with the phase intervention $(3+4i)/5$ at site 21. The distinct detector site 37 is joined to it by the oriented edge of conductance $c = 1/4$. The reference gauge copy is the exact gauge transform of the baseline. Times are the declared dimensionless action parameter.

Theorem A.5 (Resolved nonlinear charged readout)**.** *The exact classical solutions exist on* $0 \le t \le T = 1/100$ *within the principal-log chart. On the selected edge their intervention current satisfies*

$$J(t) \ge 0.015110268, \qquad \Pi(T) > 0.0001543.$$

Writing $I_i = |\psi_i|^2$*, the spatially distinct detector obeys*

$$I_{37}^{\text{intervention}}(t) - I_{37}^{\text{baseline}}(t) \le -\frac{9}{20}t^2 \qquad (0 < t \le T).$$

Thus the response magnitude is at least 0.000045 *at* T *and at least* 0.00001125 *throughout* $[1/200, 1/100]$*. For the nine initial/completed decoded intensity checkpoints across all three preparations, the exact continuous-reference error is less than* 0.000006 *per checkpoint.*

Proof. Set $S = 1/4$, $A_* = 10^{-5}$, $B = 88$, and $D = 13/250$. On the box $|\psi_i| \le S$, $|a_e| \le A_*$, differentiation of the Hamiltonian gives

$$|\ddot{\psi}_i| \le \frac{2S\sum_{e\ni i} c_e + (v_i + b_i + v_i S^2/2)S}{v_i} < B,$$
$$|\ddot{a}_e| \le \frac{2c_e S^2 + 4A_*\sum_{f\ni e}\gamma_f}{\epsilon_e} < D.$$

Exact rational enclosures of the golden weights verify every inequality and $\sum_{f\ni e}\gamma_f < 78$. Since the initial velocities vanish, $|\psi_i(t)-\psi_i(0)| \le Bt^2/2$ and $|a_e(t)| \le Dt^2/2$. Both bounds lie strictly inside the box through T; a first-exit argument and local ODE continuation prove existence there. Also $|(4/3)(\mathrm{d}a)_f| \le (16/3)A_* < 1 < 2\pi$, preserving the chart. Gauge covariance transfers the invariant bounds to the ideal gauge copy.

The current product estimate gives

$$|J(t) - 2/125| \le C_2 t^2 + C_4 t^4, \quad C_2 = \tfrac{1}{4}(2s_0 B + s_0^2 D) = 8.80052, \quad C_4 = B^2/8 = 968.$$

The magnetic force is at most $2D(78)t^2$. Integrating $\dot{\Pi} = J - \mathrm{d}^T\gamma \mathrm{d}a$ therefore bounds $|\Pi(t) - (2/125)t|$ by $(C_2 + 156D)t^3/3 + C_4 t^5/5$, yielding the stated current and feedback.

For the detector, let $u = \sum_{e\ni 37} c_e/v_{37}$; its boundary stiffness vanishes. The cubic map $z \mapsto |z|^2 z$ is $3S^2$-Lipschitz on the complex disk. The phase estimate $|e^{ia} - 1| \le |a|$ consequently gives

$$|\ddot{\psi}_{37}(t) - \ddot{\psi}_{37}(0)| \le Et^2/2, \quad E = u(2B + s_0 D) + (1 + 3S^2/2)B < 30759.$$

The quadratic Taylor remainder is at most $r = 30759t^4/24$. Put $v = v_{37}$ and

$$a_0 = -51/250, \qquad a_1 = a_0 - 1/(50v) + i/(25v).$$

These are the exact initial accelerations of the two preparations. Each intensity differs from $|s_0 + a_j t^2/2|^2$ by at most $2(s_0 + Bt^2/2)r + r^2$. Subtraction gives a quadratic coefficient $s_0 \operatorname{Re}(a_1 - a_0)$ and quartic coefficient $(|a_1|^2 - |a_0|^2)/4$. Exact rational bounds on v show that their sum with both remainders is at most $-(9/20)t^2$ for $0 < t \le T$.

Finally, decode ψ_{37} from the actual versioned reads at each initial or completed checkpoint. Interpret every stored binary64 component as its exact dyadic value. Its intensity discrepancy from the appropriate quadratic Taylor expression, plus the preceding analytic remainder, is an exact bound against the continuous IVP. The independent consumer checks all nine values, each below 6×10^{-6}, and excludes the intermediate split kick. It replays the parent custody and operations; copied public frames cannot replace the read events. □

This is a nonlinear classical current and readout result on the supplied family. The checkpoint bound does not enclose every state component or every intermediate observer operation. The exact reference preparation is distinct from its binary64 encoding. Quantum evolution, full fermion generation, a physical clock and native selection of the action or population are not supplied by this estimate.

A.4 Fermionic hypercharge transport on prepared source addresses

Fix the same 64 prepared golden addresses and 144 oriented nearest-neighbour links as the scalar execution. Give each address two Spin components of each of the five registered matter multiplets. Their internal multiplicities and integer charges are

$$(d_s) = (6, 3, 3, 2, 1), \qquad (q_s) = 6(Y_s) = (1, -4, 2, -3, 6).$$

This finite model supplies its spatial hopping action, canonical anticommutation relations, state preparation and model step. The source addresses do not select those inputs. The colour and weak indices are retained as multiplicities of an abelian transport law; the model does not execute the complete nonabelian or Yukawa dynamics.

Let $c_{s,a,v,\alpha}$ be the annihilation operators on the finite fermion Fock space, where $a = 1, \dots, d_s$ and $\alpha = 1, 2$. They satisfy $\{c_i, c_j^\dagger\} = \delta_{ij}$ and $\{c_i, c_j\} = 0$. For an oriented link $e = (u, v)$ in direction j, a unit complex link U_e and a supplied real hopping weight κ_e, put

$$K_{s,e} = i\sigma_j U_e^{q_s}, \tag{14}$$

$$\widehat{H}_e = \kappa_e \sum_{s,a} (c_{s,a,u}^\dagger K_{s,e} c_{s,a,v} + c_{s,a,v}^\dagger K_{s,e}^\dagger c_{s,a,u}). \tag{15}$$

The Pauli matrices act on Spin, not on the internal multiplicity index. The sign and link orientation in this definition fix the current convention. If $U_e \mapsto g_u U_e \overline{g}_v$ and $c_{s,a,w} \mapsto g_w^{q_s} c_{s,a,w}$, the quadratic expression is unchanged. The link is held fixed during this matter update.

Occupy one normalized spatial–Spin orbital ψ_s for every internal index a. Distinct internal indices and multiplets are orthogonal, so their exterior product is a normalized fifteen-fermion Slater state. Its one-body density is the orthogonal projector

$$P = \bigoplus_s I_{d_s} \otimes |\psi_s\rangle\langle\psi_s|.$$

The complex orbital coefficients describe this fermionic state. They are not commuting replacements for the field operators. In particular $0 \le P \le 1$ and $\operatorname{Tr} P = 15$.

For one occupied orbital write $z = (\psi_{s,u}, \psi_{s,v})$ and

$$h_{s,e} = \kappa_e \begin{pmatrix} 0 & K_{s,e} \\ K_{s,e}^\dagger & 0 \end{pmatrix}.$$

For step $\delta > 0$, take the implicit midpoint equation

$$i\frac{z' - z}{\delta} = h_{s,e}\bar{z}, \qquad \bar{z} = \frac{z' + z}{2}. \tag{16}$$

Since $K_{s,e}$ is unitary, $h_{s,e}^2 = \kappa_e^2 I$. The update is the unitary Cayley transform $(I + i\delta h_{s,e}/2)^{-1}(I - i\delta h_{s,e}/2)$. It preserves the projector property and particle number, with $P \mapsto VPV^\dagger$, and is exact over Gaussian rationals when the link, step, weight and preparation lie in that field.

Proposition A.6 (Variational current and mean Gauss preservation)**.** *Define the mean site charge and midpoint current by*

$$\rho_w = \sum_s d_s q_s \|\psi_{s,w}\|^2, \tag{17}$$

$$J_e = -2\kappa_e \sum_s d_s q_s \operatorname{Im}(\bar{\psi}_{s,u}^\dagger K_{s,e} \bar{\psi}_{s,v}). \tag{18}$$

Then J_e is the derivative of the quadratic midpoint functional of (15) under $U_e \mapsto e^{i\theta} U_e$, with the midpoint orbitals fixed. These averages need not be normalized occupied orbitals; only the step endpoints represent the normalized fifteen-fermion Slater states. The charge changes satisfy

$$\rho_u' - \rho_u = -\delta J_e, \qquad \rho_v' - \rho_v = \delta J_e.$$

Let E_e be a classical link momentum and update $E_e' = E_e - \delta J_e$. With div *positive at the initial endpoint of each oriented link, every site residual $G_w = (\operatorname{div} E)_w - \rho_w$ is unchanged. This remains true for any finite composition of these edge factors.*

Proof. For the finite canonical anticommutation relations, $[c_i^\dagger c_j, c_k^\dagger c_l] = \delta_{jk} c_i^\dagger c_l - \delta_{li} c_k^\dagger c_j$; the degree-four terms cancel with their fermionic signs. Thus the quadratic Hamiltonian acts through its one-body matrix. Differentiating its link phase at fixed state gives the displayed J_e. The identity

$$\|x'\|^2 - \|x\|^2 = 2\,\mathrm{Re}\left(\frac{x'+x}{2}\right)^\dagger (x'-x)$$

and (16) give opposite endpoint transfers. The electric update cancels those changes in G_u, G_v; all other site charges and incident momenta are unchanged. Induction proves the finite composition statement. Gauge covariance follows by conjugating the one-body matrices and their Cayley transforms. □

Write $U_e = e^{i\theta_e}$. The same first-order action is

$$S_e = \int \left(i \sum_s d_s \psi_s^\dagger \dot{\psi}_s + E_e \dot{\theta}_e - \langle \widehat{H}_e \rangle \right) dt.$$

Its variations give the orbital equation, $\dot{E}_e = -\partial_{\theta_e} \langle \widehat{H}_e \rangle$ and $\dot{\theta}_e = 0$, since this factor has no E_e dependence in its Hamiltonian. Thus frozen links and electric feedback belong to one declared action factor. The midpoint scheme discretizes these equations. Exact discrete conservation does not identify its step with the continuous Hamiltonian exponential or prove a trajectory-error estimate. The electric momentum is driven by a fermionic expectation: this is a semiclassical update, not a quantized gauge field coupled to fermion Fock space. Electric drift, plaquette and Yukawa factors are absent.

A sharp quantum-constraint boundary. At every address choose $\psi_s = (3/40, 1/10)$ before the local phase intervention. Each orbital has squared site norm $1/64$ and unit global norm. Since $\sum_s d_s q_s = 0$, the initial mean charge vanishes at every site, and $E = 0$ solves the mean Gauss constraint. Nevertheless the local charge operator has variance

$$\mathrm{Var}(\widehat{Q}_w) = \sum_s d_s q_s^2 \frac{1}{64}\left(1 - \frac{1}{64}\right) = \frac{945}{512} > 0. \tag{19}$$

Each occupied internal orbital contributes an independent Bernoulli site occupation; this proves the formula. A site phase preserves it. Consequently the product preparation with sharp classical electric data is not annihilated by the quantum Gauss operator. In ordinary hypercharge units the variance is divided by 36. Vanishing mean residuals cannot certify a physical gauge-constrained quantum state.

Execution and interpretation. The executable construction in `code/sm_fermion_current/` retains the local orbital state, incident link ports, current readback and electric feedback as versioned software records. Its independent verifier rebuilds the charge dictionary, source addresses, Cayley updates, variational currents and consumed writer/value versions. Baseline, local phase intervention and gauge-related execution are separate controls. The generation's linear, cubic and mixed anomaly sums are representation identities; a finite canonical-anticommutation implementation does not construct a chiral continuum fermion measure.

This packet has one declared generation and supplied quantum preparation. It proves neither native source selection, all-matter Standard Model evolution, a controlled spatial or temporal continuum limit, laboratory current, a physical clock, nor compatibility with a sourced curved family. The preceding scalar reduction and its continuous-readout enclosure retain their own action and preparation; their error bounds do not transfer to this fermionic transport.

A.5 A flux-dressed fermion state satisfying abelian quantum Gauss

The mean-field obstruction has a precise remedy within a separately supplied quantum link model. It changes both the gauge-field Hilbert space and the preparation. Choose one oriented link (u, v) of the same golden carrier. Its electric space is $\ell^2(\mathbb{Z})$, with

$$\widehat{E}|n\rangle = n|n\rangle, \qquad \widehat{U}|n\rangle = |n+1\rangle, \qquad [\widehat{E}, \widehat{U}] = \widehat{U}.$$

The ambient flux space is infinite. A finite reachable support is computed from the preparation; no cyclic wrap or truncated shift replaces this unitary operator.

Index the fifteen internal channels by a, retaining their charges q_a. In each channel occupy the normalized left Spin orbital $\chi = (3/5, 4/5)$ and leave its orthogonal orbital empty. The active right orbital is $-i\sigma_j\chi$, so the Pauli link maps it back to χ. The two active modes have annihilators l_a, r_a. Fix the canonical fermion order $(l_1, r_1, l_2, r_2, \ldots)$ and set

$$\widehat{h}_a = \kappa_a \left(l_a^\dagger \widehat{U}^{q_a} r_a + r_a^\dagger \widehat{U}^{-q_a} l_a \right). \tag{20}$$

Each channel retains exactly one fermion. Distinct channel Hamiltonians commute: their even fermion bilinears have disjoint mode support and the flux shifts commute. Initially all fifteen fermions are at u, the right modes are empty, and the link is in $|0\rangle$. Since $\sum_a q_a = 0$, this preparation is locally neutral.

Proposition A.7 (Exact finite-support abelian quantum Gauss state)**.** *Let $\widehat{Q}_u = \sum_a q_a l_a^\dagger l_a$ and $\widehat{Q}_v = \sum_a q_a r_a^\dagger r_a$. Every $\widehat{h}_a$ commutes with*

$$\widehat{G}_u = \widehat{E} - \widehat{Q}_u, \qquad \widehat{G}_v = -\widehat{E} - \widehat{Q}_v.$$

Its Cayley transform preserves their joint kernel. Starting from the neutral preparation, the product of these transforms is the normalized state

$$|\Psi\rangle = \sum_{x \in \{0,1\}^{15}} \left[\prod_a A_a^{1-x_a} (-iB_a)^{x_a} \right] |x\rangle_F \otimes \left| -\sum_a q_a x_a \right\rangle_E, \tag{21}$$

where $x_a = 1$ means that channel a occupies its right mode, and

$$\eta_a = \delta\kappa_a/2, \qquad A_a = \frac{1-\eta_a^2}{1+\eta_a^2}, \qquad B_a = \frac{2\eta_a}{1+\eta_a^2}.$$

Every branch, and hence the complete state, satisfies both quantum Gauss constraints. All other vertices retain zero charge and incident flux.

Proof. In $l_a^\dagger \widehat{U}^{q_a} r_a$, both left charge and electric flux increase by q_a, while right charge decreases by q_a. The adjoint reverses all three changes. This proves the two commutators on the invariant finite-support core of the electric operator. The Hamiltonians are bounded; their Cayley transforms preserve the joint Gauss sectors and the electric domain. On the one-fermion channel sector, $\widehat{h}_a^2 = \kappa_a^2 I$, which gives the displayed two coefficients. The interleaved fermion convention fixes the even-hop signs. Channel commutation gives the product expansion. Its orthogonal Fock branches have squared norms summing to $\prod_a (A_a^2 + B_a^2) = 1$. On branch x, $Q_v = \sum_a q_a x_a$, $Q_u = -Q_v$ and $E = -Q_v$. Both Gauss residuals therefore vanish individually; no cancellation of opposite residuals in an expectation is used. □

The span of these 2^{15} dressed branches is invariant under every $\widehat{h}_a$: each admissible hop changes the occupation and flux by the same signed charge. Thus the finite electric support is exact for the coupled evolution of this sector, although the standalone shift $\widehat{U}$ does not preserve it. Channel commutation also gives the continuous solution of the supplied Hamiltonian $\sum_a \widehat{h}_a$: replace A_a, B_a in (21) by $\cos(\kappa_a t), \sin(\kappa_a t)$. This is an exact finite-sector quantum history, not a time-step approximation.

For $\delta = 1/2$, choose $\kappa_a = 1$ except on the e^c channel, where $\kappa_a = 2$. These species weights are declared; they do not establish a universal physical propagation speed. The coefficients are $(A, B) = (15/17, 8/17)$ on the first fourteen channels and $(3/5, 4/5)$ on the last. All 2^{15} Fock branches occur, with electric support $\{-18, \ldots, 18\}$. The link-phase current operator is

$$\widehat{J}_a = iq_a\kappa_a \left(l_a^\dagger \widehat{U}^{q_a} r_a - r_a^\dagger \widehat{U}^{-q_a} l_a \right), \qquad \dot{\widehat{E}} = -\sum_a \widehat{J}_a.$$

Exact evaluation in (21) gives

$$\langle \widehat{E} \rangle = -\frac{18144}{7225}, \qquad \sum_a \langle \widehat{J}_a \rangle = \frac{47232}{7225}.$$

The sum of the factor-midpoint currents is instead $36288/7225$; multiplied by $-\delta$, it equals the electric change. A final-state current and an integrated discrete transfer are different readouts.

For the continuous solution with the same species weights,

$$\langle \widehat{E} \rangle_t = 6(\sin^2 t - \sin^2 2t), \qquad \langle \widehat{J} \rangle_t = -6 \sin 2t + 12 \sin 4t.$$

The electric derivative equals minus the current. At the supplied model time $t = \pi/6$ they give -3 and $3\sqrt{3}$, respectively, with electric variance $45/2$. The flux fluctuations are correlated with charge, so quantum Gauss holds throughout this history. The time parameter, action and preparation are inputs; this solution calibrates no physical clock.

The executable producer and independent verifier in `code/sm_fermion_current/quantum_link.py` and `code/sm_fermion_current/verify_quantum_link.py` bind the channel and Spin maps, CAR signs, complete branch population, flux distribution and current matrix elements. This is a supplied finite quantum state and action factor on one prepared link. It is not the uniform semiclassical Slater preparation, a measured quantum history, full gauge-field evolution, nonabelian Gauss constraints or Yukawa evolution, or a chiral continuum regulator. Native state and operation selection, source-causal attachment and laboratory realization remain additional requirements.

B A coupled fermion current in a declared hybrid gauge action

The fermionic current can be coupled to electric drift and magnetic plaquette forces on the same prepared carrier. The construction below declares a regulated Hamiltonian and a finite splitting word. Its gauge fields are classical, while its matter state belongs to finite fermion Fock space. This hybrid interpretation differs from quantizing the gauge field in the preceding one-link construction.

Take the direct-product gauge cover $SU(3) \times SU(2) \times U(1)$, with the five registered representations and integer hypercharges $q_s = 6Y_s$ in each of three declared families. The compact abelian link is $U_e = e^{i\theta_e}$ with momentum E_e. For fixed $M, \lambda > 0$, choose

$$T_\lambda(E) = \frac{2E}{\lambda} \arctan \frac{\lambda E}{2M} - \frac{2M}{\lambda^2} \log \left(1 + \frac{\lambda^2 E^2}{4M^2} \right), \tag{22}$$

$$W_f = \prod_{e \in \partial f} U_e^{\epsilon_{fe}}, \qquad V_f = \beta_f (1 - \operatorname{Re} W_f). \tag{23}$$

Here ϵ_{fe} is the oriented face boundary. The reduced Hamiltonian is

$$\mathcal{H}_{\mathrm{red}} = \sum_e T_\lambda(E_e) + \sum_f V_f + \sum_e \langle \Psi, \widehat{H}_{\mathrm{CAR},e}(U_e)\Psi\rangle,$$

where the hopping operators and Spin matrices are those defined above. In temporal gauge, its first-order action contains $\sum_e E_e\dot{\theta}_e + i\langle\Psi,\dot{\Psi}\rangle - \mathcal{H}_{\mathrm{red}}$. The link-phase variation of the same matter Hamiltonian defines its current.

This is a specified kinetic law. In particular,

$$T'_\lambda(E) = \frac{2}{\lambda}\arctan\frac{\lambda E}{2M}, \qquad T''_\lambda(E) = \frac{1}{M(1+(\lambda E/(2M))^2)} > 0.$$

Integrating $x - \arctan x = \int_0^x t^2/(1+t^2)\,dt$ gives

$$0 \le \frac{E^2}{2M} - T_\lambda(E) \le \frac{\lambda^2 E^4}{48M^3}.$$

The quadratic electric energy is therefore a controlled small-field comparison of functions. It is not the exact kinetic energy of this model, and the displayed bound is not a trajectory-error estimate. The parameter λ remains fixed when describing this action; varying it changes the law. The separate charge-one abelian Wilson term is specified on the direct-product cover. No descent of that term to the global central quotient is assumed.

Proposition B.1 (Invariant expectation-level matter sector)**.** *Extend the reduced Hamiltonian by classical nonabelian links and momenta, quadratic nonabelian electric energies, Wilson magnetic energies, the covariant Higgs kinetic and spatial terms, and a potential with vanishing first derivative at zero Higgs field. Couple the finite CAR matter through its registered representation matrices and all twenty-seven complex Yukawa coefficients of three families. Interpret every matter force as its Fock-state expectation in the canonical classical-field/quantum-matter action.*

Set the nonabelian links to identity, their electric momenta to zero, and the Higgs field and conjugate momentum to zero. In each species of one family take normal covariance $P_s = I_{d_s} \otimes |\psi_s\rangle\langle\psi_s|$, with no covariance between distinct species or families. Let the other two families be Fock vacuum, and take zero anomalous covariance $\langle c_i c_j\rangle = 0$. These conditions define an invariant hybrid subsystem with Hamiltonian $\mathcal{H}_{\mathrm{red}}$.

Proof. Every nonabelian matter charge or link-current variation inserts a traceless internal generator T_s. The remaining Spin and spatial factor is scalar in that internal index, so the expectation factors through $\operatorname{tr} T_s = 0$. This checks the omitted nonabelian matter forces, including each color and weak generator. The nonabelian electric velocities vanish at zero momenta, and their magnetic first variations vanish at identity plaquette holonomy. Scalar kinetic, spatial and gauge-current variations vanish at zero Higgs field and momentum.

In all-left-Weyl notation, the Yukawa terms have the gauge-neutral forms QHu^c, $Q\overline{H}d^c$ and $L\overline{H}e^c$, with the required Spin, color and weak contractions and their adjoints. Their Higgs first variation is a linear combination of anomalous expectations $\langle cc\rangle$ and $\langle c^\dagger c^\dagger\rangle$. Both vanish by fermion-number selection, for every value of every Yukawa coefficient. This checks the Higgs force rather than only the value of the Yukawa energy at $H = 0$.

The surviving hopping Hamiltonian is number conserving, family diagonal and scalar on each internal multiplicity space. It preserves the stated covariance form, the vanishing anomalous covariance and the two vacuum families. The field equations hence preserve every imposed condition. No stability of the zero-Higgs equilibrium is needed for this invariance statement. □

The expectation hypothesis is essential. It does not identify independent Grassmann components with one commuting orbital, and it does not annihilate the nonabelian Gauss operators. For example, in the initial uniform preparation $p = 1/64$, the weak generator $\sigma_z/2$ on the occupied quark and lepton doublets has local charge variance

$$\mathrm{Var}(\widehat{Q}_T(v)) = \sum_s \mathrm{tr}(T_s^2)p(1-p) = \frac{63}{2048} > 0.$$

The absent nonabelian mean source therefore cannot certify a full gauge-constrained quantum state with sharp zero nonabelian electric data.

Proposition B.2 (Coupled transport and conserved readout)**.** *On a finite oriented lattice with closed face boundaries, compose the following factors of the declared hybrid action:*

1. *a fermionic implicit-midpoint hopping factor, with its variational current and the same classical electric kick;*

2. *an electric factor of duration λ,*

$$U_e' = \frac{1 + i\lambda E_e/(2M)}{1 - i\lambda E_e/(2M)} U_e, \qquad E_e' = E_e;$$

3. *a magnetic factor of duration h on face f,*

$$E_e' = E_e - h\beta_f \epsilon_{fe} \operatorname{Im} W_f \quad (e \in \partial f), \qquad U_e' = U_e.$$

For Gaussian-rational initial data and rational parameters these maps are exact over Gaussian rationals. Each preserves the expectation Gauss residual at every vertex. Their recorded current and charge transfers satisfy exact discrete continuity under finite composition, and the construction is covariant under residual time-independent vertex gauge transformations.

Proof. The midpoint matter identity proved above gives opposite endpoint charge transfers and precisely the electric kick that cancels them in Gauss. The midpoint equations are applied to the orbital, link-angle and momentum variables of that same factor action; its angle is constant during the factor because its Hamiltonian has no electric-momentum dependence.

The electric factor is the exact flow of T_λ for the stated duration: exponentiating $i\lambda T_\lambda'(E) = 2i \arctan(\lambda E/(2M))$ gives the rational phase. It leaves charge and electric divergence unchanged. For the magnetic factor, $\partial_{\theta_e} V_f = \beta_f \epsilon_{fe} \operatorname{Im} W_f$. Its momentum kick is the oriented face boundary, whose vertex divergence vanishes. These facts prove Gauss preservation for every factor and hence for their composition. The unit-modulus rational phase, Hermitian Cayley matter transform and rational Wilson product also prove the asserted arithmetic closure. Abelian gauge transformations conjugate the matter updates, leave E and W_f invariant and commute with the electric drift. □

This factor word is a declared discrete transport. It is not the exact exponential of the summed Hamiltonian; exact Gauss conservation does not imply exact total-energy conservation or a continuous-trajectory error bound.

Finite execution and returned current. Use the same sixty-four prepared addresses, 144 oriented links and 108 plaquettes, with $\lambda = h = 1/2$ and $M = \beta = \kappa = 1$. A full matter sweep is followed by all electric drifts and all magnetic kicks. One subsequent matter factor reads the evolved link and momentum on edge 14. That final factor is a readout in a partial second word, not a second completed full-Hamiltonian time step.

Across baseline, local intervention, gauge copy and a disabled-drift control, the retained prefixes contain 2,444 parent events. Each of the four continuations has 255 events: 144 electric factors, 108 magnetic factors, one returning matter factor and two public readouts. The control leaves the link unchanged instead of applying electric drift; it is a counterfactual operation word, not another execution of the full declared action. In the intervention and gauge copy the return current lies in $[-0.005355160745, -0.005355160744]$, and the final edge momentum lies in $[0.002253305815, 0.002253305816]$. The difference from the disabled-drift return current obeys the exact outward enclosure

$$0 < \frac{242747271}{2^{40}} \leq J_{\text{return}} - J_{\text{disabled}} \leq \frac{30343409}{2^{37}}.$$

Thus the electric record changes the link that the later fermionic current actually consumes. Gauge-related executions agree on the invariant readouts.

The package `code/sm_fermion_current/` contains the producer `coupled.py` and independent verifier `verify_coupled.py`. They reconstruct the closed face boundaries, exact factor updates, current, Gauss residuals and consumed writer/value versions. The observer-like software patches retain local orbital state, incident ports, public records and electric feedback; declared parent labels cannot substitute for actual consumption. Local magnetic energies are exact rationals, with their large sum bounded by outward dyadic intervals.

The action, regulator, family population, prepared addresses, routing and time labels are supplied. The invariant reduction does not execute an excited Higgs or nonabelian sector, select a chiral continuum regulator, identify a laboratory current or physical clock, or establish compatibility with a sourced curved geometry. Those statements require their own common-action and physical realization maps.

B.1 Spin transport and the information needed to select exchange statistics

A one-particle spin representation and a many-particle exchange rule are different data. Likewise, a current obtained from a supplied fermionic action need not distinguish that action from a bosonic alternative on the preparations actually used. The following finite statements separate these two questions without assuming a relativistic field reconstruction.

Proposition B.3 (A spin lift does not select a two-particle exchange sector)**.** *Let $\mathcal{H}$ be a finite-dimensional complex Hilbert space with $\dim \mathcal{H} \geq 2$, and let $D : G \to U(\mathcal{H})$ be a supplied unitary representation. Suppose a central element $z \in G$ acts as $D(z) = -I$. On $\mathcal{H} \otimes \mathcal{H}$, let $\tau(v \otimes w) = w \otimes v$. Both eigenspaces*

$$\mathcal{H}_+ = \ker(\tau - I) = \operatorname{Sym}^2 \mathcal{H}, \qquad \mathcal{H}_- = \ker(\tau + I) = \bigwedge^2 \mathcal{H}$$

are nonzero invariant subspaces of $D \otimes D$. The central element acts as $+I$ on both. For every nonzero $v \in \mathcal{H}$, the vector $v \otimes v$ is a nonzero member of $\mathcal{H}_+$, whereas its antisymmetric projection is zero. Thus the one-particle spin action, including its central minus sign, does not by itself determine the exchange sign or exclude two particles in the same one-particle mode.

Proof. For any linear operator A on $\mathcal{H}$,

$$\tau(A \otimes A) = (A \otimes A)\tau,$$

as is checked on simple tensors, which span the tensor product. Hence the orthogonal projections $\Pi_\pm = (I \pm \tau)/2$ commute with $D(g) \otimes D(g)$ for every g. The restrictions are therefore unitary representations. Since $(-I) \otimes (-I) = I$, they have the stated central action. If $v \neq 0$, then $\|v \otimes v\| = \|v\|^2 > 0$ and $\tau(v \otimes v) = v \otimes v$, so $\Pi_-(v \otimes v) = 0$. For orthonormal u, v, the unit vector $(u \otimes v - v \otimes u)/\sqrt{2}$ belongs to $\mathcal{H}_-$. Both sectors therefore exist for the same supplied spin representation, but they disagree on the doubled-mode state. □

This statement applies, in particular, to a supplied two-dimensional unitary realization of a binary rotation group whose central element is $-I$. A nonsplit double cover or a unique spin structure does not remove the displayed choice of two-particle sector. The statement does not construct a relativistic bosonic spinor field satisfying locality and a positive-energy spectrum condition, and it is not a counterexample to a relativistic spin–statistics theorem. It establishes only the insufficiency of the finite one-particle representation data just specified.

Proposition B.4 (Conserved single-occupancy channels cannot select statistics)**.** *Let*

$$\mathcal{K} = \bigoplus_{\lambda \in \Lambda} \mathcal{K}_\lambda$$

be a finite orthogonal decomposition into one-particle channel spaces. In both the bosonic and fermionic Fock spaces over $\mathcal{K}$, retain the sector with exactly one particle in every channel. Use the same initial normalized channel vectors ψ_λ, the same classical registers, and the same ordered finite sequence of the following operations:

1. *evolution by a self-adjoint channel-preserving quadratic Hamiltonian, or second quantization of a specified channel-preserving one-particle unitary;*
2. *readout of expectations of channel-preserving one-body operators;*
3. *classical register updates whose values are determined by those readouts and the retained classical registers.*

The two statistics give identical one-body densities, all such readouts, and every resulting classical register value. In particular, an expectation current, mean Gauss residual, and classical feedback history formed from these data cannot determine the exchange statistics.

Proof. Order the finite channel set once. On the algebraic tensor product define

$$J_\pm\Big(\bigotimes_\lambda v_\lambda\Big) = \prod_\lambda a_\pm^\dagger(v_\lambda)\,|0\rangle,$$

where $+$ denotes the bosonic and $-$ the fermionic creation operators, and the product follows the fixed order. Orthogonality of the channel spaces and the canonical relations show that the inner product of two such vectors is $\prod_\lambda \langle v_\lambda, w_\lambda\rangle$ in both cases. The maps therefore extend to unitary identifications between $\bigotimes_\lambda \mathcal{K}_\lambda$ and the respective fixed-channel-occupation sectors.

For a one-body operator B_λ supported in channel λ, its number-conserving second quantization obeys

$$J_\pm^*\, d\Gamma_\pm(B_\lambda)\, J_\pm = I \otimes \cdots \otimes B_\lambda \otimes \cdots \otimes I.$$

In the fermionic calculation, the signs from moving annihilation and creation through the preceding occupied channels cancel. In the bosonic calculation there is no sign, and occupation one gives the same matrix coefficient. Consequently the restriction of a channel-diagonal quadratic Hamiltonian $d\Gamma_{\pm}(\bigoplus_{\lambda} h_{\lambda})$ is the same sum of tensor-slot Hamiltonians. The restrictions of second-quantized channel-preserving unitaries are likewise the same tensor products. Starting from the same tensor-product vector therefore gives the same channel vectors after every such operation. Their common one-body density is

$$P = \bigoplus_{\lambda} |\psi_{\lambda}\rangle\langle\psi_{\lambda}|, \qquad \langle d\Gamma_{\pm}(B)\rangle = \operatorname{Tr}(PB)$$

for channel-diagonal B. Equal classical inputs select equal subsequent operations and equal readouts give equal classical updates. Induction on the operation sequence proves the assertion, including adaptive mean-field feedback. A channel declared to remain empty contributes its vacuum factor and zero one-body density in both constructions. □

For the finite hypercharge-current construction, take $\lambda = (s, a)$ with $a = 1, \ldots, d_s$ and $\mathcal{K}_{\lambda} = \mathbb{C}^{64} \otimes \mathbb{C}^2$, the prepared site and Spin coordinates. There is one occupied orbital in each of the fifteen channels. Its abelian hopping and phase interventions act within each channel. Replacing the supplied CAR Fock state by the bosonic number state with one particle in each of the same channels preserves

$$P = \bigoplus_{s} I_{d_s} \otimes |\psi_s\rangle\langle\psi_s|, \qquad 0 \le P \le I, \qquad \operatorname{Tr} P = 15.$$

The projector property in this preparation is therefore not itself a statistics discriminator. The multiplicity index is an orthogonal one-particle label; repeating the same spatial–Spin orbital across different multiplicity indices does not prepare two particles in one complete one-particle mode.

The local Cayley updates use the same one-particle matrices in the two constructions. This comparison second-quantizes each one-particle Cayley unitary; it does not replace it by the Cayley transform of the full many-body Hamiltonian. The midpoint orbitals, and hence the midpoint variational functional and its current, also agree algebraically. Those midpoint averages are not normalized intermediate Fock states. Therefore the charge transfers, electric kicks, and expectation Gauss residuals coincide. The subsequent electric drifts and Wilson plaquette kicks use only the common classical registers, so they preserve this equality before the next matter factor reads the evolved link.

The retained site-charge variance supplies no distinction either. A single particle in channel (s, a) has site probability $p_{s,v} = \|\psi_{s,v}\|^2$, and its site-number projector squares to itself for both statistics. The product over channels gives

$$\operatorname{Var}(Q_v) = \sum_{s} d_s q_s^2 p_{s,v}(1 - p_{s,v})$$

in both states. The nonzero variance excludes an operator Gauss constraint with the specified classical electric numbers. Its equality does not turn that constraint into a consequence of mean Gauss.

These identical histories concern the executed abelian, channel-preserving restriction. They do not assert equivalence of the full nonabelian or Yukawa theories, of channel-mixing evolutions, or of preparations with two particles in one conserved channel. Nor do they change the validity of the CAR current calculation under its stated assumptions. They show why that current calculation alone does not select those assumptions. An exchange-sensitive comparison must specify additional preparation and readout operations outside this restriction, together with their source realization.

A finite comparison of supplied alternatives is distinct from deriving physical spin–statistics from observer consistency.

Proposition B.5 (A two-particle port discriminator)**.** *Let L, R be orthonormal modes of one declared channel, let $a = 3/5$, $b = 4/5$, and supply the one-particle unitary*

$$U = \begin{pmatrix} a & -ib \\ -ib & a \end{pmatrix}.$$

Prepare $\Psi_\pm = (L \otimes R \pm R \otimes L)/\sqrt{2}$ and apply $U \otimes U$. For the coincidence effect

$$F = |LR\rangle\langle LR| + |RL\rangle\langle RL|,$$

the antisymmetric and symmetric preparations have probabilities

$$p_- = 1, \qquad p_+ = (a^2 - b^2)^2 = \frac{49}{625}.$$

A distinguishable-particle preparation $L \otimes R$ gives $p_{\mathrm{dist}} = a^4 + b^4 = 337/625$. All three preparations have the same summed one-particle density I before and after the operation, so this contrast requires a two-particle readout.

Proof. The matrix is unitary because $a^2 + b^2 = 1$, and its determinant is one. Direct expansion gives

$$(U \otimes U)\Psi_- = \Psi_-, \qquad (U \otimes U)\Psi_+ = (a^2 - b^2)\Psi_+ - i\sqrt{2}ab\,(LL + RR).$$

The three vectors Ψ_+, LL, RR are orthonormal, so the bosonic coincidence probability is $(a^2 - b^2)^2$, with probability $2a^2b^2 = 288/625$ at each bunched port. The antisymmetric output has no bunched component. For the distinguishable product input the coincidence amplitudes are a^2 and $-b^2$ in orthogonal labelled outcomes; their squared moduli add to $337/625$. Its bunching probability at each port is $a^2b^2 = 144/625$. Finally the two orthogonal occupied orbitals give summed one-particle density I, and $UIU^\dagger = I$. □

The port labels denote a supplied invariant channel basis. They do not assert that the two physical ports carry identical raw spin-coordinate vectors: a spin-dependent hopping matrix can use different local basis vectors. The rational coefficients agree with a finite hopping factor, but this discriminator supplies its own two-particle preparation and coincidence effect. It inherits no electric flux dressing, operator Gauss constraint, clock identification or laboratory instrument from a one-particle-per-channel history.

Indistinguishability is essential. If the particles carry normalized unobserved internal vectors α, β, unchanged by the port unitary, write $\eta = |\langle\alpha, \beta\rangle|^2$. Expanding the normalized symmetrized or antisymmetrized input and tracing over those vectors gives

$$p_{\mathrm{fermion}} = \frac{337 + 288\eta}{625}, \qquad p_{\mathrm{boson}} = \frac{337 - 288\eta}{625}.$$

The cross term is the product $\langle\alpha, \beta\rangle\langle\beta, \alpha\rangle$; its sign is the exchange sign. Orthogonal hidden states give $\eta = 0$ and erase the contrast. The ideal values above require $\eta = 1$.

There is also a conditional stability bound. Let ρ be either ideal output density and $\widetilde{\rho}$ an actual normalized output density with $\|\widetilde{\rho} - \rho\|_1/2 \le \epsilon$. If an actual effect $0 \le \widetilde{F} \le I$ satisfies $\|\widetilde{F} - F\| \le \delta$, then

$$\left|\mathrm{Tr}(\widetilde{\rho}\widetilde{F}) - \mathrm{Tr}(\rho F)\right| \le \epsilon + \delta.$$

Indeed, split the difference into $\mathrm{Tr}((\widetilde{\rho}-\rho)F)$ and $\mathrm{Tr}(\widetilde{\rho}(\widetilde{F}-F))$. The positive and negative parts of the trace-zero density difference each have trace at most ϵ, so the first term has magnitude at most ϵ; the second is bounded by δ. A preparation trace-distance bound transfers unchanged through an exactly common unitary; imperfect dynamics instead require a bound on the actual output. For a common error budget $\epsilon+\delta<288/625$, the two ideal-model probability intervals are disjoint because their separation is $576/625$. These are supplied error bounds, not a calibration or a bound on finite sampling error. An observed test additionally needs authenticated preparation, indistinguishability, operation and readout evidence.

B.2 Conditional exclusion from a stable generator and a scalar deformation law

One can select exclusion without assuming it, provided the quantization class and the relation between occupation and dynamics are specified. The following theorem permits a real scalar deformation parameter rather than declaring either canonical commutation or anticommutation relations. Its phase-generator and stability hypotheses are additional physical inputs; a finite spin representation alone supplies neither.

Theorem B.6 (Stability selects exclusion within a scalar deformation class)**.** *Let $\mathcal{K}$ be a finite-dimensional complex one-particle Hilbert space. On a positive Hilbert space $\mathcal{F}$, let $v\mapsto a^\dagger(v)$ be complex-linear, with $a(v)$ its adjoint on a common dense invariant domain $\mathcal{D}$. Assume $\mathcal{D}$ is invariant under these operators and a self-adjoint Hamiltonian H, and is contained in their domains. Supply:*

1. *a common real number q such that, for every unit v,*

$$a(v)a^\dagger(v)-q\,a^\dagger(v)a(v)=I$$

   on $\mathcal{D}$;

2. *the exact occupation/phase law, for every unit v,*

$$N_v=a^\dagger(v)a(v),\qquad [N_v,a^\dagger(v)]=a^\dagger(v);$$

3. *a normalized $\Omega\in\mathcal{D}$ with $a(v)\Omega=0$ for all v and $H\Omega=E_0\Omega$;*

4. *a unit mode v_- and $\kappa>0$ such that*

$$[H,a^\dagger(v_-)]=-\kappa a^\dagger(v_-)$$

   on $\mathcal{D}$, and a finite uniform lower bound for H.

Then $q=-1$, and on $\mathcal{D}$,

$$\{a(v),a^\dagger(w)\}=\langle v,w\rangle I,\qquad \{a^\dagger(v),a^\dagger(w)\}=0,\qquad \{a(v),a(w)\}=0.$$

Here the inner product is conjugate-linear in its first argument. In particular $(a^\dagger(v))^2=0$ for every mode v. The operators extend boundedly to $\mathcal{F}$, where the same relations hold. The reference vacuum Ω is not assumed to be a ground state.

Proof. Fix a unit mode, abbreviate its operators to $a,a^\dagger,N$, and work throughout on the common invariant domain. The deformation law gives

$$Na^\dagger=a^\dagger+q\,a^\dagger N.$$

Comparison with the occupation/phase law yields $(q-1)a^\dagger N = 0$. The vector $|1\rangle = a^\dagger\Omega$ has norm one and $N|1\rangle = |1\rangle$. If $q \neq 1$, it follows that $a^\dagger|1\rangle = 0$. Taking its squared norm using the deformation law gives $0 = 1 + q$. Thus q is either 1 or -1.

If $q = 1$, induction using $aa^\dagger = I + a^\dagger a$ gives

$$|n\rangle = (a^\dagger)^n\Omega, \qquad \| |n\rangle\|^2 = n!, \qquad N|n\rangle = n|n\rangle$$

for every integer $n \geq 0$. Apply this to v_-. The Hamiltonian commutator and the vacuum eigenvalue imply $H|n\rangle = (E_0 - n\kappa)|n\rangle$. All these vectors are nonzero and belong to $\mathcal{D}$, contradicting the lower bound. This is an arbitrary-n argument, not a finite occupation scan. Hence $q = -1$.

For any unit mode, the phase identity then gives $a^\dagger N = 0$, whereas the deformation law gives $Na^\dagger = a^\dagger$. Consequently

$$0 = a^\dagger N a^\dagger = (a^\dagger)^2.$$

Linearity and scaling extend this square-zero identity to every v, including $v = 0$. Expanding it at $v + w$ proves creation anticommutation, and taking adjoints on the dense invariant domain proves annihilation anticommutation. The diagonal relation is $\{a(v), a^\dagger(v)\} = \|v\|^2 I$ for every v. Polarization at $v + w$ and $v + iw$ gives the mixed relation. Finally, for $\psi \in \mathcal{D}$,

$$\|a(v)\psi\|^2 + \|a^\dagger(v)\psi\|^2 = \|v\|^2\|\psi\|^2.$$

Both operators therefore have bounded extensions, and density extends their identities to all of $\mathcal{F}$. □

Corollary B.7 (Finite-dimensional alternative to the stability hypothesis)**.** *Retain the theorem's common scalar deformation law, linear mode fields, exact occupation/phase law and normalized annihilation vacuum, with $\mathcal{K} \neq \{0\}$. If $\mathcal{F}$ itself is nonzero and finite-dimensional, the same exclusion and CAR conclusions hold without assuming a Hamiltonian or a negative ladder.*

Proof. A dense linear domain in finite dimension is the whole space. The vacuum argument gives $q \in \{1, -1\}$. If $q = 1$, the trace of $aa^\dagger - a^\dagger a = I$ is both zero and $\dim \mathcal{F} > 0$, a contradiction. The square-zero and polarization arguments for $q = -1$ require no Hamiltonian. □

This alternative requires finite dimension of the Hilbert space carrying the full creation and annihilation algebra. A finite one-particle mode space alone is insufficient. A finite observer response space does not by itself identify that operator representation with physical matter. Applying the corollary to observer finite capacity requires precisely this additional identification, as well as the stated deformation and phase laws.

The negative-mode hypothesis has a concrete one-particle instance in the supplied finite matter hopping action. For a unitary spin-link matrix K and $\kappa > 0$, its edge matrix is

$$h_e = \kappa \begin{pmatrix} 0 & K \\ K^\dagger & 0 \end{pmatrix}, \qquad h_e^2 = \kappa^2 I.$$

For any unit spin vector χ, the vector $v_- = (-K\chi, \chi)/\sqrt{2}$ is normalized and satisfies $h_e v_- = -\kappa v_-$. The quantum dynamical law $[H, a^\dagger(v)] = a^\dagger(h_e v)$ would therefore supply the required ladder. This law is a hypothesis linking the one-particle action to its quantization. A one-particle matrix, a measured unitary step, or the existence of a spin lift does not establish that law.

There is a finite nonvacuity witness. In ordinary fermionic Fock space over the edge modes, take the empty vacuum and $H = d\Gamma(h_e)$ with its normal ordering fixed at that vacuum. The negative-energy modes can each be occupied once, and the ground energy is the sum of all negative

one-particle eigenvalues. Filling those modes gives the ground state; particles in positive modes and holes in filled negative modes both have excitation energy κ. For the two-spin edge with fifteen internal channels of the supplied matter census, there are thirty negative modes. Its ground energy is -30κ. With

$$(d_s) = (6, 3, 3, 2, 1), \qquad (q_s) = (1, -4, 2, -3, 6), \qquad \sum_s d_s q_s = 0,$$

the filled ground state has total hypercharge zero. Subtracting its constant energy gives a positive excitation Hamiltonian and preserves the field commutator. The reference empty vacuum and the filled ground state are different vectors.

For comparison, canonical bosonic Fock space on the same fixed edge action admits occupation n in one negative mode of every internal channel. These states have total hypercharge zero and energy $-15n\kappa$. Thus even the globally neutral sector has no uniform lower bound. Adding a chemical potential proportional to total hypercharge cannot repair this sequence. This neutrality statement does not impose local quantum Gauss constraints or nonabelian singlet constraints, and is not a claim about a gauge-invariant physical Hilbert space.

The hypotheses exclude several concrete alternatives. At fixed total particle number the bosonic Hamiltonian is bounded below, so stability across all occupation sectors is essential. Replacing the one-particle law by $h_e + cI$ with $c \geq \kappa$ stabilizes its full bosonic Fock Hamiltonian: its second-quantized change is cN_{total}, not a constant vacuum-energy subtraction. Positive number interactions can also bound the energy, while changing the negative ladder. Neither change is ruled out by the theorem. A positive spectral replacement such as $|h_e|$ likewise changes the supplied dynamics.

Even the phase law and stability together do not force exclusion. On a three-dimensional Hilbert space with basis $|0\rangle, |1\rangle, |2\rangle$, set

$$a^\dagger|0\rangle = |1\rangle, \qquad a^\dagger|1\rangle = \sqrt{2}|2\rangle, \qquad a^\dagger|2\rangle = 0.$$

Then $N = a^\dagger a = \operatorname{diag}(0, 1, 2)$ satisfies $[N, a^\dagger] = a^\dagger$, and $H = -\kappa N$ is bounded below, but $(a^\dagger)^2|0\rangle \neq 0$. This model violates the scalar deformation hypothesis: $aa^\dagger = \operatorname{diag}(1, 2, 0)$, whose middle entry requires $q = 1$ while the top entry then contradicts $aa^\dagger - qN = I$. A hard occupation cutoff or a more general exchange algebra is therefore a genuine alternative outside the stated class.

The theorem is a conditional Pauli-exclusion derivation within a specified quantization class. It is insensitive to spin, so it cannot attach half-integer spin to fermionic statistics or exclude a different quantization class for a different field. The common scalar deformation law, linear field structure, exact occupation/phase law, unrestricted ladder domain and stable physical generator remain supplied. The finite filled-sea construction supplies no continuum vacuum or relativistic locality theorem. Deriving these inputs and selecting the physical matter action from observer consistency remain separate requirements.

B.3 What finite source operations must supply for exclusion

Finite-dimensional response, unitary covariance and a quadratic occupation generator do not alone imply the scalar deformation law used in the stability theorem. A normalized operational interpretation can replace that law, but requires an exact identification between a field operator and an observer operation. The following constructions distinguish these requirements.

Proposition B.8 (Globally covariant finite occupation cutoff)**.** *Let $\mathcal{K} = \mathbb{C}^m$ with $m \geq 1$, and*

$$\mathcal{F}_{\leq 2} = \mathbb{C} \oplus \mathcal{K} \oplus \operatorname{Sym}^2 \mathcal{K}.$$

This positive Hilbert space has dimension $(m+1)(m+2)/2$; the two-mode case has dimension six and the four-mode spin edge has dimension fifteen. Compress the usual bosonic creation and annihilation operators to this total-occupation cutoff, writing $C(v) = P_{\le 2} b^\dagger(v) P_{\le 2}$ and $A(v) = C(v)^\dagger$. Then C is linear in v, the normalized empty vacuum is annihilated by all $A(v)$, and for every unit v,

$$N_v = C(v)A(v), \qquad [N_v, C(v)] = C(v).$$

For every genuine unitary U on $\mathcal{K}$,

$$\Gamma_{\le 2}(U) C(v) \Gamma_{\le 2}(U)^\dagger = C(Uv).$$

For any Hermitian one-particle h, its restricted second quantization $H = d\Gamma(h)$ obeys $[H, C(v)] = C(hv)$ and is bounded below. Nevertheless, $C(v)^2\Omega \neq 0$ for every nonzero v, and no scalar q satisfies $A(v)C(v) - qC(v)A(v) = I$ even for one fixed unit mode.

Proof. Annihilation lowers total occupation, so $C(v)A(v)$ is the restriction of the ordinary bosonic number operator $d\Gamma(|v\rangle\langle v|)$. It preserves total-occupation sectors and commutes with the cutoff projection. The ordinary one-particle commutator identities therefore survive compression. At total occupation two the creation operator vanishes, so both sides of the claimed raising commutators vanish there as well. The cutoff also commutes with every $\Gamma(U)$ and with $d\Gamma(h)$, proving full unitary covariance and the dynamical identity. The finite spectrum gives

$$H \ge 2 \min\{0, \lambda_{\min}(h)\} I.$$

The normalized one-mode vectors $|v\rangle$ and $|2v\rangle$ have number eigenvalues one and two, while $\|C(v)^2\Omega\|^2 = 2$. On $|v\rangle$ the proposed scalar law requires $2 - q = 1$, hence $q = 1$. On $|2v\rangle$ it instead requires $-2q = 1$. More generally its exact commutator defect is

$$A(v)C(v) - C(v)A(v) = I - (N_v + I)P_2,$$

where P_2 projects onto total occupation two. This state-dependent boundary term cannot be replaced by a scalar deformation parameter. □

For a supplied hopping matrix with eigenvalues $\pm\kappa$, the cutoff Hamiltonian has minimum -2κ and admits double occupation of its negative mode. Thus its covariance, exact quadratic phase law and stability coexist with failure of exclusion. The construction is a counterexample to this specified finite implication. It is not a model of the complete observer axioms: in particular a projective one-particle transport does not automatically lift to a projective transport on all occupation sectors, since $\Gamma(e^{i\theta} I) = e^{i\theta N_{\text{total}}}$ need not be scalar. No source production of the cutoff field operators is asserted.

Proposition B.9 (Normalized operations and an exact occupation generator)**.** *On a finite-dimensional positive Hilbert space, let C be a creation operator, $A = C^\dagger$, and $N = CA = A^\dagger A$. Assume*

$$0 \le N \le I, \qquad [N, C] = C.$$

Then $C^2 = 0$. Consequently, for a complex-linear family $v \mapsto C(v)$ satisfying these conditions for every unit v, one has $\{C(v), C(w)\} = 0$ for all v, w. No scalar deformation law is required.

Proof. Choose an orthonormal eigenbasis of N, with eigenvalues $\lambda_i \in [0, 1]$. The commutator identity gives

$$(\lambda_i - \lambda_j) C_{ij} = C_{ij}.$$

Every nonzero matrix entry therefore maps a zero-eigenvalue vector to a one-eigenvalue vector. A second application vanishes, so $C^2 = 0$. Scaling extends this conclusion from unit vectors to all modes, and expanding $C(v + w)^2 = 0$ gives the anticommutation identity. □

Under the same phase law the converse holds: if $C^2 = 0$, then $CC^\dagger C - C^2C^\dagger = C$ gives $CC^\dagger C = C$. Thus C is a partial isometry and a contraction. Within this phase-law class, contraction is equivalent to exclusion. Identifying an unscaled field operator with a normalized observer operation therefore carries the full selection content; it is not a harmless change of units.

This conclusion alone does not construct ordinary many-fermion Fock space. On $\mathbb{C}\Omega \oplus \mathcal{K}$, the operators $C(v) = |v\rangle\langle\Omega|$ satisfy the contraction and phase hypotheses for every unit v, but allow at most one particle in total. For $\dim \mathcal{K} \geq 2$, their mixed anticommutator is

$$\{C(v)^\dagger, C(w)\} = \langle v, w\rangle |\Omega\rangle\langle\Omega| + |w\rangle\langle v|,$$

which is not generally $\langle v, w\rangle I$. Additional structure is required to recover the full canonical field algebra.

For a linear family on m orthonormal modes, the exclusion test reduces to a finite set of exact operator identities. It suffices to check the contraction and unit phase conditions on every e_i and every $(3e_i + 4e_j)/5$ with $i < j$. The proposition gives $C(e_i)^2 = 0$; expanding the square at the pair vector then gives $\{C(e_i), C(e_j)\} = 0$. Linearity proves exclusion for every mode. To obtain full mixed CAR by the paired-instrument route, it suffices to check its completeness identity at the basis vectors, these real pairs, and the imaginary pairs $(3e_i + 4ie_j)/5$. The two pair identities force the symmetric and antisymmetric cross terms to vanish. These are exact matrix identities, not conclusions from finitely sampled outcome frequencies.

Basis-mode checks alone are insufficient. Set $\sigma_+ = |1\rangle\langle 0|$. Two commuting hard-core bit raisers $C_1 = \sigma_+ \otimes I$, $C_2 = I \otimes \sigma_+$ each satisfy the contraction and phase conditions, but $C = (3C_1 + 4C_2)/5$ has $C^2|00\rangle = (24/25)|11\rangle \neq 0$ and fails the unit phase law. Inserting the fermionic sign on the second raiser changes no occupation-basis transition probability. Coherent superposition tests, or source-derived identities that imply them, must therefore carry the relative-sign information.

The bound $N \leq I$ follows if the unscaled removal operator A is itself a Kraus operator of a trace-nonincreasing observer operation. Equivalently, A is a contraction; its adjoint C is a contraction too. This is an operational sufficient condition for exclusion only after the same operator's quadratic form $A^\dagger A$ is identified with the exact unit-step occupation generator. A normalized probability effect and a particle-number operator are not interchangeable merely because both are positive.

A stronger operational interface supplies the mixed relation as well. If a complete two-outcome instrument has exactly the adjoint-paired Kraus operators $A(v), C(v)$ for each unit v, trace preservation gives

$$C(v)A(v) + A(v)C(v) = I.$$

With the exact phase law and linearity, the preceding proposition and complex polarization recover all canonical anticommutation relations. The adjoint pairing, exhaustive two-outcome list and normalization for every input state are essential hypotheses; they are not consequences of probability normalization alone.

The cutoff example demonstrates the distinction. For a unit mode its operator $N_v + A(v)C(v)$ has eigenvalues at most three. Consequently

$$E_{\text{remove}} = \tfrac{1}{4}N_v, \qquad E_{\text{add}} = \tfrac{1}{4}A(v)C(v), \qquad E_{\text{idle}} = I - \tfrac{1}{4}(N_v + A(v)C(v))$$

are positive and sum to identity. Kraus operators $A(v)/2$, $C(v)/2$ and $\sqrt{E_{\text{idle}}}$ define a normalized instrument which retains the possibility of double occupation. Its removal effect is $N_v/4$, not the occupation generator N_v. In particular the phase commutator with that effect has coefficient $1/4$, not one. The extra idle outcome and the operator normalization cannot be suppressed in transferring an instrument to the exclusion theorem.

Projective readout has a different algebraic role. A Lüders outcome with Kraus operator $P = P^\dagger = P^2$ has $[PP^\dagger, P] = 0$, so a nonzero projector cannot satisfy the unit-step creation law. For the binary coordinate effects, an alternative pair $|1\rangle\langle 0|$, $|0\rangle\langle 1|$ is adjoint-paired, complete and satisfies the single-mode exclusion laws, but swaps the outcome-certain states rather than preserving them. The declared phase instrument and its swap-twisted comparison therefore illustrate different operations, despite carrying the same outcome effects. Their matrix-level semantics do not select a native particle-creation instrument. The distinction is explicit in `Lean/EventAlgebra/LuedersPhaseInstrument.lean` and `Lean/EventAlgebra/SourcePhaseInstrumentOutcomeBridge.lean`.

There is a separate obstruction in the finite history-count interface of `Lean/QFT/SourceHistoryGNSDynamics.lean`. Its eight three-bit histories carry the repair count

$$Q(x_0, x_1, x_2) = \mathbf{1}_{x_0 \neq x_1} + \mathbf{1}_{x_1 \neq x_2}.$$

The eigenspaces of its diagonal operator have dimensions two, four and two at eigenvalues zero, one and two respectively. No operator A on this full eight-dimensional carrier can obey both $Q = A^\dagger A$ and $[Q, A^\dagger] = A^\dagger$. Indeed the adjoint commutator sends A from the eigenvalue-one space into the eigenvalue-zero space. On the former space $A^\dagger A = Q$ would make A an isometry, which cannot map four dimensions into two. This excludes identifying that particular repair count with one quadratic mode occupation; it excludes neither multiple mode generators nor different source observables or larger carriers.

Counted transitions can supply a weaker positive statement. If a record basis has count labels n_x and an operator T has nonzero entry T_{yx} only when $n_y = n_x + 1$, then $[\mathrm{diag}(n_x), T] = T$ exactly, for arbitrary transition amplitudes. This follows entry by entry. It does not identify $\mathrm{diag}(n_x)$ with $TT^\dagger$: that additional equality constrains transition magnitudes, ranks and degeneracies. The native history packet fixes its diagonal counts and empirical state. Its extension to the complete matrix algebra does not authenticate these occupation-changing operations or their quantum instrument semantics.

Finite covariant response therefore does not force the scalar deformation law, while normalized operations can imply exclusion under an exact quadratic occupation/phase identification. The stated history-count interface supplies neither that identification nor an exhaustive adjoint-paired quantum instrument. These finite results leave open an observer-native construction of the required operations, a different quantization mechanism, and the physical spin–statistics attachment.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- A5 representation theory, compact Lie classification, covering groups, exterior unification, and signed graph operators [127, 99, 92, 157].
- Ad-invariant quadratic kinetic forms on reductive Lie algebras and gauge kinetic terms [152].
- Algebraic renormalization and BV/BRST cohomology [144, 95, 94, 124].
- Analytic S-matrix and resonance-pole theory [112].
- Complex-pole renormalization, Nielsen identities, FJ tadpole schemes, and finite-order electroweak calculations [147, 140, 116, 113, 107].
- DHR/DR/Tannaka reconstruction, compact-group representation theory, and Standard Model anomaly algebra [109, 110, 111, 148, 99, 96, 120].

- Finite fermion Fock space and lattice gauge current identities [98, 130].
- Finite lattice connections, Wilson-loop holonomy, vertex recharting, algebraic two-path character phases (an Aharonov–Bohm analogy only), and characters of cyclic groups [155, 91, 145].
- Finite lattice gauge theory and the Standard Model action [155, 138].
- Finite spectral theory, Kraus instruments and truncated symmetric Fock spaces [122, 132, 104, 98].
- Finite-dimensional Hamiltonian ODE enclosures and nonlinear detector Taylor bounds [139, 123].
- Finite-group equivariant current algebras, compact Lie algebras, normed approximation, and exact algebraic representation certificates [115, 129, 97].
- Finite-group heat kernels, character projectors, and lattice-gauge toy models [108, 145, 131].
- Ginsparg-Wilson fermions and determinant-line formulations [118, 136].
- Hamiltonian lattice gauge theory and constrained quantization [130, 124].
- Icosahedral Galois conjugacy and classical topological degree [102, 137].
- Icosahedral graph theory, finite group actions, equiangular tight frames, and universal optimality [100, 134, 101].
- Invariant quadratic forms on compact semisimple Lie algebras and finite Gibbs product laws [129, 93].
- Osterwalder-Schrader reconstruction [141, 142].
- Product gauge algebras and simple-GUT X/Y generators [117, 146].
- Scalar q oscillators, canonical Fock quantization and the finite trace obstruction [119, 153, 98].
- Standard Model fermion bilinears with charge-diagonal structure [143, 133].
- SU(5)-style exterior matter embeddings and Standard Model anomaly algebra [117, 92, 96, 120].
- Super-Tannakian reconstruction, exterior-algebra matter embeddings, binary icosahedral covers, and anomaly algebra [106, 92, 103, 96].
- Symmetric and exterior Fock representations [114, 98].
- Tannaka duality, representation equivalence, and commuting-diagram methods [105, 128, 110].
- Textbook electroweak Lagrangian assembly of gauge kinetic, Higgs potential, gauge mass, and Yukawa sectors [150, 143, 156].
- two-particle bunching and antibunching [125, 135].
- Weinberg and Wilczek-Zee dimension-six baryon-number-violating operator classification and the SMEFT operator basis [151, 154, 121].
- Z6 center-character descent bookkeeping for Standard Model global gauge-group forms [149, 126].

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AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.